



THE ISLAMIA UNIVERSITY OF BAHAWALPUR

Department of Information Technology

Advanced Research Methodology

Evidence, Design and Integrity in Computing Research

MASTER LECTURE HANDOUT

MS Information Technology • PhD Information Technology

Fall 2026 • 3 Credit Hours • 16 Teaching Weeks

Systematic Review • Design Science • Experiments • Measurement • Reproducibility • Integrity • Publication

This volume is the course of record for the graduate research methodology course that **PhD Regulations 2024 Sec. 10(iv)(a)** requires of every PhD scholar at this university. It is written for both tracks: MS IT scholars working towards a thesis under the *MPhil/MS Degree Regulations-2024* [42], and PhD scholars working towards a dissertation under the *PhD Degree Regulations-2024* [13]. Where the two differ — and they differ in ways that change what you must do — the difference is marked in the text and tabulated in Appendix A.

Dr. Mustafa Hameed

Department of Information Technology

How to Use This Book

A research methods course fails in a characteristic way. Students can recite the difference between a case study and an experiment, and still cannot say what would have to be true for their own study to be wrong. This book is organised to prevent that. Every chapter ends by asking what would falsify the claim it just taught you to make.

The Argument in Six Parts

- **Part I** establishes what a contribution to computing knowledge actually is, and how a vague interest becomes a question that evidence can answer.
- **Part II** is about the literature: finding it systematically, appraising it honestly, and synthesising it into something that did not exist before you read it.
- **Part III** is the catalogue of designs — design science, experiments, repository mining, case studies, surveys, qualitative work, mixed methods, benchmarking — and the grounds for choosing between them.
- **Part IV** is measurement and analysis: making a construct observable, and reasoning correctly from the numbers that result.
- **Part V** is what separates research from advocacy — reproducibility, pre-registration, ethics, integrity, and the rules governing generative AI in your own work.
- **Part VI** is production: the synopsis, the thesis, the paper, and the defence.

The Furniture in Every Chapter

Each chapter carries the same apparatus in the same order, so you can navigate by shape rather than by reading:

Element	What it is for
Learning Outcomes	What you should be able to <i>do</i> , not know, by the end
Before You Start	Which earlier chapters this one assumes
Key Terms	The vocabulary introduced here
Anchor Reading	The one paper to read for this chapter, with a DOI
Regulation	A binding IUB or HEC rule, quoted, with its clause number
Worked Example	A calculation carried through to a number
Common Pitfalls	The mistakes that get papers rejected and theses deferred
Four Traditions	How the chapter's idea looks to each way of knowing
Template	A worksheet to photocopy and fill in
Try It Yourself	A hands-on task with open-source tools
Checkpoint	Self-test; sketches in Appendix E
Chapter Summary	What to retain if you retain nothing else
Review Questions	Harder, discussion-grade questions

Two Conventions That Matter

Worked studies are illustrative. Every study in a *Worked Study* box is constructed for teaching. The numbers are invented and are labelled as such. No result in this book is attributed to a real published study unless it carries a citation.

Open access is claimed only where it is true. A reading marked  OA has a publisher version that is free to read. An author copy posted on a personal page is not open access, and this book does not pretend otherwise — which is itself part of the lesson.

If You Are Reading This to Write a Synopsis

Read Chapter 3, Chapter 5 and Chapter 30 first, then Appendix C for the worksheets and Appendix A for the deadlines and thresholds that bind you. Come back for the rest when your design is chosen.

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PART I

Foundations of Inquiry in Computing

Before any method can be chosen, three things have to be settled: what would count as a contribution, what you believe about how knowledge is produced, and what exactly you are asking. Most failed graduate projects failed here, months before the first line of code, and no amount of later sophistication repairs them. This part is short and unglamorous, and it is the part that decides whether the rest of the work has anywhere to go.

What Part 1 Covers

- **Chapter 1 — What Counts as a Contribution.** Claim, evidence, falsifier; eight contribution types and what each owes its reader.
- **Chapter 2 — Philosophy of Science.** The commitments your methods already make on your behalf, and how to keep them consistent.
- **Chapter 3 — Research Questions and Constructs.** FINER, PICOC, operationalisation, and the alignment table that exposes an unanswerable question.
- **Chapter 4 — The Publication System.** Where computing work belongs, what the regulations require of it, and how to avoid destroying it in a predatory venue.

What Counts as a Contribution in Computing Research

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 1 — Foundations of Inquiry in Computing

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Distinguish** research from development, and say which one a given piece of work is.
- **Classify** a computing paper into one of eight contribution types, and name the evidence each type owes its reader.
- **State** any proposed study as a claim–evidence–falsification triple.
- **Diagnose** the four ways a proposal fails to be a contribution at all.
- **Relate** your own topic to the relevance criteria your dissertation will be judged against.

🔗 Before You Start

None. This is where the course starts. It assumes you have a degree in computing and a vague idea of what you want to work on — which is exactly the situation this chapter is designed to fix.

📖 Key Terms in This Chapter

contribution • claim • evidence • falsification • artefact • empirical study • replication • benchmark • delta • novelty • significance • validation

1.1 The Sentence That Is Not Yet Research

Almost every graduate scholar arrives with a sentence like one of these:

- “I want to work on federated learning for healthcare.”
- “My topic is blockchain in supply chain management.”
- “I will apply deep learning to intrusion detection.”

None of these is a research topic. Each is a *subject area* — a region of the literature, not a question about it. You could spend three years inside any of them and produce nothing a dissertation examiner would accept, because none of them contains a claim that could turn out to be wrong. The distance between that sentence and a dissertation is the distance this chapter covers. It is shorter than it looks, and it is crossed by asking three questions in order.

📖 Research contribution

A **contribution** is a claim about the world, supported by evidence gathered in a way that could have failed to support it, which somebody who already knows the field did not know before.

Every clause in that sentence is doing work. *A claim*: not a topic, not an activity, not a system. *Supported by evidence*: not by plausibility or enthusiasm. *Could have failed*: if no possible result would have changed your conclusion, you have not tested anything. *Did not know before*: the field, not you.

1.1.1 Research is not development, and the difference is not difficulty

Building a working federated-learning system for three hospitals is hard. It may take a year, require real skill, and produce something genuinely useful. It is still not, by itself, research — and confusing the two is the single most common reason a synopsis is sent back by a Departmental Research Committee.

	Development	Research
Succeeds when	The thing works	The claim is tested, whichever way it comes out
A negative outcome	Is a failure	Is a result, if the design was sound
Asks	Can I build this?	Is this true, and how would I know?
Ends with	A deliverable	A claim plus its evidence plus its limits
Is judged by	Users	Peers who know the alternatives

The same project can be either. “I built a federated-learning framework for three hospitals” is development. “Federated training on non-IID clinical data partitioned by hospital loses less accuracy under FedProx than under FedAvg, and the gap widens with partition skew” is a claim — one that a careful experiment could refute. The system is now the *instrument*, not the result.

The instrument and the result

Design science research (Chapter 9) complicates this usefully: there, the artefact genuinely *is* a contribution. But even then it contributes by demonstrating something — that a class of problem is solvable this way, that a design principle transfers, that a trade-off can be relaxed. The artefact still has to carry a claim. What is never a contribution is an artefact accompanied only by the assertion that it was built.

1.2 Eight Kinds of Contribution

Computing research does not have one shape. A paper claiming a new algorithm owes its reader completely different evidence from a paper claiming that developers misunderstand static analysis warnings. Knowing which kind of thing you are producing tells you what you must show.

Type	The claim takes the form	Evidence it owes
Technique or algorithm	“This method achieves X under conditions C, better than the best prior method on dimension D”	Correctness argument or proof; comparison against a <i>current</i> baseline on data the baseline was not tuned on; ablation showing which component earns the gain
Artefact / system	“A system with properties P is achievable, and building it revealed design principle Q”	The artefact, plus an evaluation matched to the claim (Chapter 9); the principle stated separately from the code
Empirical study	“In population P, X is associated with Y” or “treatment T changes outcome O”	A design that could have shown no effect; effect size with an interval, not only a p-value; threats to validity (Chapter 24)

Type	The claim takes the form	Evidence it owes
Theory or model	“These phenomena are explained by mechanism M”	Derivation; predictions that differ from rival explanations; at least one confrontation with data
Methodology	“Research of type R is better conducted this way”	Demonstration on real studies; comparison with existing practice; honest statement of where it does not apply
Dataset or benchmark	“This resource enables questions that could not be asked before”	Construction and sampling documented; datasheet [20]; baseline results; known biases and limits
Replication	“The original finding does / does not hold under conditions C”	Faithfulness to the original protocol, or a documented and justified deviation; the same analysis, pre-specified
Survey / review	“The state of knowledge on Q is S, and the gaps are G”	A protocol, a reproducible search, and a synthesis that is more than a list (Chapter 5)

! Owing the wrong evidence

Most rejections come from delivering the evidence of one type while claiming another.

- A *system* paper that reports only accuracy numbers, with no design principle a reader could reuse — the artefact was built, but nothing was learned.
- A *technique* paper whose baseline is the 2016 method because the 2024 one was harder to install. The comparison is not wrong, it is irrelevant.
- An *empirical* paper reporting $p < 0.05$ on 12 participants with no effect size and no power analysis (Chapter 18).
- A *survey* that tabulates 60 papers without ever answering a question. A table is not a synthesis.

📖 Anchor Reading: Shaw on what reviewers actually look for

Mary Shaw, *Writing Good Software Engineering Research Papers*, ICSE 2003 [54].

Why it matters. Shaw analysed what ICSE accepted and rejected and extracted the implicit rules: every paper answers a *kind* of question, produces a *kind* of result, and validates it in a *kind* of way — and reviewers reject papers where those three do not match each other. It remains the clearest short statement of what a computing contribution is.

What to extract. Her three tables of question types, result types and validation types. Write down which cell of each your own project sits in. If you cannot, your project is not yet defined.

1.3 The Claim–Evidence–Falsification Triple

Here is the operational test. Before anything else — before the literature review, before choosing a method, certainly before writing code — write three lines:

Claim	One sentence, in the indicative, that could be true or false.
Evidence	What you would have to observe for a reasonable sceptic to accept it.
Falsifier	What you could observe that would force you to withdraw it.

If the third line is empty, you do not have a research project. This is the single most useful diagnostic in the course, and it costs five minutes.

Worked Example: From a subject area to a testable claim

Round 0. “I want to work on federated learning for healthcare.” No claim. No falsifier. Not research.

Round 1. “I will build a federated learning system for hospitals in Punjab.” Now there is an activity, still no claim. The falsifier would be “the system does not compile”, which nobody would publish.

Round 2. “Federated learning preserves privacy while maintaining accuracy.” A claim at last — but not yours, and not testable as stated. Preserves privacy against which adversary? Maintains accuracy relative to what? This is a slogan from the abstract of somebody else’s paper.

Round 3. “On clinical tabular data partitioned by hospital, FedProx loses less accuracy than FedAvg as label skew increases.”

- **Claim:** stated, comparative, conditional on a data regime.
- **Evidence:** accuracy of both algorithms across a controlled range of skew, on more than one dataset, with repeated runs and intervals.
- **Falsifier:** the two curves are within each other’s confidence intervals across the whole skew range, or FedAvg wins at high skew.

Round 4 — the one that makes it a contribution. Round 3 may already be known. A literature search (Chapter 5) will tell you. Suppose it is known for image data but not for tabular clinical data with missingness that is itself hospital-dependent. *That* is the delta, and now the claim is yours:

“On clinical tabular data where missingness is correlated with the partitioning hospital, the FedProx advantage over FedAvg reported for image benchmarks does not hold, because the proximal term penalises exactly the local adaptation that hospital-specific missingness requires.”

Note what happened: the claim acquired a *mechanism*. It now predicts something specific, which makes it far easier to refute — and therefore far more worth testing. All numbers here are illustrative; this is a shape to imitate, not a result to cite.

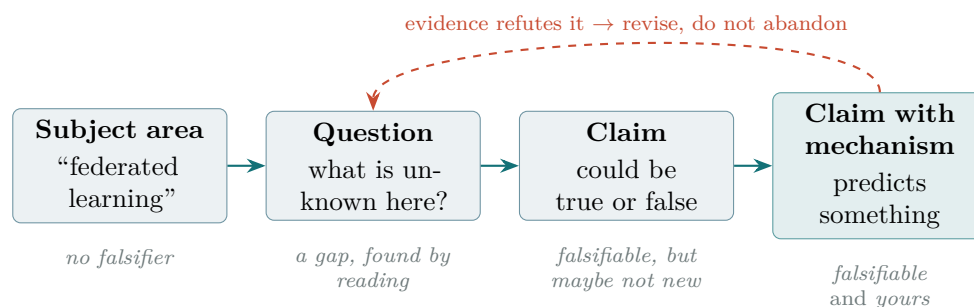


Figure 1.1: From a subject area to a claim worth testing. The dashed return path is not failure: a refuted claim that was sharp enough to refute has told you something, and the revised question is better than the one you started with.

1.4 Novelty, and the Word “Delta”

Reviewers and examiners talk about the **delta**: the difference between what was known before your work and what is known after. Three things are worth knowing about it.

The delta is measured against the literature, not against your own prior ignorance.

Work is not novel because it is new to you. This is what Chapter 5 is for, and it is why the review comes before the design.

A small, well-established delta beats a large, unsupported one. A convincing demonstration that a widely believed result fails under one common condition is a better contribution than an unvalidated framework claiming to solve everything. Examiners have seen many of the latter.

Negative and null results are deltas. If a well-powered study finds no effect where the field assumed one, that is knowledge. It is harder to publish, which is a defect of the publication system (Chapter 26) rather than of the result. Pre-registration is partly a device for protecting such findings.

What your dissertation is judged on — PhD Regulations 2024, §14

§14(i): the dissertation “must be an original and innovative contribution to knowledge that contributes to solving socioeconomic problems”.

§14.1, on choosing the research area, requires that it:

- “Correspond to the community needs at regional and local levels and comply with the priority national research agenda”;
- “Reflect the basic and pure research”;
- “Signify emerging areas of research that coincide preferably with sustainable development goals (SDGs)”.

Relevance is a constraint, not a decoration

It is tempting to treat §14.1 as a paragraph to add to the synopsis after the real work is planned. Read it instead as a filter applied at the start: of the several defensible claims you could test, prefer the one whose answer would change a decision somebody in this region actually makes.

That is also good research strategy, independent of the regulation. A claim tied to a real setting comes with a real population to sample, real practitioners to interview, and a real baseline to beat — which is to say, it comes with its evidence already half-specified. A claim floating free of any setting has to invent all of that.

1.5 Four Ways to Have No Contribution

The four non-contributions

1. “We applied X to Y.” Applying an existing method to a new dataset is not a contribution unless something about Y made the outcome uncertain. The test: before running it, could you have confidently predicted the result? If yes, you have done an exercise, not a study. The rescue is to find what makes Y hard — and that becomes the mechanism in your claim.

2. Benchmark chasing. “Our model reaches 94.2%, beating the previous 94.0%.” Without an ablation showing *which* change earned the 0.2, and an interval showing it is not run-to-run noise, this is not a finding. Chapter 22 takes this apart.

3. The unevaluated tool. A tool, a framework or an architecture, presented with a feature list and screenshots. The claim is implicitly “this is good”, and nothing in the paper could have shown it was not.

4. The survey that is a list. Sixty papers, tabulated by year and venue, with a closing paragraph saying more research is needed. A review contributes when it *synthesises* — reconciles disagreements, explains why results differ, and identifies a gap precisely enough that somebody could fill it.

Four Traditions — a contribution Across Ways of Knowing

Empirical

A contribution is a measured effect, reported with its uncertainty and the conditions under which it was observed. The delta is a number somebody did not have.

Design Science

A contribution is an artefact plus the design knowledge it demonstrates. The delta is a class of problem now known to be solvable this way, and the principle that made it so.

Qualitative

A contribution is an account that renders a setting intelligible — concepts, categories or a theory grounded in what participants actually do. The delta is understanding, and it is judged by whether it illuminates, not by whether it generalises.

Formal / Analytical

A contribution is a proved property: a bound, an impossibility result, a correctness argument, a complexity class. The delta is a theorem, and the evidence is the proof.

Try It Yourself: Classify five abstracts

Take five recent papers from a venue in your area — IEEE Xplore, the ACM Digital Library or arXiv will do. For each, in no more than three lines:

1. Which of the eight contribution types is it? (Some are two.)
2. Write its claim as a single indicative sentence. If you cannot find one in the abstract, look in the introduction; if it is not there either, note that.
3. What would have falsified it?

Then do the same for your own intended project. The discomfort you feel doing step 3 for your own work is the point of the exercise. Keep this sheet — it becomes portfolio piece P1 (Appendix H).

Checkpoint — Test Yourself

1. A colleague proposes: “I will develop a machine learning model to predict student dropout at our university.” Is this a contribution as stated? What single addition would make it one?
2. Give an example of a result in computing that would be worth publishing even though the effect measured was zero.
3. Why does a technique paper owe an ablation, while an empirical study owes a threats-to-validity section? What is each protecting the reader against?
4. Your literature search shows your intended claim was published in 2019. Name three legitimate contributions still available to you.

 Chapter Summary

- A contribution is a claim, plus evidence that could have failed to support it, plus something the field did not already know.
- Development asks whether a thing can be built; research asks whether a claim is true. The same project can be either. An artefact contributes by carrying a claim, never by existing.
- Eight contribution types recur in computing, and each owes its reader a different kind of evidence. Most rejections are a mismatch between the type claimed and the evidence supplied.
- Write every project as claim, evidence, falsifier. An empty falsifier line means there is no project yet.
- The delta is measured against the literature, which is why the review comes first. A sharp small delta beats a vague large one.
- Your dissertation is additionally judged on relevance to community needs, the national research agenda and the SDGs. Treat that as a filter applied early, not a paragraph added late.

 Review Questions

1. Shaw argues that reviewers reject papers where question type, result type and validation type do not match. Construct a plausible computing paper in which each is individually reasonable but the three are mismatched, and say what a reviewer would object to.
2. A design science researcher and an empiricist disagree about whether a novel architecture, evaluated only by a proof-of-concept implementation, is a contribution. Argue both positions, then state the condition under which you think each is right.
3. “Negative results are contributions” is widely asserted and rarely acted on. Identify two structural reasons the publication system resists them, and one mechanism that partially corrects for this.
4. **PhD Regs 2024 §14.1** requires research areas to align with community needs and the national research agenda. Some would argue this constrains basic research, which the same clause also requires. Is there a genuine tension? Defend your answer with reference to a specific topic in your own area.

Philosophy of Science for Computer Scientists

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 1 — Foundations of Inquiry in Computing

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Define** ontology, epistemology and methodology, and explain how a choice at one level constrains the next.
- **Locate** a computing study within five research paradigms from its methods alone.
- **Distinguish** deduction, induction and abduction, and identify which one design science actually uses.
- **Justify** your own paradigm choice in the terms a Departmental Research Committee will use.
- **Recognise** the two paradigms already encoded in the IUB dissertation quality criteria.

🔗 Before You Start

Chapter 1 — particularly the idea that different contribution types owe different evidence. This chapter explains *why* they differ.

📖 Key Terms in This Chapter

ontology • epistemology • axiology • methodology • paradigm • positivism • post-positivism • interpretivism • critical realism • pragmatism • deduction • induction • abduction • falsification • demarcation

2.1 Why This Chapter Is Not Optional

Computing scholars tend to regard philosophy of science as decoration — a paragraph to insert in Chapter Three of the thesis, copied from a textbook, declaring the study “positivist” because that word appeared in a paper the supervisor liked. Examiners read a great many such paragraphs and are not fooled by any of them.

There is a practical reason to take this seriously, and it has nothing to do with sounding scholarly. **Your methods already commit you to a philosophical position, whether or not you state one.** If you run a controlled experiment, you have assumed that the effect you are measuring exists independently of your measuring it, and that it is stable enough to reappear for the next investigator. If you conduct unstructured interviews and build categories from them, you have assumed something quite different — that the phenomenon of interest is partly constituted by how participants understand it. Those assumptions are not compatible, and a thesis that runs both without noticing produces a Chapter Three that contradicts Chapter Five.

📋 The four questions, in order

- **Ontology** — what kinds of thing exist? Is “technical debt” a real property of a codebase, or a metaphor practitioners use to talk about their discomfort?

- **Epistemology** — what can count as knowledge of those things, and what is the relationship between the knower and the known? Can I know about technical debt without participating in the team that has it?
- **Axiology** — what is the role of values? Should the researcher be neutral, or is neutrality itself a position?
- **Methodology** — given the answers above, what procedures are legitimate?

The order matters: each answer constrains the next. Choosing a method first and back-filling a philosophy is the standard error, and it is visible.

2.2 Five Paradigms, and What They Look Like in Computing

Paradigm	What exists	What counts as knowledge	Typical computing work
Positivism	A single reality, independent of observers, governed by regularities	Verified observation; law-like generalisation	Algorithm complexity; performance benchmarking; anything with a ground truth that nobody disputes
Post-positivism	A single reality, but knowable only imperfectly and probabilistically	Conjectures that survive attempts at refutation; results always provisional	Most empirical software engineering and HCI experimentation; the working position of Chapter 10 and Chapter 19
Interpretivism / constructivism	Multiple realities, constructed by participants	Understanding of meaning in context; the researcher is part of the instrument	Studies of how teams adopt tools; why a process fails in one organisation and works in another (Chapter 14)
Critical realism	A real world with underlying causal mechanisms, only partly observable	Explanation by mechanism: what generative structure would produce what we see?	Case study research seeking <i>why</i> an intervention worked here and not there (Chapter 12)
Pragmatism	Whatever is useful for the problem; the ontological question is deferred	What works, judged against a purpose	Design science (Chapter 9); most mixed-methods work (Chapter 15)

💡 The honest position of most computing research

Very little computing research is strictly positivist, though a great many theses claim to be. Positivism proper holds that we can verify universal laws by observation — a position largely abandoned in the philosophy of science by the mid-twentieth century.

What most quantitative computing research actually practises is **post-positivism**: there is a real effect; our measurement of it is imperfect; we advance by proposing something that could be refuted and failing to refute it; every conclusion carries an interval rather than a full stop. That is the position underlying null hypothesis testing, confidence intervals, replication, and the entire apparatus of Chapter 24.

If you write “this study adopts a positivist paradigm” and then report a confidence interval, you have described one philosophy and practised another. Write post-positivist, and mean it.

2.3 Three Ways to Reason, and the One Nobody Teaches

Mode	Moves from	In computing research
Deduction	General rule $\rightarrow$ particular prediction	Theory says caching helps read-heavy workloads; therefore this workload should speed up; test it. Conclusion is certain <i>if</i> premises hold
Induction	Particular observations $\rightarrow$ general rule	Forty repositories show the pattern; perhaps it holds generally. Conclusion is probable, never certain, and the next repository can refute it
Abduction	Surprising observation $\rightarrow$ best explanation of it	Latency spikes every 40 minutes; what mechanism would produce exactly that? Generates the hypothesis that the other two then test

Abduction is the mode that actually starts most research, and it is almost never named in methods chapters. You notice something that does not fit, and you reach for an explanation that would make it unsurprising. That explanation is a conjecture, and it earns nothing until deduction turns it into a prediction and evidence tests the prediction.

💡 Design science is abductive, not inductive

This matters for how you write up design science work (Chapter 9). A designer confronting a problem does not induce a solution from observations; they conjecture an artefact whose properties, *if* the conjecture about the problem is right, would resolve it. The artefact is a hypothesis in executable form. Evaluation is the deductive test of it.

Writing this honestly — “we conjectured that the bottleneck was X, designed Y on that assumption, and evaluated whether Y removed the bottleneck” — is far more convincing than the usual retrospective narrative in which the correct design appears fully formed and the evaluation confirms it.

2.4 Falsification, and Its Limits

Popper’s answer to the problem of induction is the intellectual backbone of Chapter 1’s falsifier line: no number of confirming observations establishes a universal claim, but one genuine counter-example refutes it, so science advances by proposing bold conjectures and trying hard to break them [48].

This is the right working attitude, and it is why “what would falsify this?” is the most useful question in the course. But three qualifications matter for a computing researcher.

Falsification is rarely clean in practice. When an experiment contradicts a prediction, you may reject the hypothesis — or the measurement instrument, the sample, the implementation, or an auxiliary assumption. This is the Duhem–Quine problem, and it is entirely familiar to anyone who has had a benchmark disagree with a paper and spent a week finding the configuration difference. Rejecting a claim honestly requires ruling out those alternatives, which is precisely what a threats-to-validity section does.

Fields do not abandon a framework on one refutation. Kuhn’s account of normal science, anomaly and paradigm shift [37] describes what actually happens: anomalies accumulate and

are tolerated until an alternative framework is available. You will see this in your own literature. A result that contradicts a dominant approach is not, on its own, going to displace it, and a dissertation that claims otherwise will be met with scepticism.

Not all good computing research is falsificationist at all. An impossibility proof is not refuted by evidence; a qualitative study generating a grounded theory is not trying to test a conjecture; an artefact demonstrating feasibility establishes an existence claim. Falsification is the right discipline for the empirical traditions, not a universal test of scholarly worth.

2.5 Is Computing a Science?

The demarcation question — what separates science from non-science — lands awkwardly on our discipline, and it is worth having a position because examiners occasionally ask.

Computing is at least three things at once. Where it proves theorems about computability and complexity, it behaves like mathematics: knowledge is a priori and evidence is proof. Where it builds systems to specification, it behaves like engineering: success is fitness for purpose. Where it studies what happens when people, organisations and machines interact — which is most of software engineering, information systems and HCI — it behaves like an empirical science, with all the messiness that implies.

The practical consequence is that *you must say which one your study is*, because the standard of evidence differs. Confusion here is a real source of examiner disagreement: a formal researcher may find an empirical study unrigorous because it proves nothing, while an empiricist may find a formal result irrelevant because it was never confronted with data. Both are applying their own tradition's standard, and both are wrong to apply it universally. The *Four Traditions* box in every chapter of this book exists to keep that distinction visible.

Two paradigms, written into the regulations — PhD §14.3

The regulations do not use the word paradigm, but they encode the distinction exactly, by giving *separate* quality criteria for the two kinds of work.

§14.3(A), qualitative research asks whether the research “illuminate[s] the subjective meaning, actions and contexts of those being researched”; whether the design was adapted responsively to real-life settings; whether the description is detailed enough for a reader to interpret meaning and context; and whether “subjective perceptions and experiences [are] treated as knowledge in their own right”.

§14.3(B), quantitative research asks about reliability (“are the results repeatable?”), validity (“does it measure what it says?”), internal validity, external validity — both ecological and population — and replicability.

§14.3(C) then requires *appropriateness of the methods to the aims of the study*. That clause is the one that penalises a mismatch between your stated paradigm and your actual practice.

Three ways the philosophy chapter goes wrong

Paradigm as decoration. A paragraph asserting a paradigm, followed by a study that ignores it. The test: delete the paragraph. If nothing about the rest of the thesis would have to change, it was decoration.

Method-first, philosophy-backfilled. Choosing a survey because surveys are familiar, then justifying it philosophically afterwards. Detectable because the justification never rules anything out — it is compatible with every method the student might have chosen.

“Mixed methods” as a way of avoiding the question. Combining qualitative and quantitative work is legitimate and often excellent (Chapter 15). But pragmatism is a

considered position about deferring ontological commitment in favour of what serves the problem — not a licence to hold incompatible assumptions simultaneously without noticing. A good mixed methods study says explicitly what each strand is for and how the two are brought together.

Worked Example: Locating a study from its methods alone

A paper reports: semi-structured interviews with 14 developers at three firms, transcripts coded openly then axially, categories member-checked with participants, no statistical testing, findings presented as five themes with extended quotations.

Ontology. Multiple constructed realities — the study is interested in how these developers understand their situation, not in a rate or an effect. **Epistemology.** Knowledge is co-produced; the interviewer is part of the instrument, which is why member checking matters. **Axiology.** Values are acknowledged rather than eliminated; the researcher's position is disclosed. **Paradigm.** Interpretivist.

Now notice what would be a category error: asking whether these 14 developers are “representative” and computing a margin of error. Statistical representativeness is a positivist criterion imported into a study that never claimed it. The right questions are whether the sample produced the kind of knowledge needed to understand the structures at work, whether the description is thick enough to interpret, and whether saturation was reached (Chapter 14) — which is, almost word for word, **PhD Regs 2024 §14.3(A)**.

Four Traditions — philosophical commitment Across Ways of Knowing

Empirical

Post-positivist by default. A real effect exists; our estimate of it is uncertain; we report the uncertainty and expect to be replicated.

Design Science

Pragmatist. Truth is deferred in favour of utility for a purpose; the artefact is an abductive conjecture and evaluation is its test.

Qualitative

Interpretivist or critical realist. Either meaning is constructed and must be understood in context, or real mechanisms underlie observed events and the task is to explain them.

Formal / Analytical

Rationalist. Knowledge is a priori, evidence is proof, and empirical observation is irrelevant to the truth of the theorem — though very relevant to whether anyone should care about it.

Try It Yourself: Read the philosophy off three papers

Choose three papers in your area that use visibly different methods — ideally one experiment, one qualitative study, one formal or design paper.

For each, *without reading any explicit statement of paradigm*, infer from the methods alone: what the authors take to exist, what they accept as evidence, and what role they give themselves. Then find whatever paradigm claim the paper makes, and compare.

You will frequently find either no statement at all, or one that does not match the practice. Note which, and why you think it happened. Bring the mismatch to the seminar.

? Checkpoint — Test Yourself

1. A study measures the effect of code review on defect density across 200 projects and reports a 95% confidence interval. Which paradigm is being practised, and why is it not, strictly, positivism?
2. Explain the Duhem–Quine problem using an example from benchmarking.
3. Which mode of reasoning generates a hypothesis, and which two test it? Why is the generating one so rarely described in methods chapters?
4. Your supervisor says the study must be “objective”. Under interpretivism, what does that request mean, and what would you offer instead?

✓ Chapter Summary

- Your methods commit you to a philosophy whether you state one or not. The purpose of stating it is to keep the thesis internally consistent.
- Ontology constrains epistemology, which constrains methodology. Choose in that order; back-filling is visible to examiners.
- Five paradigms cover nearly all computing research. Most quantitative computing work is post-positivist, not positivist, and should say so.
- Abduction generates hypotheses; deduction and induction test them. Design science is abductive, and reads better when written that way.
- Falsification is the right discipline for empirical work but is complicated by the Duhem–Quine problem and does not apply to formal or purely exploratory work.
- **PhD Regs 2024 §14.3** already encodes the qualitative / quantitative split as two separate quality checklists, and §14.3(C) penalises a mismatch between the paradigm you claim and the one you use.

✍ Review Questions

1. A thesis states a positivist paradigm, uses a survey instrument validated in another country without re-validation, and reports significance tests without effect sizes. Identify the philosophical inconsistency and the two methodological consequences that follow from it.
2. Critical realism is sometimes proposed as the natural paradigm for software engineering case study research. Construct the argument for that claim, then give the strongest objection to it.
3. Is a formal proof of an algorithm’s correctness a scientific contribution under Popper’s demarcation criterion? Does your answer embarrass the criterion, or the question?
4. **PhD Regs 2024 §14.3(A)** treats “subjective perceptions and experiences as knowledge in their own right”. Reconcile this with §14.3(B)’s demand for replicability, given that a single dissertation may contain both strands.

Research Questions, Hypotheses and Constructs

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 1 — Foundations of Inquiry in Computing

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Classify** a research question into one of six types and name the designs each type admits.
- **Apply** the FINER criteria to reject or repair a candidate question.
- **Decompose** a question into PICOC elements and use them to build a search string.
- **Operationalise** a construct into variables, measures and an instrument, and say where the definition could be contested.
- **Construct** an alignment table linking every research question to its data and its planned analysis.
- **Decide** whether your study should state hypotheses at all.

🔗 Before You Start

Chapter 1 for the claim–evidence–falsification triple, and Chapter 2 for why the question type constrains the method. This chapter turns both into a procedure.

📖 Key Terms in This Chapter

research question • FINER • PICOC • construct • operationalisation • variable • independent and dependent variable • moderator • mediator • confounder • null and alternative hypothesis • directional hypothesis • alignment table

3.1 The Alignment Chain

Everything in a competent study is a chain, and a study fails at its weakest link rather than at its most sophisticated one. A perfectly executed mixed-effects model cannot rescue a measure that does not capture the construct the question asked about.

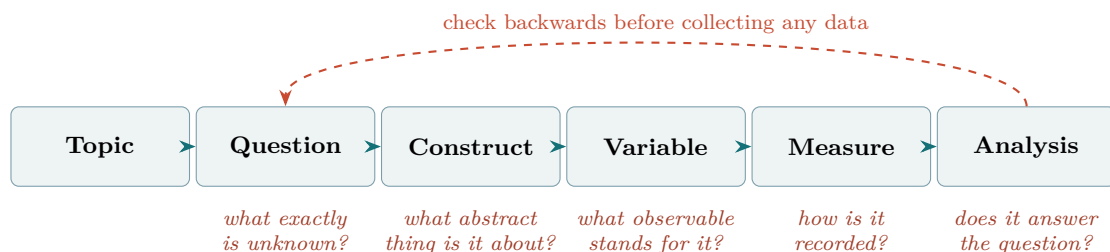


Figure 3.1: The alignment chain. Build it left to right, then verify it right to left: given this analysis, on this measure, of this variable, standing for this construct — have I answered the question I asked? Most fatal design errors are visible on that backward pass, and only on it.

3.2 Six Kinds of Question

The type of question you ask determines the designs available to you. Asking a causal question and then running a correlational study is not a small mismatch — it is the flaw that makes a study unpublishable.

Type	Form	Designs that can answer it
Exploratory	What is going on here? What forms does X take?	Qualitative study, grounded theory, exploratory case study (Chapter 14, Chapter 12)
Descriptive	How prevalent is X? What is the distribution of Y?	Survey with probability sampling; repository mining; systematic mapping (Chapter 13, Chapter 11, Chapter 5)
Relational	Is X associated with Y?	Correlational study, observational repository analysis (Chapter 11)
Causal	Does X <i>cause</i> a change in Y?	Controlled experiment; quasi-experiment with an identification strategy (Chapter 10, Chapter 11)
Design	Can an artefact with properties P be built, and what does building it teach?	Design science research with a matched evaluation (Chapter 9)
Evaluative / comparative	Which of A and B performs better under conditions C?	Benchmarking; controlled comparison; classifier comparison across datasets (Chapter 16, Chapter 22)

Anchor Reading: Matching method to question

Steve Easterbrook, Janice Singer, Margaret-Anne Storey and Daniela Damian, *Selecting Empirical Methods for Software Engineering Research*, in *Guide to Advanced Empirical Software Engineering*, Springer, 2008 [15].

Why it matters. It is the clearest treatment of the point this chapter turns on: the method is chosen by the question and the philosophical stance, not by preference or familiarity. It also connects method choice back to the paradigms of Chapter 2 explicitly, which most methods texts do not.

What to extract. Their mapping from question type to method, and their discussion of what each method cannot tell you. Write the second one into your threats section before you start.

3.3 FINER: Five Tests a Question Must Pass

A question can be well-formed and still be a bad choice for *you*, in *this* degree, with *these* resources. The FINER criteria [29] catch that early, when changing course is still cheap.

	Test	What kills a question here
Feasible	Subjects, time, expertise, money, access	300 developers needed; you can recruit 20. A dataset that requires an agreement no one will sign. A study needing 18 months when you have 9
Interesting	To the field, not only to you	Nobody outside your supervisor’s office would change any belief on hearing the answer
Novel	Adds to what is known	Answered in 2019 and again in 2022. Only Chapter 5 can tell you this
Ethical	Approvable, and defensible	Deception without debriefing; scraping personal data; a design whose control arm withholds something people need (Chapter 27)
Relevant	To practice, policy or subsequent research	Fails PhD Regs 2024 §14.1 : no connection to community need, the national research agenda, or an SDG

! Feasibility is checked with numbers, not optimism

“Feasible” is the criterion students assess by feel, and it is the one that ends degrees. Do the arithmetic before committing: an a-priori power calculation (Chapter 18) for a quantitative design, or an estimate of interviews-to-saturation for a qualitative one, multiplied by the realistic recruitment rate you can demonstrate — not the one you hope for.

If the arithmetic says the study is infeasible, that is a finding, arrived at in week 10 rather than month 20. Portfolio piece P3 asks you to record exactly such a decision, and Appendix H gives credit for it.

3.4 PICOC: Decomposing the Question

PICOC breaks a question into the parts that drive both the design and, later, the literature search [46]. It originates in clinical research, and in computing the Comparison and Context elements do most of the work.

	Element	Example (illustrative)
Population	Who or what is studied	Professional developers performing code review
Intervention	The technique, tool or treatment	An LLM-based review assistant
Comparison	The alternative it is measured against	Unassisted review; static analysis alone
Outcome	The effect that matters	Defects correctly identified; false positives raised
Context	Setting, domain, constraints	Industrial teams, safety-critical C code, time-boxed reviews

💡 The Comparison element is where most computing questions leak

“Does an LLM assistant improve code review?” has no C. Improve compared with what — no review at all, unassisted review, a linter, a second human reviewer? Each comparison makes it a different study with a different conclusion, and a paper that leaves C implicit will be attacked on exactly this point.

The same omission is what makes a literature search return thousands of irrelevant hits: without C, there is no term to constrain the query (Chapter 5).

Template: C1 — Problem, gap and question

Copy this into your portfolio (P1) and fill it in. One page, no more.

1. The problem, in the world. Who has it, and what does it cost them?

2. What is currently known. Two or three sentences, with citations.

3. The gap. What specifically is *not* known? “Little research exists” is not a gap; name the missing condition, population or comparison.

4. The question. One sentence, ending in a question mark.

5. Question type. Exploratory / descriptive / relational / causal / design / evaluative.

6. Why the answer matters. What decision changes?

Template: C2 — FINER and PICOC worksheet

PICOC

Population

Intervention

Comparison

Outcome

Context

FINER — for each, write the evidence, not a yes.

Feasible

Interesting

Novel

Ethical

Relevant

3.5 From Construct to Measure

A **construct** is the abstract thing you care about: code quality, developer productivity, trust in an automated system, technical debt, usability. None of these can be observed. What you observe is a **measure**, and the distance between the two is where studies quietly go wrong.

Operationalisation

Operationalisation is the explicit statement of the rule by which an abstract construct becomes a recorded observation.

It is a claim, and it can be wrong. “Developer productivity = commits per week” is an operationalisation; it is also, demonstrably, a bad one, because it rewards splitting work and penalises the reading and design that good development requires. Chapter 17 treats validity of measures properly; the point here is that the choice must be *visible* so it can be argued with.

Worked Example: Narrowing a topic in four rounds

Round 1 — the topic. “Code review and software quality.” Unbounded. Thousands of papers.

Round 2 — add a population and a context. “Code review practices in small software houses in Pakistan.” Now bounded, but still not a question: there is no outcome and nothing that could come out either way.

Round 3 — add an outcome and a comparison. “Do teams in small software houses that use checklist-guided code review find more defects than teams using unguided review?” A causal-comparative question, with a falsifier: no difference, or fewer.

Round 4 — confront feasibility, and revise honestly. A between-teams experiment needs many teams. Suppose you can access four. The power calculation (Chapter 18) says four teams cannot detect any plausible effect. Three legitimate moves:

- **Change the unit of analysis.** Randomise *reviews* within teams rather than teams, using a within-subjects design (Chapter 10). Many more units, at the cost of a possible carryover effect that must then be controlled.
- **Change the question type.** Ask an exploratory question instead: how do reviewers in these firms actually decide what to comment on? Four teams is ample for that (Chapter 14).
- **Change the outcome.** If defects found is too noisy at this scale, a construct closer to the intervention — review coverage, comment specificity — may be measurable with the units available.

All three are honest. Proceeding with four teams and reporting $p = 0.31$ as “no significant difference” is not: an underpowered null result is not evidence of no effect (Chapter 19).

3.5.1 Variables and their roles

Role	Definition	Example
Independent	What you manipulate or contrast	Review style: checklist vs unguided
Dependent	What you measure as the outcome	Defects correctly identified
Moderator	Changes the <i>size</i> of the effect	Reviewer experience: the checklist may help novices more
Mediator	Explains <i>how</i> the effect happens	Attention to error-prone constructs
Confounder	Associated with both, and not controlled	Task difficulty, if harder code happened to get checklists
Control	Held constant to remove it as an explanation	Same codebase, same time box

Confounders are the reason Chapter 10 exists: randomisation is the only technique that handles confounders you have not thought of.

3.6 Hypotheses, and When Not to Have Them

A **hypothesis** is a specific, testable prediction derived from the question. Stating one commits you in advance, which is its whole value — see Chapter 26 on why a prediction registered before the data is worth more than the same prediction offered after.

H_0 (null)	Checklist and unguided review do not differ in mean defects identified.
H_1 (alternative, directional)	Checklist-guided review identifies more defects than unguided review.

A directional hypothesis is stronger and should be used when theory genuinely predicts a direction — but it must be committed to beforehand, because choosing the direction after seeing the data is a way of halving your p-value without gaining any evidence (Chapter 26).

⚠ Exploratory studies should not invent hypotheses

Not every study has hypotheses, and manufacturing them is a common way to damage otherwise good work.

An exploratory qualitative study is trying to discover what the relevant categories *are*; it cannot predict relations between categories it has not yet found. A systematic mapping study characterises a literature. A design science project asks whether an artefact can be built and what it teaches. None of these needs a null hypothesis, and bolting one on invites an examiner to ask why it was not tested properly.

State research questions. Add hypotheses only where you are genuinely predicting, in advance, a specific relation — and then test exactly those, reporting anything else you find as exploratory.

3.7 The Alignment Table

This is the single most valuable page in a synopsis, and the one examiners turn to first. It forces every question to have a plan, and exposes any question that does not.

	Question	Data	Source	Analysis
RQ1	How do reviewers decide what to comment on?	12 interviews	4 firms, purposive	Thematic analysis, two coders
RQ2	Does a checklist change comment specificity?	240 reviews	Within-subjects, randomised	Mixed-effects model, reviewer as random effect
RQ3	Does the effect depend on experience?	As RQ2	As RQ2	Experience $\times$ condition interaction

💡 How to use it as a diagnostic

Read each row across and ask the four killing questions.

1. Can this *analysis* produce an answer to this *question*? A thematic analysis cannot answer “how much”; a regression cannot answer “what forms does this take”.
2. Does this *data* exist, and can you get it? Write the number, and where it comes from.
3. Does the analysis assume something the data cannot supply — independence, when the same reviewer contributes 20 reviews?
4. Is any row empty? An empty cell is not an oversight, it is an unanswerable question. Delete the question or fix the row.

An alignment table built after data collection is a formality. Built before, it is the cheapest possible way to discover that your study cannot work.

❗ Four questions that cannot be answered as written

The compound question. “How do developers use static analysis tools and what are the barriers to adoption and how can these be overcome?” Three questions, and the third is not a research question at all — it asks for a recommendation, which is a conclusion, not a finding.

The question with no possible negative. “How can machine learning improve intrusion detection?” presupposes that it does. Ask whether, and under what conditions.

The question already answered. Detectable only by Chapter 5, which is why the review precedes the design.

The question whose construct cannot be measured in your setting. “Does technical debt reduce team morale?” — both constructs are contested, neither has an agreed measure, and you have one semester. Either commit to defending an operationalisation properly (Chapter 17), or choose constructs with validated instruments.

🔗 Four Traditions — the research question Across Ways of Knowing

📊 Empirical

The question names variables and a population, and its answer is a quantity with an interval. It must be settled before data collection, because after that it is no longer a prediction.

🔧 Design Science

The question is a design problem: can an artefact meeting requirements R be constructed for context C, and what generalisable design knowledge does constructing it yield?

 Qualitative

The question is open and expected to evolve. What is going on here, and how do participants understand it? Fixing it too early forecloses exactly what the study exists to discover.

 Formal / Analytical

The question is a conjecture about a formal object: does this property hold, what is the bound, is the problem decidable? It is settled by proof, and no sample size applies.

 Try It Yourself: Build your own alignment table

Using template C1 and C2, take your own topic through the four rounds of the worked example. Then draw the alignment table with a row per research question.

Now apply the backward pass from Figure 3.1: start at the analysis column and read right to left. For at least one row you will find that the analysis does not answer the question as stated. Fix the row — by changing the question, the data or the analysis — and note in one sentence which you changed and why. That sentence is the beginning of portfolio piece P3.

 Checkpoint — Test Yourself

1. Convert this into a PICOC question: “I want to study whether containers help deployment.”
2. A study asks “Does pair programming improve code quality?” and measures quality as lines of comments per function. State the operationalisation, and give the strongest objection to it.
3. Why is a directional hypothesis chosen after seeing the data worth less than the same hypothesis chosen before, even though the sentence is identical?
4. Give a research question in your own area for which stating a null hypothesis would be inappropriate, and explain why.

 Chapter Summary

- Topic, question, construct, variable, measure, analysis form a chain. Build it forwards; verify it backwards before collecting anything.
- Six question types, each admitting different designs. A causal question answered by a correlational study is a fatal mismatch, not a minor one.
- FINER rejects questions early and cheaply. Feasibility is settled with a power calculation or a saturation estimate, not with optimism.
- PICOC decomposes the question and later drives the search string. The Comparison element is the one computing questions most often omit.
- Operationalisation is a claim that can be wrong. Make it visible so it can be argued with.
- The alignment table — question, data, source, analysis — is the cheapest diagnostic in the course. An empty cell means an unanswerable question.
- State hypotheses only where you genuinely predict in advance. Exploratory work states questions instead.

 Review Questions

1. Easterbrook et al. argue method follows from question and stance. Give a case in computing where the same question could legitimately be answered within two different paradigms, and say what each answer would and would not establish.

2. A reviewer objects that your operationalisation of “maintainability” as cyclomatic complexity is invalid. Construct the strongest version of their objection, then the strongest defence available to you, and say what additional evidence would settle it.
3. The alignment table exposes underpowered rows before data collection. Why, then, do so many published studies contain them? Give one structural and one psychological reason.
4. Under what circumstances is it legitimate to change a research question *after* data collection has begun? Distinguish carefully between the legitimate cases and HARKing (Chapter 26).

The Publication System and Where Work Belongs

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 1 — Foundations of Inquiry in Computing

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Explain** why computing publishes in conferences when most disciplines do not, and what follows for your publication plan.
- **Interpret** the HJRS W, X, Y categories and state exactly what your degree requires of you.
- **Evaluate** a journal for legitimacy using three independent checks.
- **Critique** the impact factor and the h-index using the arguments the research-metrics community itself makes.
- **Describe** the peer review process from the reviewer's side, and why that view improves your writing.

🔗 Before You Start

Chapter 1, for what a venue is being asked to accept. Read this chapter early even though you will publish late: **PhD Regs 2024 §15(iv)** constrains *when* your paper counts, and that constraint has to be planned for, not discovered.

📖 Key Terms in This Chapter

peer review • single- and double-blind review • rebuttal • HJRS categories • impact factor • quartile • h-index • preprint • predatory journal • paper mill • open access • gold, green and diamond • APC • DOI • ORCID

4.1 Computing Publishes Differently

In most sciences the conference is a place to present work in progress and the journal is the archival record. In computing, for large parts of the field, it is the other way round: the top conferences are selective, archival, and carry more prestige than most journals [63]. ICSE, CHI, NeurIPS, SIGCOMM and their peers run full peer review with acceptance rates below 25%, and their proceedings are the record.

This is a genuine disciplinary difference, and it creates a specific problem for a scholar under Pakistani regulations, because the requirement in **PhD Regs 2024 §15** is written in terms of *journals*.

📌 The publication requirement — PhD Regulations 2024, §15

“For award of PhD degree, a PhD researcher shall be required to publish research articles meeting the following criteria:

- (a) One research article in W category journal or two research articles in X category journals, for Science disciplines;
- (b) One research article in X category journal or two research articles in Y category journals, for Social Science disciplines.”

And further: the researcher “shall be the first author”; the article “shall be relevant to the PhD research work”; it “shall be published *after approval of the research synopsis*”; and it “shall be published in a relevant research journal”.

There is no publication requirement for the MS IT degree ([42], §47–48). MS scholars who publish do so for their own reasons, which are good ones, but not to satisfy a regulation.

⚠ Three traps in §15 that cost scholars a year

“After approval of the research synopsis.” A paper accepted before your synopsis is approved does not count towards the requirement, however good it is. If you arrive with work in progress, sequence the synopsis first.

“First author.” A paper on which your supervisor is first author does not count for you. Agree authorship order in writing, at the start — Chapter 28 gives the criteria to base that conversation on.

Conference papers are not journal articles. A CORE A conference paper may be a far better contribution than a W-category journal article and still not satisfy §15. Plan for both: target the conference for the field, then a substantially extended journal version for the regulation. Extension must be genuine — Chapter 28 on text recycling and salami slicing.

4.2 Reading the Journal Categories

The HEC Journal Recognition System [27] assigns Pakistani-relevant categories to journals, and these are what §15 refers to. Categories are reassigned periodically, so a journal that was X when you submitted may be something else when you are examined.

Category	Broadly	What it means for you
W	The strongest tier recognised	One paper here satisfies the Science-discipline requirement outright
X	Strong, widely indexed	Two papers here satisfy it; one satisfies the Social Science requirement
Y	Recognised	Two satisfy the Social Science requirement; insufficient alone for Science
Z / unlisted	Not recognised for this purpose	Does not count, whatever the journal claims about itself

ⓘ Check the category on the day you submit, and again on acceptance

Categories move. Verify the journal’s current HJRS status before submitting, record a dated screenshot with your synopsis file, and check again at acceptance. A scholar who submits to an X-category journal and is examined after it has been downgraded has a problem that is entirely avoidable with two minutes of record-keeping.

4.3 How Peer Review Actually Works

Understanding review from the inside is the fastest way to improve your writing, because it tells you what the reader is trying to do.

Stage	What happens
Desk check	An editor or chair rejects out-of-scope, unformatted or obviously unsound submissions without review. A substantial fraction of submissions die here, mostly avoidably
Assignment	Two to four reviewers, chosen for topic fit. They are unpaid, busy, and reading your paper late at night
Review	Each reviewer forms a recommendation and writes it up. Most decide provisionally within the first two pages
Discussion	Reviewers see each other’s reports and argue. This is where a paper with one hostile review is often saved — or lost
Rebuttal	Many conferences let authors respond before the decision. Chapter 32 on how to write one
Decision	Accept, minor or major revision, or reject. For journals, revision cycles can repeat

💡 What a reviewer is looking for, in order

Reviewers are not looking for reasons to accept. They are looking, quickly, for the answers to four questions — and if a paper does not supply them in the abstract and introduction, it starts from behind.

1. **What is the claim?** If a reviewer cannot state your claim after the introduction, nothing later will rescue the paper.
2. **Is it new?** Which is to say: does the related work section demonstrate you know what has been done, and locate your delta in it?
3. **Is the evidence adequate to the claim?** The Chapter 1 match between contribution type and evidence type.
4. **Do the authors know the limits?** A candid threats section builds far more confidence than an absent one. Reviewers find the flaw regardless; the only question is whether you found it first.

Blinding varies. **Double-blind** review hides author identity from reviewers and is now common in software engineering and machine learning; **single-blind** hides only the reviewer; **open** review publishes reports and sometimes names. Each trades bias against accountability differently, and none eliminates bias.

4.4 Metrics, and Why the Metrics Community Distrusts Them

You will be asked for your h-index. You will be told to publish in “high-impact” journals. It is worth knowing that the strongest criticisms of these numbers come from bibliometricians, not from disgruntled authors.

Metric	What it is	What it is not
Journal Impact Factor	Mean citations in year t to items published in $t-1$ and $t-2$	A property of any individual paper. Citation distributions are heavily skewed: most papers in a high-IF journal are cited far below its IF
Quartile (Q1–Q4)	The journal's rank within a subject category	Comparable across categories, or stable — category assignment changes the quartile
h-index	h papers with at least h citations each	Comparable across fields or career stages; it cannot decrease, and it rewards longevity over quality
Citation count	How often a work was cited	A measure of correctness or importance. Work is cited for being wrong, for being convenient, and for being famous

Anchor Reading: The Leiden Manifesto

Diana Hicks, Paul Wouters, Ludo Waltman, Sarah de Rijcke and Ismael Rafols, *Bibliometrics: The Leiden Manifesto for research metrics*, Nature 520, 2015 [26].

Why it matters. Ten principles from the people who build these indicators, on how they should and should not be used. Principle 1 — quantitative evaluation should *support* qualitative expert assessment, not replace it — is the one most often violated, in Pakistan and everywhere else.

What to extract. The two principles that bear directly on you: normalise for field before comparing anything, and never use journal-level metrics to judge an individual article. Pair it with DORA [14], which commits signatories to the second.

Living with a system you have criticised

There is an obvious tension: this section argues journal-level metrics should not judge individual work, and §15 requires you to publish in a journal-category system. Both things are true, and the resolution is not cynicism.

Treat the category requirement as a *floor* that must be cleared, and venue *fit* as the thing that actually determines whether your work is read. A W-category journal whose readership has no interest in your subfield will satisfy the regulation and be cited by nobody. A well-fitted X-category journal in your area, plus a second X paper, satisfies the same regulation and reaches the people whose work you want to influence.

4.5 Predatory Venues

A **predatory** journal charges a fee and provides no meaningful peer review, while presenting itself as legitimate. The volume is substantial [55], the solicitation emails are relentless, and graduate scholars under publication pressure are the target market.

The cost is not the fee

Publishing in a predatory journal is worse than not publishing. The paper will not satisfy §15, because such venues are not HJRS-recognised. It cannot be resubmitted elsewhere, because it is already published — so the work is destroyed, not merely wasted. And the line stays on your CV permanently, where every future committee will see it.

4.5.1 Three independent checks

Use all three. Any one alone can be fooled.

Check	What to do
HJRS	Is the journal listed, and in which category? For your degree this is decisive regardless of anything else
DOAJ [12]	For open-access journals: is it indexed? DOAJ applies criteria on peer review, editorial process and transparency
Think. Check. Submit. [60]	Work its checklist. Do you recognise the editorial board? Is the peer review process described? Are fees stated clearly <i>before</i> submission? Is the publisher a member of COPE?

! Warning signs, in rough order of reliability

- An unsolicited email praising a paper of yours, and promising review in 72 hours. Genuine journals do not work this way.
- A fee disclosed only after acceptance.
- An editorial board of people who cannot be found, or who did not know they were on it.
- Scope so broad it spans unrelated disciplines in one title.
- A claimed impact factor from an organisation nobody has heard of.
- Spelling errors on the journal’s own home page.

Related, and increasingly serious: **paper mills** sell authorship on fabricated manuscripts. Buying authorship is fabrication and falls under **PhD Regs 2024 §32** as academic misconduct, with the consequences Chapter 28 describes.

4.6 Preprints and Open Access

A **preprint** is a manuscript posted publicly before or during peer review — arXiv is the computing norm. It establishes priority, makes work citable immediately, and costs nothing. Most computing venues permit it; check the specific venue, and note that a preprint is *not* a publication for the purposes of §15.

Route	How it works	What to watch
Gold	Publisher makes it free; author or funder pays an APC	APCs of USD 2000–5000 are common and rarely fundable locally
Green	Publisher version is paywalled; author deposits an accepted manuscript in a repository	Free. Check the embargo and which version is permitted
Diamond	Free to read <i>and</i> free to publish; funded institutionally	Legitimate and growing; verify via DOAJ like any other
Hybrid	A subscription journal offering paid open access per article	Widely criticised; pays twice for the same content

💡 Green open access is the practical route

For a scholar without a grant, green is almost always the answer: publish in the best-fitting recognised journal, then deposit the accepted manuscript in an institutional or subject repository. It costs nothing, is permitted by most publishers after an embargo, and makes the work readable by people who cannot pay. Register an **ORCID** identifier first, so that everything you deposit accumulates against one unambiguous identity — particularly valuable where names are commonly shared.

📍 Four Traditions — where work is published Across Ways of Knowing

📊 Empirical

Journals and archival conferences with full review. Increasingly expects data and analysis scripts to be deposited alongside (Chapter 25).

🔧 Design Science

Information systems journals and design-oriented conferences. Evaluation is what reviewers scrutinise, and the artefact itself is often archived as a badged artefact.

🗨️ Qualitative

Journals that allow the length qualitative reporting requires; a four-page limit cannot carry the thick description **PhD Regs 2024 §14.3(A)** asks for.

√* Formal / Analytical

Theory conferences and journals where proofs are refereed in detail and the archival version carries full appendices.

🔧 Try It Yourself: Assess five venues

Pick five journals or conferences plausibly relevant to your topic — include at least one that has emailed you an invitation.

For each, record: HJRS category and the date you checked; DOAJ listing if open access; whether it passes Think. Check. Submit.; the review model; typical time to first decision; APC if any; and whether papers you have actually cited appear there.

That last column matters more than the others combined. A venue that has published nothing you cite is probably not where your readers are. Rank the five for your first paper and justify the ordering in three sentences.

❓ Checkpoint — Test Yourself

1. A PhD scholar in a Science discipline publishes one X-category paper before synopsis approval and one after. What does §15 require of them now?
2. Why does a high journal impact factor tell you little about a specific paper in that journal?
3. Give two reasons a top-tier conference paper may be a better contribution than a W-category journal article, and one reason it may still not help you graduate.
4. You are invited to submit to a journal that promises review within one week and lists an “International Impact Factor” of 6.2. State the three checks you would run and what each would likely show.

✅ Chapter Summary

- Computing treats top conferences as archival, but §15 is written in terms of journals. Plan for both; a genuinely extended journal version of a conference paper is the standard route.
- PhD: one W or two X papers for Science, first author, after synopsis approval. MS: no publication requirement at all.
- Verify the HJRS category at submission and at acceptance, and keep dated evidence.

- Reviewers ask four questions in order: what is the claim, is it new, is the evidence adequate, do the authors know the limits. Write so the first two are answerable from the abstract.
- Journal-level metrics do not measure individual papers; the Leiden Manifesto and DORA say so from inside the field. Clear the category floor, then choose on fit.
- Predatory publication destroys work rather than wasting it. Check HJRS, DOAJ and Think. Check. Submit. — all three.
- Green open access plus an ORCID is the practical, free route to being read.

Review Questions

1. Computing's conference-centric culture is defended as fast and selective, and criticised as producing rushed, unreplicated work. Argue both sides, then state which you find more persuasive for your own subfield and why.
2. §15 requires publication before submission of the dissertation. Give the strongest argument for this requirement and the strongest argument against it, considering effects on both quality and on scholars' timelines.
3. Double-blind review reduces some biases and introduces others. Name one of each, and propose a practical modification that would reduce the second without sacrificing the first.
4. If journal-level metrics should not be used to evaluate individual researchers, what should be? Design an evaluation scheme for a hiring committee that is both fair and actually operable at scale, and identify what it would cost.

PART II

Finding and Synthesising Knowledge

You cannot know whether your question is novel, and you cannot state a delta, without knowing what has already been established. That is a claim about a literature — and like any other claim it needs a method, or it is just an impression formed from whatever you happened to read. This part treats the review as a study in its own right: with a protocol, a reproducible search, an appraisal stage, and a synthesis that produces something that did not exist in any single paper.

What Part 2 Covers

- **Chapter 5 — Systematic Literature Review.** Protocol, search strings, screening, quality appraisal, extraction, synthesis, PRISMA.
- **Chapter 6 — Grey Literature and Scoping Reviews.** Snowballing, multivocal reviews, and when the practitioner literature is the evidence.
- **Chapter 7 — Reference Management and Bibliometrics.** Zotero and BibTeX in practice; mapping a field; why the metrics community distrusts its own metrics.
- **Chapter 8 — Reading and Appraising a Paper.** The three-pass method, critical appraisal, and reproducing one claim to see whether it holds.

Systematic Literature Review and Systematic Mapping

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 2 — Finding and Synthesising Knowledge

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Distinguish** a systematic review, a systematic mapping study and a narrative review, and choose between them.
- **Write** a review protocol before conducting the search.
- **Construct** a search string from PICOC and adapt it to the syntax of four digital libraries.
- **Screen** studies against explicit criteria and report inter-rater agreement.
- **Appraise** study quality with an instrument appropriate to the designs under review.
- **Synthesise** findings without resorting to vote counting.
- **Report** the whole process as a PRISMA 2020 flow.

🔗 Before You Start

Chapter 3 for PICOC, which drives the search string, and for the question types — a review's questions have types too. Chapter 1 for why a review that does not synthesise is not a contribution.

📖 Key Terms in This Chapter

systematic review • systematic mapping study • protocol • primary study • search string • digital library • inclusion and exclusion criteria • screening • inter-rater agreement • Cohen's kappa • quality assessment • data extraction • synthesis • vote counting • publication bias • PRISMA

5.1 A Review Is a Study

The narrative review — read widely, form an impression, write it up — is how most people first encounter the literature, and it has a fatal property: nobody can check it. Its conclusions depend on which papers the author happened to find, which is unrecorded, and on which they found convincing, which is unexamined. Two competent people reviewing the same field this way routinely reach different conclusions, and there is no way to adjudicate.

A **systematic review** treats the literature as data. It has a protocol written in advance, a search that another person could re-run, explicit criteria applied by more than one screener, an appraisal stage, and a synthesis that is answerable to what was found rather than to what was remembered. It is a study whose subjects happen to be papers, and every methodological concern in this book applies to it: bias, reliability, validity, reproducibility.

	Systematic review	Mapping study	Narrative review
Question	Specific: what is the effect of X on Y?	Broad: what has been researched, and where are the gaps?	Whatever the author chooses
Search	Exhaustive within a stated scope	Exhaustive, usually broader	Unstated
Appraisal	Quality assessed, may exclude	Usually classification only	None
Synthesis	Answers the question	Categorises the field; counts by facet	Argues a position
Output	An answer with confidence in it	A map, and identified gaps	An orientation
Use it when	The question is answerable from existing evidence	The area is unfamiliar or immature	Writing a thesis background section, honestly labelled

💡 Which one does your synopsis need?

Almost always a mapping study first, then a systematic review if the map shows enough primary studies to answer a specific question.

A scholar entering an unfamiliar area cannot yet write the specific question a systematic review requires. The map produces it: once you can see that 60 papers address microservice testing but only four evaluate anything in production, the specific question writes itself — and you have the evidence that it is a genuine gap, which is exactly what Chapter 1 demands of a delta.

📄 Anchor Reading: The two guideline documents to work from

Barbara Kitchenham and Stuart Charters, *Guidelines for Performing Systematic Literature Reviews in Software Engineering*, EBSE-2007-01, 2007 [34].  OA

Matthew Page and colleagues, *The PRISMA 2020 statement*, BMJ 372:n71, 2021 [43].  OA

Why they matter. Kitchenham and Charters is the procedure the software engineering community actually follows; PRISMA 2020 is the reporting standard reviewers and editors increasingly require, and its flow diagram is expected in any serious review. They are complementary: the first tells you what to do, the second what to report.

What to extract. From Kitchenham, the three-phase structure — plan, conduct, report — and the protocol contents. From PRISMA, the 27-item checklist and the flow diagram, and from PRISMA-S [49] the requirement to report the full search string for *every* database, not just one.

5.2 The Protocol Comes First

The protocol is written, reviewed by your supervisor, and fixed *before* the search runs. This is not bureaucracy: it is the same device as pre-registration (Chapter 26), and it exists to stop you deciding what counts as relevant after you have seen which papers support the conclusion you were hoping for.

📄 Template: C3 — Systematic review protocol

1. Rationale. Why is this review needed now? What existing reviews are there, and why are they insufficient?

2. Review questions. Numbered. Each must be answerable from published primary studies.

3. PICOC. From template C2.

4. Search strategy. Databases, date range, and the exact string per database (attach as an appendix).

5. Inclusion criteria. What must a study have to be included?

6. Exclusion criteria. Stated separately, not as negations of the above.

7. Screening procedure. Who screens, at which stages, how disagreement is resolved, and how agreement is measured.

8. Quality assessment instrument. Which, and whether low quality excludes or is analysed as a moderator.

9. Data to extract. The fields of template C4.

10. Synthesis method. Narrative, thematic, or meta-analysis — decided now, not after seeing the data.

5.3 Building the Search

5.3.1 From PICOC to a string

Take the PICOC elements from template C2. For each element that will constrain the search — usually P, I and C — list synonyms, abbreviations, spelling variants and older terminology. Join synonyms with OR, join elements with AND.

Worked Example: A search string, built and then repaired

Question. What techniques are used to test microservice-based systems, and which have been evaluated in industrial settings?

Attempt 1.

```
microservice AND testing
```

Returns far too much, and misses more. “Microservices” (plural) is not matched by every engine; “micro-service” and “micro service” are used in older papers; “testing” misses “test”, “validation”, “verification”.

Attempt 2.

```
("microservice*" OR "micro-service*" OR "micro service*")
AND
("test*" OR "verification" OR "validation" OR "quality assurance")
```

Better, but “validation” pulls in a large machine-learning literature about model validation that has nothing to do with the question.

Attempt 3 — add the context element.

```
("microservice*" OR "micro-service*" OR "micro service*")
AND
("test*" OR "verification" OR "integration testing"
OR "end-to-end testing" OR "contract testing")
AND
("software" OR "system*" OR "architecture")
```

Then calibrate. Assemble a *quasi-gold standard*: five to ten papers you already know must be included. Run the string. If it does not return all of them, it is not yet good enough — inspect each miss and find the term that would have caught it. Record this calibration in the protocol; it is the only evidence a reviewer has that your search was adequate.

All counts and terms here are illustrative.

Every database has different syntax, and the differences bite

- **IEEE Xplore** supports nested parentheses and * wildcards, but caps command-search string length — long strings must be split and the results unioned.
- **ACM Digital Library** changed its search interface and syntax; strings written for the old one silently return different results.
- **Scopus** and **Web of Science** use field codes (TITLE-ABS-KEY, TS=). Omitting the field code searches full text and returns thousands of irrelevant hits.
- **SpringerLink** handles Boolean operators inconsistently across content types.
- **Google Scholar** should not be used as a primary source: results are not reproducible between users or dates, and there is no export of a complete result set. Use it for snowballing (Chapter 6), not for the systematic search.

Record the exact string, the database, the fields searched, the date, and the number of hits — separately for each. PRISMA-S requires it [49], and without it your review is not reproducible.

5.4 Screening

Stage	What is read	Typical retention
Deduplication	Metadata	Overlap between IEEE, ACM and Scopus is substantial; a reference manager does this
Title screen	Title only	Fast, high recall, deliberately generous
Abstract screen	Title and abstract	Where most exclusion happens
Full-text screen	The whole paper	Slowest; the only stage where you can apply criteria about method or data
Snowballing	Reference lists and citing papers	Adds papers the string missed (Chapter 6)

5.4.1 Criteria that can actually be applied

An inclusion criterion is usable when two people applying it to the same paper reach the same answer. “Studies of high quality” is not usable. “Studies reporting at least one quantitative outcome measure with a sample size stated” is.

💡 Measuring agreement, and what the number means

Have a second screener — your supervisor, a fellow scholar — independently screen a sample, typically 10–20% of records. Report Cohen’s κ , which corrects for agreement expected by chance:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

where p_o is observed agreement and p_e the agreement expected if both screeners were guessing at their observed marginal rates.

Landis and Koch’s conventional bands [39] — 0.41–0.60 moderate, 0.61–0.80 substantial, above 0.81 almost perfect — are widely quoted and were explicitly described by their authors as arbitrary. Treat them as a rough signal, not a threshold to be passed.

A low κ is useful information: it means the criteria are ambiguous. The fix is to repair the criteria and re-screen, not to discuss individual papers until you agree. Chapter 14 covers κ and its alternatives in more depth.

5.5 Quality Appraisal

Not every published study deserves equal weight. Appraisal asks whether a study’s design supports its conclusions — not whether you like its findings.

The instrument must match the designs under review. A checklist built for randomised trials cannot appraise a case study. In computing, a practical approach is a short list of design-appropriate questions, each scored 1 / 0.5 / 0:

-
- | | |
|----|--|
| Q1 | Are the aims and research questions clearly stated? |
| Q2 | Is the design appropriate to those aims (Chapter 3)? |
| Q3 | Is the sample or subject selection described and justified? |
| Q4 | Are the measures defined well enough to be reproduced (Chapter 17)? |
| Q5 | Is the analysis appropriate, and are assumptions checked (Chapter 19)? |
| Q6 | Are threats to validity identified and discussed (Chapter 24)? |
| Q7 | Are the data or artefacts available (Chapter 25)? |
-

⚠️ Decide the role of quality scores in the protocol

There are two defensible uses, and you must choose in advance:

- **As an exclusion threshold** — studies below a stated score are excluded. Simple, but discards evidence and the threshold is arbitrary.
- **As a moderator** — all studies are included, and you check whether high- and low-quality studies give different answers. More informative, and it turns a methodological worry into a finding.

What is not defensible is scoring quality after synthesis and using it to explain away studies that disagreed with your conclusion.

5.6 Extraction and Synthesis

📋 Template: C4 — Data extraction form

One row per included study. Pilot it on five papers before extracting all of them; you will always find fields you cannot populate.

Study ID / citation	_____
Year, venue, venue type	_____
Research question(s)	_____
Study type	_____
Population / subjects, N	_____
Intervention	_____
Comparison	_____
Outcome measures	_____
Effect size / result	_____
Context (industrial / academic / simulated)	_____
Quality score	_____
Threats acknowledged	_____
Artefacts available	_____
Relevance to RQ1 / RQ2 / RQ3	_____

5.6.1 Three ways to synthesise, and one to avoid

Narrative synthesis groups studies and describes patterns in prose. It is the default in software engineering, because primary studies are rarely commensurable enough for anything stronger. Done well it explains *why* studies disagree — different populations, different operationalisations, different baselines. Done badly it is a list.

Thematic synthesis codes findings across studies and builds themes, using the machinery of Chapter 14. Appropriate when the primary studies are themselves qualitative.

Meta-analysis pools effect sizes statistically, weighting by precision. It requires that studies measure comparable constructs with reported effect sizes and variances, which in computing is uncommon — but where it is possible, it is far stronger than anything else. The **metafor** package in R is the standard open-source tool.

! Vote counting is not synthesis

“Twelve studies found a positive effect, five found none, therefore the effect is real.” This reasoning is wrong, and it is everywhere.

It ignores sample size, so a study of 12 participants counts as much as one of 1,200. It ignores effect size, so a trivial effect counts as much as a large one. And it is biased against exactly the truth it claims to establish: several underpowered studies finding nothing are perfectly compatible with a real effect, and a set of small studies finding an effect is compatible with publication bias (Chapter 26) and no effect at all.

If effect sizes are reported, pool them. If they are not, say so, describe the pattern, and treat the absence of pooled evidence as itself a finding about the state of the field.

5.7 Reporting: the PRISMA Flow

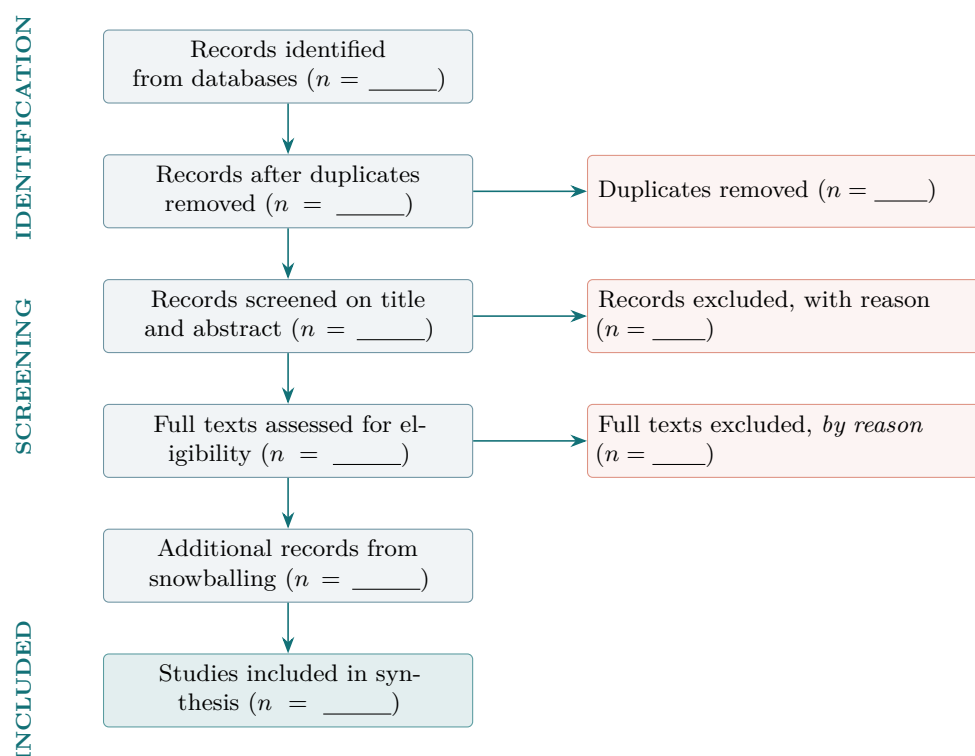


Figure 5.1: The PRISMA 2020 flow, as a form to fill in. The single most scrutinised box is the third exclusion node: reporting *how many* full texts were excluded is routine, reporting them *broken down by reason* is what distinguishes a review a reader can trust. Reproduced in fill-in form as template C5.

💡 What the numbers reveal to a reader

An experienced reviewer reads your flow diagram diagnostically.

A very high title-screen exclusion rate — 4,000 down to 60 — suggests the search string was too broad and probably imprecise. A very low one suggests it was too narrow and may have missed relevant work. A large snowballing yield relative to the database yield says the search string was inadequate, and honest reviews say so. And full-text exclusions with no stated reasons are the clearest possible signal that criteria were applied after the fact.

Why this chapter matters for your synopsis — PhD §13, MS §8

The synopsis is defended before the Departmental Research Committee, and for PhD scholars additionally evaluated by an external referee from a national university, chosen from a panel of six experts supplied by your supervisor (**PhD Regs 2024 §13(ii)**).

The reviewer's first question is always whether the gap is real. A protocol, a search string and a PRISMA flow answer that question with evidence. An assertion that “little work exists in this area” invites the referee to name three papers that do, and that is a bad afternoon.

Four Traditions — reviewing the literature Across Ways of Knowing

Empirical

Pool effect sizes where the primary studies permit it; where they do not, report why not, and treat the incommensurability as a finding about the field's measurement practices.

Design Science

Review artefacts and design knowledge rather than effects: what has been built, for which problem class, evaluated how convincingly.

Qualitative

Thematic or meta-ethnographic synthesis, building second-order constructs across studies. Aims at interpretation, not at a pooled estimate.

Formal / Analytical

Review results and their assumptions: which bounds are known, under which models, and where the open problems sit. Exhaustiveness matters more than sampling.

Try It Yourself: Run the first two stages for real

Using template C3, write a protocol for a review on your own topic. Then:

1. Assemble a quasi-gold standard of five papers you know must be included.
2. Build a search string from your PICOC. Run it on IEEE Xplore, ACM DL and Scopus. Record the exact string, fields, date and hit count for each.
3. Check recall against your gold standard. Repair and re-run. Record how many iterations it took — reviewers of your synopsis will ask.
4. Deduplicate in Zotero or JabRef and screen the first 100 titles.
5. Have a colleague screen the same 100. Compute κ .

If κ is below about 0.6, rewrite the criteria rather than arguing about the papers. This is portfolio piece P2.

Checkpoint — Test Yourself

1. Why must the protocol be fixed before the search is run? Name the specific bias this prevents.

2. Your string returns 12,000 records. Give three distinct repairs and say what each risks.
3. A review reports “143 full texts assessed, 38 included”. What is missing, and why does its absence matter?
4. Explain why nine studies finding an effect and four finding none is not evidence that the effect is real.

Chapter Summary

- A systematic review is a study whose subjects are papers; it has a protocol, a reproducible search and an appraisal stage, and it can be checked.
- Map first when the area is unfamiliar; the map generates the specific question a review needs.
- Write the protocol before searching. It is pre-registration for a review.
- Build the string from PICOC, calibrate it against a quasi-gold standard, and record string, fields, date and hits per database.
- Screen with criteria two people can apply identically; report κ and repair the criteria, not the disagreement.
- Decide the role of quality scores in advance — exclusion threshold or moderator.
- Synthesise; do not vote count. If effect sizes exist, pool them; if not, explain the disagreement.
- Report as a PRISMA 2020 flow, with full-text exclusions broken down by reason.

Review Questions

1. Systematic reviews were imported into software engineering from evidence-based medicine. Identify two features of computing research that make the transfer imperfect, and say how the guidelines accommodate — or fail to accommodate — each.
2. A reviewer excludes all studies scoring below 5 on a 7-point quality instrument, and the remaining studies agree. A critic argues this manufactured the agreement. Assess the criticism and describe an analysis that would settle it.
3. Google Scholar has the best coverage of any source and is excluded from systematic searches. Explain the tension, and propose a defensible policy for its use.
4. Publication bias means the literature you can find is not the research that was done. What can a reviewer in computing actually do about this, given that trial registries do not exist for our field?

Snowballing, Scoping Reviews and Grey Literature

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 2 — Finding and Synthesising Knowledge

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Conduct** backward and forward snowballing from a defensible start set, and state a stopping rule.
- **Choose** between a systematic review, a scoping review, a rapid review and a multivocal review.
- **Justify** the inclusion of grey literature, and appraise it with criteria designed for it.
- **Explain** how publication bias distorts a database-only search, and what partially corrects it.

🔗 Before You Start

Chapter 5. This chapter is the set of techniques that repair the database search, and the set of review types that are appropriate when a full systematic review is not.

📖 Key Terms in This Chapter

snowballing • backward and forward snowballing • start set • scoping review • rapid review • grey literature • multivocal literature review • publication bias • file drawer problem • saturation of search

6.1 Why the Database Search Is Never Enough

A Boolean string matches terminology, and terminology is unstable. Authors in adjacent communities describe the same phenomenon differently; a foundational paper may predate the term now used for its subject; a highly relevant study may use “serverless” where you searched for “function-as-a-service”. No string catches all of this, and a review that relies on one alone has a coverage problem it cannot see.

Snowballing exploits a different signal: what authors themselves considered relevant.

📖 Backward and forward snowballing

Backward snowballing follows the reference list of an included paper to find earlier work it drew on. **Forward snowballing** finds later papers that cite it, using Google Scholar, Scopus or Semantic Scholar.

Together they traverse the citation graph in both directions from a **start set** of known-relevant papers, iterating until no new relevant papers appear.

📖 Anchor Reading: Snowballing as a systematic procedure

Claes Wohlin, *Guidelines for snowballing in systematic literature studies and a replication in software engineering*, EASE 2014 [65].

Why it matters. It makes snowballing a documented procedure rather than incidental reading, and shows it can produce results comparable to a database search — sometimes better, because it is insensitive to terminology drift.

What to extract. How to choose a start set that is diverse in community, venue and year; the iteration procedure; and the discipline of recording, at each iteration, how many papers were examined and how many included.

6.1.1 The procedure

-
- | | |
|--------|--|
| Step 1 | Choose the start set. Diverse in authorship, venue, community and publication year. A start set of five papers by one group traverses only that group's citation neighbourhood. |
| Step 2 | Backward pass. For each paper, scan the reference list. Screen on title, then on the <i>context of the citation</i> — read the sentence that cites it. This is faster and more accurate than reading abstracts. |
| Step 3 | Forward pass. For each paper, list citing works and screen the same way. |
| Step 4 | Iterate. New included papers form the next iteration's set. |
| Step 5 | Stop when an iteration yields no new included papers. Record the iteration count and the yield at each. |
-

💡 Reading the citation context is the trick

The single most useful habit in snowballing: do not screen a reference by its title. Screen it by *what the citing paper says about it*. “Similar to the approach of Ahmed et al. [12], we partition...” tells you far more about the relevance of [12] than its title does, and takes two seconds.

6.2 Four Kinds of Review, and When Each Is Right

Type	Question it answers	Effort	Use when
Systematic review	What is the effect of X on Y?	Months; two screeners	The question is specific and enough primary studies exist
Systematic mapping	What has been researched, and where are the gaps?	Weeks to months	Entering an area; generating the question a review will answer [45]
Scoping review	What is the extent and nature of the evidence?	Weeks	The concept itself is unclear and needs charting before it can be questioned
Rapid review	What does the evidence say, quickly?	Days to weeks	A decision must be made on a deadline; limitations declared explicitly
Multivocal (MLR)	What do researchers <i>and</i> practitioners know?	As a review, plus grey literature appraisal	The topic is practice-led and the academic literature lags it

⚠ A rapid review is legitimate; a sloppy review called rapid is not

Rapid reviews shorten the process deliberately and say exactly where. Acceptable shortcuts: fewer databases, a restricted date range, single screening with a sampled check, no formal quality scoring. What is never acceptable, in any review type: an undocumented search, criteria invented during screening, or a synthesis that does not report what was excluded. Speed is bought from coverage, not from transparency.

6.3 Grey Literature

Grey literature is everything produced outside commercial and academic publishing: technical reports, standards, white papers, engineering blogs, conference talks, documentation, Stack Overflow, theses, and preprints.

For large parts of computing this is not a marginal supplement. Industrial practice frequently runs years ahead of the academic literature — container orchestration, site reliability engineering, and much of modern MLOps were documented in engineering blogs and internal reports long before they were studied. A review of such a topic that excludes grey literature is not conservative; it is describing a different, later world.

📖 Anchor Reading: When and how to include the practitioner literature

Vahid Garousi, Michael Felderer and Mika Mäntylä, *Guidelines for including grey literature and conducting multivocal literature reviews in software engineering*, Information and Software Technology 106, 2019 [19].

Why it matters. It supplies both halves of the problem: a decision procedure for whether grey literature is needed at all, and a quality instrument designed for sources that have no peer review to lean on.

What to extract. Their seven questions for deciding to include grey literature — of which the decisive ones are whether the topic is practice-led, whether the academic literature is sparse, and whether practitioner voice is itself part of the phenomenon — and their tiered classification of grey sources by expertise and outlet control.

6.3.1 Appraising a source with no peer review

Peer-reviewed papers arrive with a weak quality signal already attached. Grey literature does not, so appraisal has to do more work.

Criterion	Question to ask
Authority	Is the author identifiable, and do they have demonstrable standing — a role, a track record, an organisation?
Methodology	Is it stated how the claims were arrived at? A post reporting measured latencies with a described setup differs entirely from one asserting a conclusion
Objectivity	Is there a commercial interest? A vendor blog showing its own product winning a benchmark is evidence about the vendor's claims, not about the product
Date	Is it dated, and is it current for a fast-moving topic?
Novelty	Does it add something the peer-reviewed sources do not?
Impact	Is it cited, referenced or discussed by others in the field?
Outlet type	Tier 1: books, reports, theses. Tier 2: annual reviews, presentations, wikis. Tier 3: blogs, emails, tweets

! Three ways grey literature is misused

Cited as if it were a study. “Blogs report that microservices improve scalability [17]” treats an opinion as a finding. Attribute it: “a practitioner account describes...”. The claim is about what practitioners believe, which may be exactly what you want — but say so.

Included without appraisal. If peer-reviewed studies are quality-scored and blog posts are not, the grey literature has been given an easier standard than the academic literature, which is the opposite of the correct treatment.

Uncitable because it moved. Blogs vanish and get edited. Archive every grey source — the Internet Archive’s Wayback Machine will do — and cite the archived URL with the access date. A reference an examiner cannot follow is not a reference.

6.4 Publication Bias

Studies finding an effect are more likely to be submitted, more likely to be accepted, and published sooner than studies finding nothing. The consequence is that the literature you can find is not the research that was done, and a perfectly executed review of a biased literature returns a biased answer with false confidence. The unpublished null results are the **file drawer**.

In medicine this is partly controlled by trial registries: a study registered before it runs can be chased even if never published. Computing has no equivalent, which makes the problem harder for us, not easier.

? What you can actually do about it

1. **Search for the unpublished.** Theses, technical reports and preprints are where negative results in computing usually end up. This is a strong argument for grey literature even in a review that is otherwise academic.
2. **Look at the shape of the evidence.** If every small study finds a large effect and the only large study finds none, that pattern is what publication bias looks like. Where enough effect sizes exist, a funnel plot makes it visible.
3. **Say so.** A review that names publication bias as a limitation and explains which direction it would push the conclusion is more useful than one that ignores it.
4. **Do not add to it.** Pre-register and report your own null results (Chapter 26).

🔍 Four Traditions — extending the search Across Ways of Knowing

📊 Empirical

Snowballing improves coverage of comparable studies; grey literature matters mainly as a source of unpublished nulls and of effect sizes omitted from published abstracts.

🔧 Design Science

Practitioner sources often *are* the state of the art. Design knowledge frequently appears first in engineering blogs and standards, and a DSR review that ignores them misdescribes the problem space.

💬 Qualitative

Practitioner writing is data about how a community understands its own work, and can be analysed as documents rather than merely cited as evidence.

√* Formal / Analytical

Snowballing is the primary method: results are connected by citation far more reliably than by keyword, and terminology in theory papers is comparatively stable.

🔧 Try It Yourself: Snowball from three seeds

Take three papers from your quasi-gold standard in Chapter 5, chosen to be from different groups and different years.

1. Backward pass: screen every reference by citation context. Record examined and included counts.
2. Forward pass: use Google Scholar or Semantic Scholar. Record the same.
3. Run a second iteration on whatever you included.
4. Compare the result with your database search. How many papers did snowballing find that the string missed? How many did the string find that snowballing missed?

The overlap tells you how good your string was. A large snowballing-only set means the string needs repair — go back and repair it, then report both in the PRISMA flow.

? Checkpoint — Test Yourself

1. Why is a start set of five papers by the same research group a poor one, even if all five are highly relevant?
2. Give a computing topic for which excluding grey literature would produce a misleading review, and say why.
3. A vendor benchmark shows their database is fastest. Under what circumstances, if any, would you include it in a review, and how would you describe it?
4. Publication bias inflates apparent effects. Name a situation in computing research where it could plausibly *deflate* them instead.

📋 Chapter Summary

- Boolean search matches terminology; snowballing follows what authors themselves judged relevant. Use both, and report the yield of each.
- Screen references by citation context, not by title.
- Mapping, scoping, rapid and multivocal reviews are legitimate types with different purposes. Speed is bought from coverage, never from transparency.
- In practice-led computing topics grey literature is not a supplement; excluding it describes a different world.
- Appraise grey sources on authority, methodology, objectivity, date, novelty, impact and outlet tier — and archive them, or they will vanish.
- Publication bias means the findable literature is not the research done. Search for theses and preprints, look at the shape of the evidence, declare the limitation, and do not add to the problem yourself.

🔪 Review Questions

1. Wohlin reports snowballing producing results comparable to database search. If that is so, why do guidelines still require the database search? Give the strongest argument for keeping both.
2. Grey literature inclusion is contested: critics say it imports unreviewed claims into the evidence base. Argue the case against inclusion as strongly as you can, then say what would answer it.

3. Computing has no trial registry. Design one that could plausibly be adopted by a software engineering conference, and identify the incentive problem that would most likely defeat it.
4. A multivocal review finds that practitioners and researchers disagree about whether a technique works. Is that a limitation of the review, or a finding? Defend your answer.

Reference Management and Bibliometrics

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 2 — Finding and Synthesising Knowledge

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Operate** an open-source reference manager as the single source of truth for a review.
- **Write** BibTeX entries that will still resolve in five years.
- **Distinguish** co-citation, bibliographic coupling and co-word analysis, and say what each reveals.
- **Produce** a field map from a Scopus or Web of Science export using open-source tools.
- **Judge** when a bibliometric analysis is a contribution and when it is a substitute for reading.

🔗 Before You Start

Chapter 5 for the review workflow this tooling supports, and Chapter 4 for the critique of citation metrics as evaluation — this chapter uses them as *data* instead.

📖 Key Terms in This Chapter

reference manager • BibTeX and biblatex • citation key • DOI • bibliometrics • co-citation • bibliographic coupling • co-word analysis • co-authorship network • science mapping

7.1 Never Type a Citation Again

Hand-typed references are wrong. Not occasionally — systematically. Volume numbers drift, initials get dropped, a conference is recorded as a journal, and the error propagates because the next author copies your reference list rather than the source. In a review with 80 primary studies, hand management guarantees a corrupted bibliography and hours lost to reformatting when you change venue.

Tool	Licence	Why you might choose it
Zotero	Open source (AGPL)	Browser connector captures metadata and the PDF in one click; group libraries make two-screener review practical; exports BibTeX. The default recommendation
JabRef	Open source (MIT)	Native BibTeX: the <code>.bib</code> file <i>is</i> the database, so there is no export step to go stale. Best if you write in L ^A T _E X and want one file under version control
Mendeley	Proprietary	Widely used; owned by a publisher, with the data-ownership questions that raises

💡 A workflow that survives a three-year degree

1. **One library, from day one.** Every paper you read goes in immediately, including ones you reject — with a note saying why. In month 20 you will not remember, and you will re-read it.

2. **Collections per review stage.** 01-identified, 02-screened, 03-included, and 99-excluded with the reason in each item's note. Your PRISMA counts then come out of the library rather than out of a spreadsheet you forgot to update.
3. **Fix metadata on import.** Automatic capture is good, not perfect. Check the DOI, the venue and the year while the paper is in front of you.
4. **Export refs.bib into the repository** and commit it with the thesis. The `refs.bib` shipped with this handout is exactly that file for this course.
5. **Back it up somewhere that is not your laptop.**

7.1.1 BibTeX entries that keep working

Rule	Why
Stable keys	<code>authoryearfirstword</code> , lowercase. Never renumber; a key is an identifier, not a position
DOI on everything	The only durable link. URLs rot; DOIs resolve
Correct entry type	<code>@inproceedings</code> for conference papers, <code>@article</code> for journals. Getting this wrong misformats the whole reference
Protect capitals	<code>{PRISMA}</code> , <code>{IEEE}</code> , <code>{FAIR}</code> — otherwise many styles lowercase them
Full author lists	Let the style truncate to <i>et al.</i> , not the data. Truncating in the <code>.bib</code> loses information permanently
Archived URL + <code>urldate</code>	For anything grey (Chapter 6)

! The two errors that reach print

Citing something you have not read. You copy a reference from another paper's list; that paper misdescribed it; now you have too. If you cite it, open it. If you genuinely cannot obtain it, cite it as reported in the source you read — “[8], as cited in [23]” — which is honest and rare enough to be noticed favourably.

Letting the manager write the sentence. A reference manager gets the formatting right and the *meaning* not at all. It cannot tell you that the paper you are citing for a claim actually concluded the opposite under conditions you did not read. Chapter 8 is about that.

7.2 Bibliometrics as a Research Method

Chapter 4 argued that citation metrics are poor instruments for *evaluating* people. That is a separate question from using citation data as *evidence about a field* — which is legitimate, well-established, and occasionally a contribution in its own right.

Technique	Two items are linked when	What it shows
Co-citation	Both are cited by the same later paper	The intellectual base a community shares. Backward-looking; identifies foundational works and schools of thought
Bibliographic coupling	Both cite the same earlier paper	Current research fronts. Forward-looking; groups work being done now, before citation has accumulated
Co-word / co-occurrence	Two terms appear together in titles, abstracts or keywords	The conceptual structure and its vocabulary; useful for finding search terms you missed (Chapter 5)

Technique	Two items are linked when	What it shows
Co-authorship	Two authors share a paper	Collaboration structure; which groups exist and whether they talk

💡 The distinction that matters most

Co-citation and bibliographic coupling look similar and answer opposite questions.

Co-citation needs *time*: two papers only become co-cited once enough later work has cited both. It therefore describes the settled past, and a brand-new paper cannot appear in a co-citation map at all.

Bibliographic coupling needs *no* time: two papers published last month that cite the same twenty references are strongly coupled immediately. It is the right tool for “what is happening now”, which is usually what a scholar choosing a topic actually wants to know.

7.2.1 Doing it with open-source tools

Tool	What it is	Use
VOSviewer	Free (not open source)	Point-and-click maps from Scopus, WoS, Dimensions, Lens exports. The fastest route to a publishable map
bibliometrix / biblioshiny	R package, open source	Scriptable and reproducible, which VOSviewer is not. <code>biblioshiny()</code> gives a browser interface over it
CiteSpace	Free	Emphasises temporal evolution and burst detection

Listing 7.1: A minimal reproducible field map. The whole point of scripting it is that the figure in your thesis can be regenerated when you add 40 more records.

```
library(bibliometrix)

# Export from Scopus as BibTeX, or from Web of Science as plain text.
M <- convert2df("scopus_export.bib", dbsource = "scopus", format = "bibtex")

results <- biblioAnalysis(M)
summary(results, k = 20)           # top authors, sources, countries, growth

# Conceptual structure: which terms occur together?
CS <- conceptualStructure(M, field = "ID", method = "MCA",
                          minDegree = 8, clust = "auto")

# Research fronts: bibliographic coupling of documents.
NetMatrix <- biblioNetwork(M, analysis = "coupling",
                           network = "documents", sep = ";")
networkPlot(NetMatrix, n = 40, type = "fruchterman",
            labelsize = 0.7, cluster = "louvain")
```

⚠️ A map is a hypothesis-generator, not a finding

A co-word map will show clusters. Clusters are produced by any clustering algorithm on any data, including noise, and their number depends on parameters you chose. The map tells you where to look; it does not tell you what is there.

Every cluster you report should be validated by *reading* several papers in it and confirming that the label you gave it is the label they would recognise. A bibliometric section that

reports six clusters without demonstrating that anybody read them is the most sophisticated possible way of not doing a literature review.

💡 When bibliometrics is itself the contribution

A bibliometric study can be a legitimate contribution — of the “survey / review” type in Chapter 1 — when it answers a question about the field that could not be answered by reading. Has research on this topic shifted from academia to industry? Do the two communities studying this problem cite each other? Has the geographic distribution of authorship changed as the problem became commercially important?

Those are real questions with real answers. What is not a contribution is a descriptive count of papers per year with a growth curve, which is true of almost every topic and tells nobody anything.

📍 Four Traditions — mapping a field Across Ways of Knowing

📊 Empirical

Bibliometrics is an observational study of the literature: it has a population, a sampling frame with known coverage limits, and confounders — notably that database coverage differs systematically by region and venue.

🔧 Design Science

Maps the artefact space: which classes of solution have been built, by whom, and where practitioner and academic communities diverge.

💬 Qualitative

Treats the corpus as documents to be read closely; the map is a sampling device for choosing which documents deserve that reading.

√* Formal / Analytical

Citation graphs are objects in their own right, and the algorithms that cluster them have properties worth stating — resolution limits, parameter sensitivity, instability across runs.

🔧 Try It Yourself: Map your own field

1. Run your Chapter 5 search on Scopus or Web of Science and export the full records, including references, for at least 200 documents.
2. Load them in **bibliometrix** and produce three artefacts: a co-word map, a bibliographic coupling map of documents, and a table of the ten most locally cited papers.
3. Read the abstracts of three papers from each cluster and name the cluster yourself. Compare your labels with the automatic ones.
4. Extract from the co-word map at least two terms that were *not* in your search string. Add them and re-run the search from Chapter 5.

Step 4 is the reason this chapter sits inside Part II rather than being an optional extra: the map repairs the search.

❓ Checkpoint — Test Yourself

1. Why can a paper published this month appear in a bibliographic coupling map but not in a co-citation map?

2. Give two reasons a reference list should live in a manager rather than a document, beyond convenience.
3. A thesis reports a co-word analysis identifying five research themes. What single piece of evidence would most increase your confidence in those five labels?
4. Scopus and Web of Science have different coverage. How would that difference bias a bibliometric study of research output by country?

Chapter Summary

- Use an open-source reference manager from the first paper, with collections mirroring the review stages, so PRISMA counts fall out of the library.
- Stable keys, DOIs, correct entry types, protected capitals, full author lists, archived URLs for grey sources.
- Cite only what you have opened. If you truly cannot, say “as cited in”.
- Co-citation shows the settled intellectual base; bibliographic coupling shows current fronts and works immediately; co-word shows vocabulary and repairs your search string.
- **bibliometrix** is scriptable and therefore reproducible; VOSviewer is faster but its figures cannot be regenerated automatically.
- A map generates hypotheses. Validate every cluster label by reading, or the analysis is a way of avoiding the literature rather than reviewing it.

Review Questions

1. Chapter 4 argued citation counts are poor measures of individual quality, yet this chapter uses citation data as evidence about fields. Is this inconsistent? Set out the conditions under which aggregate citation analysis is sound even though individual-level use is not.
2. Database coverage is not neutral: Scopus and Web of Science index journals from some regions and languages far more completely than others. What does this imply for a bibliometric claim about research productivity in South Asia?
3. Design a bibliometric study that would answer a question about your own field that could not be answered by reading fifty papers. State the question, the data, the technique, and the threat to validity you would have to address.
4. Clustering a citation network requires parameter choices that change the number of clusters. What reporting practice would make a bibliometric result as trustworthy as a well-reported statistical one?

Reading and Critically Appraising a Paper

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Apply** the three-pass method and stop at the right pass for your purpose.
- **Interrogate** a paper’s claim, evidence and limits rather than summarising it.
- **Take** notes that remain useful two years later.
- **Write** a referee report that is useful to authors and editor alike.
- **Reproduce** one claim from a paper and interpret what happens when you cannot.

🔗 Before You Start

Chapter 1 for the contribution types and the evidence each owes — appraisal is largely checking that match. Chapter 5 for where appraisal sits in a review.

📖 Key Terms in This Chapter

three-pass method • critical appraisal • claim extraction • referee report • reproduction • annotated bibliography • charitable reading

8.1 Reading Linearly Is the Problem

The default way to read a paper — start at the abstract, proceed to the references, highlighting as you go — is slow, and it produces a summary rather than a judgement. Worse, it allocates the same effort to a paper you will discard as to the one your thesis will build on. A graduate scholar who must screen 900 records and deeply understand 30 of them cannot afford that. The alternative is to read in passes of increasing depth, deciding after each whether to continue.

📄 Anchor Reading: The three-pass method

S. Keshav, *How to read a paper*, ACM SIGCOMM Computer Communication Review 37(3), 2007 [33].  OA

Why it matters. Two pages, and it changes how much literature you can handle. It is the most immediately actionable thing in this part of the book.

What to extract. The three passes and their time budgets, and his description of using the method to do a literature survey — which is a different use of the same technique.

Pass	Budget	What you read	What you should be able to say
First	5–10 min	Title, abstract, introduction, section headings, conclusions, references (scanning for ones you know)	Category, context, correctness (plausible?), contributions, clarity. Enough to decide whether to read on

Pass	Budget	What you read	What you should be able to say
Second	About 1 hour	The whole paper, ignoring proofs and detailed derivations. Look carefully at every figure and table	The paper's claim and the evidence offered for it, in your own words. Enough to summarise it to somebody else
Third	4–5 hours	Everything, re-deriving the work: what would <i>you</i> have done, and where do the authors' choices differ?	The paper's hidden assumptions, its unstated limitations, and what a reviewer should have objected to

💡 Which pass, for which purpose

- **Screening for a review** (Chapter 5): first pass only, on hundreds of papers. Anything more is a failure of discipline.
- **Building the related-work section:** second pass, on perhaps 50. You need the claim and the evidence, not the derivation.
- **The three or four papers your own work stands on:** third pass, without exception. If your contribution is a delta from a paper, you must understand that paper better than its reviewers did.
- **Refereeing:** second pass minimum, third pass if you will recommend rejection. You owe an author whose work you are rejecting the effort of having understood it.

8.2 From Summarising to Interrogating

A summary says what the paper claims. An appraisal says whether to believe it. The difference is a set of questions asked in a fixed order.

- 1 **What is the claim?** State it in one sentence, in your own words. If you cannot, that is a finding about the paper.
- 2 **What kind of contribution is this?** (Chapter 1) The type determines what evidence is owed.
- 3 **Is the evidence the right kind?** A causal claim from correlational data fails here, however elegant the analysis.
- 4 **Is the evidence enough?** Sample size, power, number of datasets, number of runs (Chapter 18).
- 5 **What is the comparison?** Against what baseline, and is it current and fairly tuned?
- 6 **What is measured, and does it capture the construct?** (Chapter 17)
- 7 **What would have falsified this?** If nothing, it was not a test.
- 8 **Which threats do the authors name, and which do they not?** The unnamed ones are the interesting ones (Chapter 24).
- 9 **Could I reproduce this?** Are data, code and parameters available (Chapter 25)?
- 10 **What does this let *me* do?** Method to borrow, baseline to beat, gap to fill, or nothing.

⚠️ Read charitably before you read critically

There is a failure mode that looks like rigour: dismantling a paper for flaws that are, on a fair reading, either acknowledged or irrelevant to its claim.

The discipline is to first construct the strongest version of the paper's argument — including filling gaps the authors left implicit — and only then attack it. A criticism that survives that treatment is worth making. One that does not was a misreading, and offering it in a seminar or a referee report costs you credibility.

This matters most when you disagree with a paper's conclusion. That is exactly when your reading is least reliable.

Worked Example: Second-pass appraisal of a plausible paper

The paper (constructed for this example). “Static Analysis Adoption Reduces Post-Release Defects: Evidence from 200 Open-Source Projects.” The authors identify 200 GitHub projects, 100 that added a static analysis tool to CI and 100 that did not, and report that adopters have 23% fewer post-release issues labelled **bug** in the following year ($p < 0.01$).

1–2. Claim and type. Claim: adopting static analysis causes fewer post-release defects. Type: empirical, and specifically causal.

3. Right kind of evidence? No. This is observational: projects chose whether to adopt. The claim is causal; the design supports association only. This is the paper's central problem and everything else is secondary.

5. Comparison. Non-adopters — but adoption is not random. Teams that add static analysis to CI are plausibly teams that already have CI, code review, tests and maintainers who care. Any of those could produce the effect. This is confounding by *selection*, and Chapter 11 exists to address it: a difference-in-differences design using each project as its own control before and after adoption would be far stronger.

6. Construct. The outcome is issues labelled **bug**. Does that measure defects, or does it measure *labelling practice*? Teams disciplined enough to adopt static analysis are plausibly also more disciplined about triaging and labelling — which could push the measured count in either direction. The measure may be correlated with the treatment through a path that has nothing to do with defects.

7. Falsifier. As designed, almost nothing: any difference would have been reported as an effect, and no difference would have been reported as a null. The design does not discriminate between the causal hypothesis and the selection hypothesis at all.

8. Threats named? If the paper discusses selection bias, states the labelling problem, and softens the claim to an association, it is a decent descriptive study honestly reported. If it says “reduces” in the title and recommends adoption in the conclusion, the claim outruns the evidence.

10. What it gives me. A dataset construction method worth borrowing, and a clearly identified gap: nobody has done this with an identification strategy. That gap is a thesis.

Notice what the appraisal produced

The worked example did not end in “this paper is bad”. It ended in a research opportunity that is specific, defensible, and directly derived from a named weakness in existing work.

That is what appraisal is for. Chapter 1 said the delta is measured against the literature; this is the mechanism by which you find it. A student who appraises thirty papers this way finishes with a list of gaps, not a pile of summaries.

8.3 Notes That Survive

! Highlighting is not note-taking

A highlighted PDF is unsearchable, tells you nothing about *why* a passage mattered, and is worthless in month 20 when you cannot remember which of 200 papers had the good threats section. Highlighting feels like work and produces nothing.

Write, for every paper you take past the first pass, a fixed short record — in your reference manager’s note field, so it lives with the citation:

Claim	One sentence, your words, not theirs
Type	From Chapter 1
Method	Design, subjects, N, measures
Result	Effect size if given; “not reported” if not, which is itself worth recording
Weakness	The strongest objection you can make after a charitable reading
Use to me	Baseline / method / gap / background / discard
Quote	At most one, verbatim, with a page number, for anything you might quote later
RQ relevance	Which of your research questions it bears on

! The one-quote rule protects you from yourself

Record verbatim quotations *as* quotations, in quotation marks, with a page number, at the moment you take the note. The most common route to inadvertent plagiarism is not dishonesty — it is a paraphrase written into notes six months ago that was actually a quotation, and is now indistinguishable from your own prose. Chapter 28 sets out what the consequences of that look like under **PhD Regs 2024 §32**, and they are severe enough that the five seconds of discipline is worth it every time.

8.4 Refereeing, and Why You Should Volunteer

Reviewing other people’s work is the fastest way to improve your own, because it puts you on the side of the table that decides. Ask your supervisor to sub-review a paper for a conference. Most are glad to arrange it.

Section	What goes in it
Summary	The paper’s claim and contribution, in your words. Authors read this first to check you understood them — and a mistaken summary tells the editor to discount the rest
Strengths	Genuine ones, specifically. Every paper has some
Major concerns	Numbered. Each states the problem, why it threatens the claim, and what would resolve it
Minor points	Typos, unclear figures, missing citations
Recommendation	With the reasoning, aimed at the editor

? Two rules for a review you would be proud to sign

Every criticism carries an action. “The evaluation is weak” is useless. “The evaluation compares against a 2016 baseline; the 2023 method [ref] is stronger and available, and the claim of state-of-the-art performance cannot stand without it” is actionable and fair.

Write as though signed. Whether or not the venue is open, write nothing you would not put your name to. Anonymity is there to let you be honest, not to let you be careless or unkind — and the field is smaller than it looks.

8.5 The Deepest Reading Is Reproduction

You do not fully understand an empirical or computational result until you have tried to reproduce it. Take one claim — not the whole paper — and attempt it from the artefacts provided.

What happens	What it means
It reproduces	You now understand the method at a level no amount of reading gives you, and you have a working baseline for your own work
It reproduces after a struggle	Note exactly what was underspecified. That gap is publishable as a reproducibility study, and it is a real contribution (Chapter 25)
It does not reproduce	Before concluding anything: check your own setup twice, then write to the authors politely. Most non-reproduction is environment, version or seed, not error — and authors are usually helpful
There are no artefacts	That is itself a finding about the field, and one worth counting in a review (Chapter 5, quality item Q7)

 **Four Traditions — appraising a paper Across Ways of Knowing**

 **Empirical**

Check the design against the claim, the power against the effect, and the measure against the construct. Ask what the confidence interval was, not whether *p* crossed a threshold.

 **Design Science**

Ask what the artefact demonstrates beyond its own existence, whether the evaluation matches the claimed contribution, and whether the design knowledge is stated separately from the implementation.

 **Qualitative**

Ask about the researcher’s position, whether the description is thick enough to interpret, whether interpretations are traceable to data, and whether disconfirming cases were sought.

 **Formal / Analytical**

Check the proof. Then check the theorem statement — most errors in formal papers are in the assumptions, not the derivation, and an unrealistic assumption makes a correct theorem irrelevant.

 **Try It Yourself: Three passes and a referee report**

Choose a recent paper central to your topic.

1. First pass, 10 minutes. Write the five C’s: category, context, correctness, contributions, clarity.
2. Second pass, one hour. Write the eight-field note record.
3. Third pass, as long as it takes. Work through the ten appraisal questions and identify one assumption the authors did not state.
4. Write a full referee report as if for the venue that published it, with a recommendation.

5. Then attempt to reproduce one claim, and record what happened.

Bring the report to the week 5 seminar. Comparing reports on the same paper is the fastest way to discover how differently competent readers weigh the same evidence — which is also why reviewers disagree.

Checkpoint — Test Yourself

1. You have 900 records to screen and 30 papers to understand deeply. How many of each pass do you perform, and why is doing second passes during screening a mistake?
2. Why should you construct the strongest version of a paper's argument before criticising it? Give a practical, not moral, reason.
3. A paper reports $p = 0.03$ and no effect size. What do you write in the Result field of your note, and why does it matter later?
4. You cannot reproduce a result. List, in order, the four things you do before concluding the paper is wrong.

Chapter Summary

- Read in three passes and stop at the right one. Screening is first-pass work; the papers your delta is measured against get the third pass.
- Appraisal is ten questions, not a summary. The type of contribution determines what evidence is owed, and most weaknesses are a mismatch.
- Read charitably first. A criticism that survives the strongest version of the argument is worth making; one that does not was a misreading.
- Appraisal produces research opportunities, not verdicts. Thirty appraised papers give you a list of gaps.
- Keep a fixed eight-field note record in the reference manager. Highlighting produces nothing.
- Mark quotations as quotations, with page numbers, at the moment you write them. This is the main defence against inadvertent plagiarism.
- Referee something. Every criticism carries an action, and write as though signed.
- Reproduce one claim from the paper your work depends on. Whatever happens, you learn something worth having.

Review Questions

1. The three-pass method allocates effort by expected value. Construct a case where it would fail — a paper whose importance is invisible at first and second pass — and say what would catch it.
2. Peer review is criticised as slow, inconsistent and unable to detect fraud. Given that, why does the system persist, and what would a credible replacement have to provide?
3. You referee a paper whose conclusion you believe is wrong but whose methods you cannot fault. What do you recommend, and what do you write?
4. Reproduction is described here as the deepest form of reading, yet almost no graduate scholar does it. Identify the incentives that produce this, and propose one change to a course like this one that would shift them.

PART III

Research Designs for Computing

This is the catalogue. Each design answers some questions and is incapable of answering others, and the skill this part teaches is not knowing all eight — it is choosing correctly under constraint, and being able to say what your choice cannot establish. Read it with your own question from Chapter 3 in front of you. Every chapter ends by telling you what it forecloses.

💡 What Part 3 Covers

- **Chapter 9 — Design Science Research.** Artefacts as contributions; Hevner's guidelines, the DSRM process, and FEDS evaluation.
- **Chapter 10 — Controlled Experiments.** Randomisation, factorial and within-subjects designs, blocking, and human subjects in computing.
- **Chapter 11 — Quasi-Experiments and Repository Mining.** Identification strategies when you cannot randomise, and the confounders in observational software data.
- **Chapter 12 — Case Study Research.** Units of analysis, propositions, chain of evidence, and why a case study is not a small survey.
- **Chapter 13 — Survey Research.** Sampling frames, instrument construction, piloting, and the nonresponse problem.
- **Chapter 14 — Qualitative Methods and Coding.** Interviews, grounded theory, thematic analysis, coding reliability and saturation.
- **Chapter 15 — Mixed Methods.** Convergent and sequential designs, joint displays, and integration that is more than juxtaposition.
- **Chapter 16 — Benchmarking and Performance Evaluation.** Workload characterisation, statistically defensible measurement, and simulation validity.

Design Science Research

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 3 — Research Designs for Computing

🕒 Learning Outcomes

By the end of this chapter you will be able to:

- **Distinguish** design science research from software development and from applied engineering.
- **Classify** an artefact as a construct, model, method or instantiation.
- **Apply** the DSRM process model to structure a project and a thesis.
- **Position** a contribution as routine design, exaptation, improvement or invention.
- **Select** an evaluation strategy using FEDS, and justify it against risk.
- **State** the design knowledge your artefact produces, separately from the artefact.

🔗 Before You Start

Chapter 1 — especially the point that an artefact contributes by carrying a claim, never by existing. Chapter 2 for pragmatism and for why design science reasons abductively.

📖 Key Terms in This Chapter

design science research • artefact • construct • model • method • instantiation • design knowledge • relevance cycle • rigour cycle • design cycle • exaptation • formative and summative evaluation • naturalistic and artificial evaluation • FEDS

9.1 The Tradition That Fits Computing Best, and Is Done Worst

Most graduate computing projects build something. That makes design science research the natural home for a great deal of our work — and also makes it the tradition most often invoked as cover for a project that is simply development with a literature review attached.

📖 Design science research

Design science research produces knowledge by building and evaluating an artefact intended to solve a class of problem, and by articulating what the building and evaluating taught that was not known before.

Two clauses do the work. *A class of problem*: not this hospital's data pipeline, but the class of problems of which it is an instance. *What it taught*: the **design knowledge** — a principle, a trade-off, a demonstrated feasibility — stated so that somebody solving a different instance can use it without your code.

	Development / consultancy	Design science research
Target	This client's problem	A class of problems this instance represents
Success	The stakeholder is satisfied	A knowledge claim survives evaluation
Output	A working system	An artefact <i>and</i> transferable design knowledge
Novelty	Not required	Required, and positioned against prior artefacts
Failure	A failure	A result, if the design rationale was explicit

⚠ The question that separates them

Ask: *if my artefact were deleted tomorrow, what would remain?*

If the answer is “nothing”, it was development. If it is “the finding that this class of problem yields to this principle, and the evidence for it”, it was research. Write that sentence into your synopsis. It is what an examiner is looking for and what most design-oriented theses never quite say.

9.2 What Counts as an Artefact

March and Smith's four categories remain the clearest taxonomy [40], and naming yours precisely tells you what to evaluate.

Type	What it is	Example, and what evaluation asks
Construct	Vocabulary and concepts	A taxonomy of failure modes in federated learning. Does it carve the space usefully? Can two people classify the same case identically?
Model	A representation of a problem and solution space	A formal model of trust propagation. Does it capture what matters, and does it predict?
Method	A process or algorithm	A procedure for partitioning clinical data across sites. Does it work, and is it better than the alternative on a stated dimension?
Instantiation	A working system embodying the above	The deployed framework. Does it demonstrate feasibility in a realistic setting?

💡 Most theses claim an instantiation and contribute a method

A common and easily fixed problem. The thesis presents a system, evaluates the system, and buries the actual contribution — a novel scheduling method, a new way of decomposing the problem — inside the implementation chapter.

Separate them explicitly. The method is the contribution; the instantiation is the evidence that the method is realisable. Written that way, the work generalises, and a reader who will never run your code can still use your result.

9.3 Hevner's Guidelines, Read as Questions

The seven guidelines [25] are the most cited framing in the field. They are more useful as a checklist of questions to answer in the synopsis than as a description of process.

1	Design as an artefact	What exactly is produced, and which of the four types is it?
2	Problem relevance	Whose problem, and what does it cost them now?
3	Design evaluation	By what method, against what criteria, and how do those criteria follow from the problem?
4	Research contributions	What is now known that was not? State it as a sentence, not as a system
5	Research rigour	Which existing theory and methods were drawn on — the rigour cycle
6	Design as a search	What alternatives were considered and rejected, and on what grounds?
7	Communication	Can both a technical and a managerial audience take something from it?

! Guideline 6 is the one that is skipped, and it is the one that matters

Design is a search through a space of possible solutions. A thesis that presents one design, fully formed, with no account of what else was considered, has hidden the research. The alternatives you rejected — and *why* — are design knowledge, and often the most transferable part of the work.

Keep a design log from day one: each significant decision, the options, the criterion that decided it, and what you would need to have known to decide differently. It costs minutes a week and becomes a thesis chapter.

9.4 The DSRM Process

Peffers and colleagues give the process model most theses are structured around [44].

Entry point varies: problem-centred, objective-centred, design-centred, or from an observed solution in use

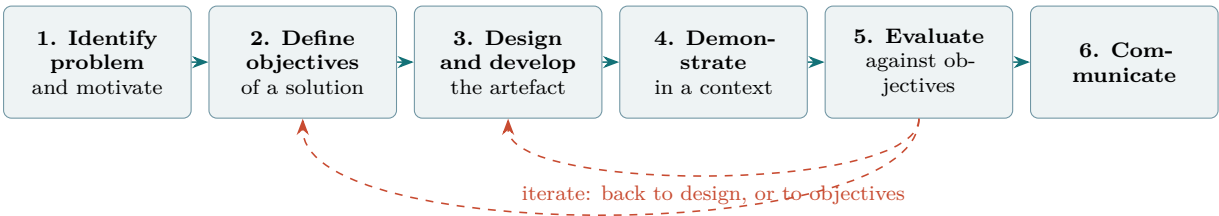


Figure 9.1: The DSRM process. The iteration arrows are the substance: a design science project that ran straight through once, with no loop back from evaluation, has either been extraordinarily lucky or has not reported what actually happened.

! Report the iterations, do not hide them

Theses routinely present DSRM as though it ran once, cleanly. Examiners know it did not. The loops are where the design knowledge is: version 1 failed because of X, which revealed that assumption Y was wrong, and version 2 addressed it. That narrative is more convincing, more useful to a reader, and easier to write than a retrospective fiction of a design that was right first time.

9.5 Positioning the Contribution

Gregor and Hevner’s two-by-two [23] settles the question examiners ask: is this novel enough? Position it on solution maturity against problem maturity.

	Problem is new	Problem is known
Solution is new	<i>Invention.</i> Rare; a genuinely novel solution to a genuinely new problem	<i>Improvement.</i> The most common good thesis: a better solution to a known problem. Requires comparison against the current best
Solution exists	<i>Exaptation.</i> Extending a known solution into a new problem domain. Legitimate research <i>if</i> the transfer was non-obvious and something was learned in making it work	<i>Routine design.</i> Not a research contribution. Competent professional work

⚠ Exaptation is where most “applied” theses live, and where they are won or lost

“Applying deep learning to crop disease detection in Punjab” is exaptation. Whether it is research depends entirely on one thing: **was the transfer non-obvious, and did making it work require solving something?**

If the answer is that you fine-tuned a standard architecture on a local dataset and it worked as expected, that is routine design, and no amount of DSR vocabulary changes it (Chapter 1, non-contribution 1).

If the answer is that local field conditions — class imbalance across smallholdings, imagery from low-cost handsets, labels from non-expert farmers — broke the standard approach in a specific way, and your artefact addresses that, then the exaptation is genuine. The contribution is the characterisation of *why* it broke plus the design response. Say that in the abstract.

9.6 Evaluation: FEDS

Evaluation is where design science theses most often fail, because “demonstration” is mistaken for evaluation. Showing that your system runs on one example establishes existence, not merit. The FEDS framework [61] organises the choice along two dimensions.

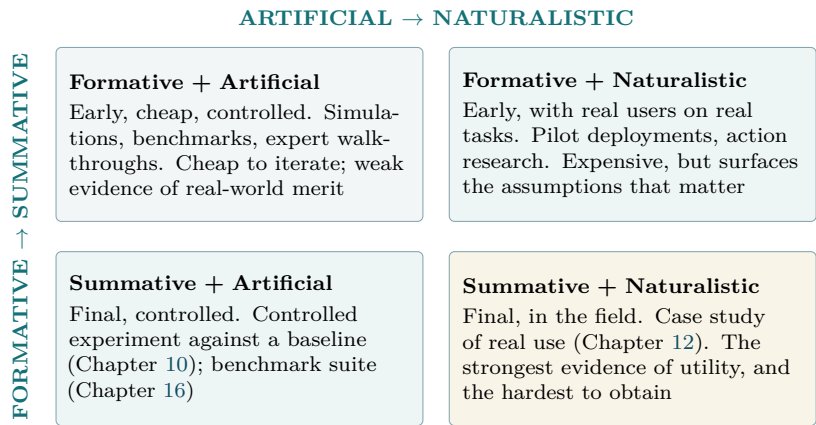


Figure 9.2: The FEDS evaluation space. A thesis normally moves diagonally: cheap artificial formative evaluation early to fix the design, then the most naturalistic summative evaluation the constraints allow.

Strategy	Choose it when
Human risk and effectiveness	The main uncertainty is whether people will actually use it or benefit. Move to naturalistic evaluation early, even if crude
Technical risk and efficacy	The main uncertainty is whether the technique works at all. Stay artificial and rigorous for longer; naturalistic evaluation late
Quick and simple	Low risk, small artefact, limited resources. One or two evaluation episodes, honestly limited
Purely technical	There are no human users — an algorithm, a protocol. Artificial evaluation throughout is appropriate, not a compromise

💡 Choose the strategy from the risk, and say so

The reason FEDS is worth using is that it makes evaluation choice *arguable*. A thesis that says “we evaluated by simulation” invites the objection that simulation is unrealistic. A thesis that says “the dominant uncertainty in this design was technical efficacy rather than user acceptance, so we followed a technical risk strategy: three rounds of artificial formative evaluation, then a summative benchmark against the current best method, with a single naturalistic case study to check external validity” has pre-empted the objection by answering it.

❗ Four evaluations that are not evaluations

- **Demonstration.** “The system successfully processed the dataset.” Establishes that it runs.
- **Self-comparison.** Comparing version 2 against version 1 of your own artefact. Shows you improved it; says nothing about the state of the art.
- **The satisfied-user survey.** Five students rate the interface 4.2 out of 5. Who selected them, what were they comparing against, and what would a score of 2 have caused you to conclude?
- **Criteria chosen after the results.** If the evaluation criteria were not written into the protocol before the artefact was built, they were selected to be passed.

📌 What the regulations ask of a built artefact — PhD §14

§14(i) requires an “original and innovative contribution to knowledge that contributes to solving socioeconomic problems” — which is, almost exactly, Hevner’s problem relevance guideline.

§14.3(C), appropriateness of methods, requires that a dissertation “justify the use of methods and techniques to achieve study objectives” and “show evaluation of obtained results in relation with study objectives”. For design science that is a direct instruction: your evaluation criteria must be traceable to your DSRM step 2 objectives.

§14.3(D) then asks for relevance to policy and practice, including that “the study output should be significant enough to be published or patented”.

🌀 Four Traditions — building an artefact Across Ways of Knowing

📊 Empirical

Treats the artefact as a treatment: what is the effect of using it, compared with the alternative, measured how? This is the summative-artificial quadrant, and Chapter 10 supplies the machinery.

 Design Science

The artefact is the vehicle of the contribution, and design knowledge is the contribution. Rigour comes from the design rationale and the evaluation, not from statistics alone.

 Qualitative

Asks how the artefact is understood and appropriated by the people using it — often revealing that it is used for something other than its designed purpose, which is itself a finding.

 Formal / Analytical

Asks what can be proved about the artefact: correctness, complexity, guarantees under stated assumptions. Complements empirical evaluation rather than replacing it.

 Try It Yourself: Position and plan an evaluation

For an artefact you intend to build — or one from a paper you admire:

1. Name the artefact type: construct, model, method or instantiation. If you name two, say which is the contribution.
2. Place it in the Gregor–Hevner two-by-two, and write one sentence defending the placement against the next cell down.
3. Write the design knowledge sentence: “if the artefact were deleted, what remains is...”
4. Identify the dominant risk — technical or human — and choose a FEDS strategy.
5. Write the evaluation criteria *now*, before building, and trace each to a DSRM step 2 objective.

Step 5 is the one that must be done in this order. Feed it into portfolio piece P3.

 Checkpoint — Test Yourself

1. A scholar builds a recommender for a university library and reports that librarians liked it. Which FEDS quadrant is this, and what is the strongest objection?
2. Distinguish exaptation from routine design with an example from your own area where the boundary is genuinely unclear.
3. Why must evaluation criteria be written before the artefact is built? Name the specific bias.
4. Your artefact is a method, and you built a system to test it. Which is the contribution, and how should the thesis chapters be ordered to make that clear?

 Chapter Summary

- Design science produces knowledge by building and evaluating an artefact for a *class* of problem, and stating what was learned.
- If the artefact were deleted, what remains is the contribution. Write that sentence.
- Four artefact types; most theses claim an instantiation and actually contribute a method. Separate them.
- Hevner’s guideline 6 — design as search — is the most skipped and the most transferable. Keep a design log.
- DSRM’s iteration loops are the substance. Report them.
- Position with Gregor and Hevner: invention, improvement, exaptation, routine design. Applied theses are usually exaptation, and stand or fall on whether the transfer was non-obvious.

- Choose evaluation from risk using FEDS, and write the criteria before building. Demonstration, self-comparison and satisfied-user surveys are not evaluations.

Review Questions

1. Design science is sometimes accused of being unfalsifiable: an artefact that fails is redesigned rather than refuted. Is this a fair charge? Construct the criticism at its strongest and then answer it.
2. Naturalistic evaluation gives stronger evidence of utility but weaker control. For a specific artefact in your area, describe an evaluation programme that obtains both, and state its cost.
3. Gregor and Hevner treat exaptation as a legitimate contribution. Where exactly is the line between exaptation and routine design, and who decides — the author, the reviewer, or the problem?
4. **PhD Regs 2024 §14.3(D)** requires that research output be “significant enough to be published or patented”. What does this imply for a design science thesis whose artefact is released as open source, and is the requirement well-suited to such work?

Controlled Experiments in Software Engineering and HCI

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 3 — Research Designs for Computing

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **State** the three features that make a study an experiment.
- **Explain** what randomisation buys that no other technique can.
- **Choose** between between-subjects, within-subjects and factorial designs, and name what each costs.
- **Counterbalance** a within-subjects design and say which threat it removes.
- **Judge** the student-versus-professional question on evidence rather than reflex.
- **Specify** an experiment completely enough to be pre-registered.

🔗 Before You Start

Chapter 3 for causal questions and variable roles, and Chapter 2 for why this design belongs to the post-positivist tradition. Chapter 18 follows immediately: an experiment you cannot power is not an experiment you can run.

📖 Key Terms in This Chapter

experiment • manipulation • randomisation • control • experimental unit • subject and object • treatment • factor and level • between-subjects • within-subjects • crossover • carryover effect • counterbalancing • Latin square • blocking • factorial design • interaction • pilot study

10.1 Three Features, and Only the Third Is Optional

📋 Controlled experiment

A study in which the researcher (1) **manipulates** at least one independent variable, (2) **randomly assigns** experimental units to treatments, and (3) **controls** other variables that could affect the outcome.

Manipulation and randomisation are what license a causal conclusion. Control improves precision, and where randomisation is absent but manipulation remains, you have a **quasi-experiment** (Chapter 11) — weaker, often the only option, and requiring an explicit argument that assignment was not confounded.

💡 What randomisation actually buys

This is the most misunderstood idea in empirical research, and it is worth being precise.

Randomisation does not make groups *equal*. With 20 subjects per group they will differ on experience, on the day they slept badly, on everything. What randomisation does is make the assignment **independent of every variable, including the ones you have not thought of and cannot measure**.

That is the whole trick, and nothing else achieves it. Matching handles confounders you have identified. Statistical adjustment handles confounders you have measured. Only randomisation handles the confounder you have never heard of — which, in a field as young as ours, is most of them. It converts an unknown bias into known, quantifiable noise, and the noise is what the statistics in Chapter 19 are built to handle.

10.1.1 Vocabulary that prevents confusion

Experimental unit	The thing assigned to a treatment. Randomise it, count it, and analyse at this level. Getting this wrong is the most common analysis error in the field
Subject	The person taking part
Object	The material worked on — the code, the document, the dataset
Factor	An independent variable being manipulated
Level / treatment	A value of a factor
Response	The measured outcome

⚠ If you randomise teams, your sample size is teams

Suppose you assign four teams of ten developers each to two conditions and analyse 40 individual results as if independent. You have reported a sample size of 40 when the design supports 4. Members of a team share a codebase, a manager, a culture and a coffee machine; their outcomes are correlated.

This inflates significance dramatically, and it is exactly the error that makes underpowered studies look conclusive. The remedies are to analyse at the unit of randomisation, or to model the nesting explicitly with a mixed-effects model (Chapter 20). Deciding which *before* collecting data is part of portfolio piece P5.

10.2 Three Designs, and the Price of Each

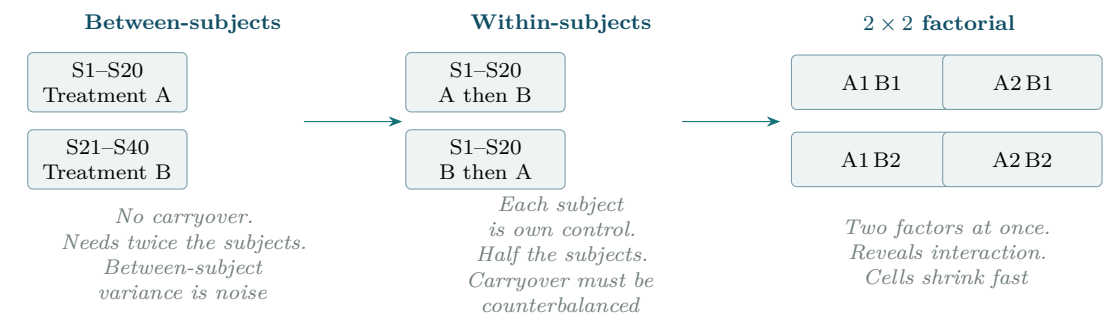


Figure 10.1: Three experimental designs. The choice is almost always forced by how many subjects you can actually recruit, which is why Chapter 18 should be read before this decision is fixed rather than after.

Design	Use when	Principal threat
Between-subjects	Treatments cannot be un-learned; subjects are plentiful	Individual differences become error variance, so power is low. Randomise well, and consider blocking on experience

Design	Use when	Principal threat
Within-subjects (crossover)	Subjects are scarce — almost always, in software engineering	<i>Carryover</i> : having done task A changes performance on B. Also fatigue and learning. Counterbalance, and use different but equivalent objects
Factorial	You suspect two factors interact	Cells shrink: a 2×2 with 40 subjects has 10 per cell. Interactions need substantially more power than main effects
Blocked	A known nuisance variable dominates, e.g. experience	None specific; blocking on a variable that turns out irrelevant costs little, so block when in doubt

10.2.1 Counterbalancing

In a within-subjects design every subject sees every treatment, so order becomes a confound. Counterbalancing distributes order systematically.

Two treatments	Half the subjects do A then B, half B then A. Order becomes a factor you can test rather than a bias you carry
Three or more	Full counterbalancing needs $k!$ orders. A Latin square uses k orders in which each treatment appears once in each position
Objects too	Rotate the objects (the code, the task) against the treatments as well, or the effect of treatment is confounded with the difficulty of the material

Worked Example: A 3×3 Latin square for treatments and objects

Three review techniques (T1, T2, T3), three code artefacts (O1, O2, O3), and subjects assigned to three groups. Each group sees every technique and every artefact, and no group sees the same pairing.

	Session 1	Session 2	Session 3
Group 1	T1 on O1	T2 on O2	T3 on O3
Group 2	T2 on O3	T3 on O1	T1 on O2
Group 3	T3 on O2	T1 on O3	T2 on O1

Each technique appears once per session position, so learning and fatigue are balanced across techniques. Each technique meets each artefact exactly once, so artefact difficulty does not favour any technique.

What it does *not* handle: a genuine interaction between technique and artefact — if T1 is uniquely suited to O1 — which a Latin square averages away rather than detects. Detecting that needs a full factorial design and correspondingly more subjects.

10.3 Human Subjects: the Students Question

Every software engineering experiment using students is challenged on the same ground: professionals would behave differently. The question deserves a better answer than reflex in either direction, and there is evidence to give one [17, 57].

Position	The considered version
Students are acceptable	For tasks where the relevant skill is one students possess; for testing a theory rather than estimating an industrial effect; for early formative work. Graduate students in particular are often comparable to junior professionals
Students are not acceptable	Where the task depends on experience the population lacks — architectural judgement, legacy system familiarity, organisational negotiation. Here students are not a noisy proxy, they are a different population

💡 How to answer it in your thesis

Do not claim students are equivalent to professionals. Do the following instead, and the objection loses most of its force:

1. **Characterise your subjects** rather than labelling them. Years of programming, industrial months, familiarity with the language and with the task type. “Students” spans a first-year undergraduate and a part-time MS scholar with six years in industry.
2. **State the target population** the inference is to, explicitly.
3. **Argue the specific task.** Is the skill it demands one your subjects have?
4. **Put it in threats to external validity** (Chapter 24) with a direction: would professionals plausibly show a larger or smaller effect, and why?

10.4 Pilot First. Always

A pilot with three or four subjects, analysed exactly as the real study will be, is the cheapest insurance in empirical research.

What a pilot catches	What it costs you to find later
Instructions that are ambiguous	Every subject interpreted the task differently; the data is uninterpretable
A task too long or too short	Floor or ceiling effects; no variance to analyse
Instrumentation that fails	Timing not logged; the primary outcome is missing
An outcome measure with no variance	The whole study, wasted
An analysis script that cannot run on the data shape	Discovered at the worst possible moment

⚠️ Write and run the analysis script on pilot data

Run your full analysis on the pilot data — or on simulated data with the same shape — before collecting anything real. If the script produces a number, the design and the analysis are compatible. If it errors, or produces something you cannot interpret, you have found a design flaw for the price of an afternoon.

This also forces the analysis to be specified in advance, which is precisely what pre-registration requires (Chapter 26). Pilot data must not be pooled into the final analysis.

 Template: C7 — Experiment design sheet

Research question / hypothesis	_____
Factor(s) and levels	_____
Experimental unit	_____
Subjects: population, N, recruitment	_____
Objects and how they are assigned	_____
Design (between / within / factorial)	_____
Randomisation procedure	_____
Counterbalancing / blocking	_____
Response variable(s) and instrument	_____
Expected effect size, and its source	_____
Power and required N (Chapter 18)	_____
Planned analysis	_____
Exclusion rules, decided now	_____
Ethics approval / consent	_____

 Five ways experiments in our field go wrong

- **Analysis at the wrong level.** Randomising teams, analysing individuals.
- **No power analysis.** Twelve subjects, an effect of $d = 0.4$, and a null result reported as “no difference”.
- **An outdated or badly tuned baseline.** Your treatment beats a comparison nobody would use.
- **Optional stopping.** Collecting until the result is significant is a way of guaranteeing significance (Chapter 26).
- **Exclusions decided after seeing the data.** Removing outliers that inconveniently weaken the effect. Write the rule in advance and apply it blind to condition.

 Four Traditions — the controlled experiment Across Ways of Knowing

 Empirical

Its home. Manipulation plus randomisation is the strongest available warrant for a causal claim, and the price is artificiality.

 Design Science

Experiments serve as summative artificial evaluation of an artefact (Chapter 9); the artefact is the treatment.

 Qualitative

Complements rather than competes: an experiment establishes that an effect occurred, and a qualitative strand explains why participants behaved as they did. This is a standard explanatory sequential design (Chapter 15).

 Formal / Analytical

Experiments do not settle formal questions, but they do tell you whether an asymptotic result matters at realistic input sizes — which is often the more useful question.

 Try It Yourself: Design one experiment completely

Take a causal question from your alignment table (Chapter 3) and fill in template C7 completely — every row, no blanks.

Then submit it to the two tests that catch most flawed designs:

1. **The unit test.** What exactly is randomised? Is your stated N the number of those things?
2. **The null test.** Write the sentence you would publish if the result were exactly zero. If that sentence is not publishable, the study is not worth running as designed — and you have learned this before spending six months on it.

 Checkpoint — Test Yourself

1. Randomisation does not equalise groups. What does it do, and why can no amount of statistical adjustment substitute for it?
2. You have 16 professional developers and three techniques to compare. Which design, and what specific threat does it introduce?
3. In the Latin square worked example, what is averaged away rather than detected, and what design would detect it?
4. Why must outlier exclusion rules be fixed before data collection, even though the rule itself may be entirely reasonable?

 Chapter Summary

- Manipulation plus randomisation licenses a causal claim; control adds precision.
- Randomisation makes assignment independent of *unknown* confounders. Nothing else does this.
- Analyse at the unit of randomisation, or model the nesting. Randomising teams and counting individuals inflates significance.
- Between-subjects costs subjects; within-subjects costs carryover and must be counterbalanced; factorial costs cell size but reveals interaction.
- Latin squares balance order and material; they average away treatment-by-object interactions rather than detecting them.
- Characterise subjects rather than labelling them, state the target population, and put the inference limit in threats with a direction.
- Pilot, and run the real analysis script on pilot or simulated data before collecting anything.

 Review Questions

1. Controlled experiments are high in internal validity and low in external validity. For a specific question in your area, design a two-study programme in which each study's weakness is covered by the other, and say what it would cost.
2. The students-versus-professionals debate has run for two decades without resolution. Is it resolvable in principle? If yes, what evidence would settle it; if no, what should the field do instead?
3. Optional stopping is prohibited in frequentist analysis but is acceptable under some sequential and Bayesian designs. Explain why the difference exists, and whether it offers a practical route for resource-limited software engineering research.
4. An experiment finds no effect with adequate power. Describe how you would write this up so that it is publishable, and identify the features of the reviewing system that make that difficult.

Quasi-Experiments and Repository Mining

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 3 — Research Designs for Computing

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Explain** the selection problem and why it, not sample size, is the central difficulty in observational research.
- **Choose** an identification strategy — difference-in-differences, interrupted time series, regression discontinuity or matching — and state the assumption it rests on.
- **Design** a repository mining study and enumerate its data quality threats.
- **Judge** when an observational claim may legitimately be phrased causally.

🔗 Before You Start

Chapter 10, especially what randomisation buys. This chapter is about what to do when you cannot have it. Chapter 3 for confounders.

📖 Key Terms in This Chapter

quasi-experiment • selection bias • identification strategy • counterfactual • difference-in-differences • parallel trends • interrupted time series • regression discontinuity • propensity score matching • instrumental variable • natural experiment • mining software repositories • survivorship bias • tangled commit

11.1 The Problem Is Selection, Not Size

Most interesting questions in software engineering cannot be answered by experiment. You cannot randomly assign companies to adopt microservices, or developers to have ten years of experience. What you can do is observe what happened where the choice was made by someone else — and the difficulty is that the choice was not made at random.

📄 The selection problem

When units choose their own treatment, the treated and untreated groups differ in every respect that influenced the choice — including respects you cannot observe.

A comparison of outcomes then confounds the effect of the treatment with the effect of *being the kind of unit that selects into it*. More data does not help: a million observations estimate the confounded quantity with great precision.

💡 What an identification strategy is

An **identification strategy** is an argument — not a statistical technique — that the comparison you are making approximates the counterfactual: what would have happened to the treated units had they not been treated.

Every strategy below rests on an assumption that cannot be tested directly. Your job in the thesis is to state that assumption plainly, give whatever indirect evidence for it you can, and

say what would happen to your conclusion if it failed. A quasi-experimental study whose assumption is stated and defended is respectable research. One that runs a regression and says “controls were included” is not.

11.2 Four Strategies

Strategy	How it identifies	The assumption it rests on
Difference-indifferences	Compares the <i>change</i> in treated units with the <i>change</i> in untreated units over the same period	Parallel trends: absent the treatment, both groups would have changed by the same amount. Partly checkable by inspecting pre-treatment trends
Interrupted time series	Models the series before an intervention and tests for a change in level or slope at the intervention point	No other change occurred at the same moment, and the pre-trend would have continued
Regression discontinuity	Compares units just above and just below a threshold that determines treatment	Units cannot precisely manipulate their position relative to the threshold; nothing else changes at it
Propensity score matching	Matches treated to untreated units with similar probability of being treated, given observed covariates	No unobserved confounding — everything driving selection was measured. Strong, untestable, and usually false

⚠ Matching is the weakest of the four, and the most used

Propensity score matching is popular because it is easy to run and produces balanced-looking tables. But it only balances what you *measured*. If teams that adopt a practice differ in engineering culture, and you did not measure engineering culture, matching on repository age and star count changes nothing about that.

Difference-in-differences is usually available in software repository data — you almost always have a time series — and it is far stronger, because each unit acts as its own control and any time-invariant unmeasured difference cancels. When you have a before and an after, prefer it.

📊 Worked Example: Repairing the static analysis study

Chapter 8 appraised a constructed paper claiming that adopting static analysis reduces post-release defects, comparing 100 adopting projects with 100 non-adopting ones. The design supports association only. Here is the repair.

The counterfactual we need. What would have happened to the adopting projects’ defect rates had they not adopted?

Difference-in-differences. For each adopting project, identify the adoption date. Measure the outcome for 12 months before and 12 months after. Do the same for non-adopters over a matched calendar window. The estimate is

$$\hat{\delta} = (\bar{Y}_{\text{adopt}}^{\text{post}} - \bar{Y}_{\text{adopt}}^{\text{pre}}) - (\bar{Y}_{\text{control}}^{\text{post}} - \bar{Y}_{\text{control}}^{\text{pre}}).$$

What this fixes. Any *time-invariant* difference between adopters and non-adopters — better developers, stricter culture, a more testable architecture — differences out. That is precisely the confounding the original design could not address.

What it does not fix. Anything that changed *at the same time* as adoption. If teams adopt static analysis as part of a broader quality push that also introduced code review and CI, difference-in-differences attributes the whole package to static analysis. This must be stated as the principal threat, and can be partly probed by checking whether other practices appear in the same commits.

Checking parallel trends. Plot both groups' monthly defect rates for 24 months before adoption. If the curves are roughly parallel, the assumption is plausible. If adopters were already improving faster, it is not, and the estimate is biased — report the plot either way.

The construct problem remains. Issues labelled **bug** still measure labelling as much as defects (Chapter 17). Difference-in-differences helps here too, but only if labelling practice did not change at adoption — which, if the quality push was real, it probably did. Consider a second outcome less sensitive to labelling, such as post-release reverts.

All figures illustrative; this is a design, not a result.

11.3 Mining Software Repositories

Software development leaves an unusually rich trace: version control, issue trackers, code review, CI logs, package registries. Mining it is cheap, scales to thousands of projects, and is entirely unobtrusive. It is also, as data, far messier than it appears.

Anchor Reading: What repository data will and will not tell you

Eirini Kalliamvakou, Georgios Gousios, Kelly Blincoe, Leif Singer, Daniel M. German and Daniela Damian, *The promises and perils of mining GitHub*, MSR 2014 [31].

Why it matters. It is the standard reference for the data quality threats in this design, and it is specific: not “be careful” but a catalogue of named perils with evidence for each.

What to extract. The list of perils, and the corresponding mitigation for each one you cannot avoid. Cite it in your threats section and demonstrate you handled the ones that apply.

Threat	What goes wrong	What to do
Not a project	Many repositories are course assignments, dotfiles, forks or personal experiments	Filter on activity, contributor count, issue use and commit history length; report the filter
Survivorship	Abandoned and deleted projects vanish. You study the ones that survived	Be explicit that inference is to surviving projects; consider archived snapshots
Tangled commits	A single commit mixes a bug fix, a refactor and a formatting change	Do not treat one commit as one change; consider untangling heuristics and report their error rate
Bot activity	Dependabot and CI bots generate large volumes of commits and issues	Detect and exclude bot accounts; report how many were removed
Missing links	Only some commits reference the issue they fix, and not at random	Acknowledge that linked-fix datasets are biased towards disciplined projects
Label semantics	bug means different things across projects	Validate a sample manually; report agreement
Activity is not effort	Commit counts measure committing, not work	Choose constructs closer to the question (Chapter 17)

! Generalising from GitHub to software engineering

Open-source projects on one platform are not a random sample of software development. They differ in ownership, incentives, review practice, team stability and above all in what is *visible*: much industrial engineering leaves no public trace.

This is a limit on the population your inference reaches (Chapter 24), and the fix is not more repositories — it is stating the population honestly. “In active open-source projects on GitHub between 2019 and 2024” is a defensible scope. “In software projects” is not.

11.4 When May You Say “Causes”?

Design	Language it licenses
Randomised experiment	“causes”, “increases”, “reduces” — within the studied population
Strong quasi-experiment with a defended assumption	“is associated with, and the difference-in-differences estimate is consistent with a causal effect of X on Y under parallel trends”
Matching on observed covariates	“is associated with, after adjusting for observed characteristics” — and say what is unobserved
Plain correlation	“is associated with”. Nothing more, in the title as well as the text

💡 The title is part of the claim

A paper whose body carefully says “associated with” and whose title says “improves” has made a causal claim. Titles are what get read, cited and repeated. **PhD Regs 2024 §14.3(C)** requires that “the obtained results should support the study objectives” — a title outrunning the design is exactly the mismatch that clause is aimed at.

🌀 Four Traditions — observational study Across Ways of Knowing

📊 Empirical

The natural home. The entire methodological effort goes into constructing a credible counterfactual, and the reporting standard is that a reader can judge the identifying assumption for themselves.

🔧 Design Science

Repository data characterises the problem space before an artefact is designed, and can supply the naturalistic evaluation baseline afterwards.

🗨️ Qualitative

Trace data shows what happened but never why. Pairing mining with interviews of the developers involved is one of the strongest mixed designs available in our field (Chapter 15).

√* Formal / Analytical

Treats the artefacts as formal objects — dependency graphs, call graphs, change histories — where properties can be computed exactly rather than estimated.

🔧 Try It Yourself: Turn a correlation into a design

- Find a published claim in your area that is correlational but written causally — they are not hard to find.
1. State the counterfactual the claim requires.

2. Choose an identification strategy that could plausibly recover it from available data.
3. Write the identifying assumption in one sentence, and say what evidence you could offer for it.
4. Name the confounder that would survive your strategy, and say which direction it would bias the estimate.
5. Rewrite the paper's title so that it matches what the original design supports.

Checkpoint — Test Yourself

1. Why does increasing the sample size not solve selection bias?
2. What does difference-in-differences remove that matching does not, and what does it fail to remove?
3. Give two reasons a dataset of commits linked to issues is a biased sample of bug fixes.
4. Under what circumstances, if any, would you use propensity score matching in preference to difference-in-differences?

Chapter Summary

- The obstacle in observational research is selection, not sample size. Precision does not fix bias.
- An identification strategy is an argument about a counterfactual, and every one rests on an untestable assumption. State it, defend it, and say what happens if it fails.
- Prefer difference-in-differences where a before and after exist: it removes all time-invariant unmeasured differences. Matching removes only what you measured.
- Repository data is cheap and scales, and carries specific known threats — non-projects, survivorship, tangled commits, bots, missing links, label semantics. Handle the ones that apply and report how.
- GitHub is not a sample of software engineering. State the population your inference actually reaches.
- Match the causal strength of your language — in the title as well as the text — to the design.

Review Questions

1. Parallel trends cannot be verified for the post-treatment period, only made plausible for the pre-treatment period. Does this make difference-in-differences unscientific? Answer with reference to Chapter 2.
2. Mining studies routinely analyse thousands of projects and report very small p-values. Explain why this is not reassuring, and what should be reported instead.
3. Design a natural experiment in software engineering: identify a real event that assigned something close to randomly, and say what it could identify.
4. Industrial software leaves little public trace, so the mining literature describes open source. What does this systematically distort in the field's collective picture of software engineering?

Case Study Research

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 3 — Research Designs for Computing

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Distinguish** a case study from an example, a pilot and a small survey.
- **Define** the unit of analysis and choose between holistic and embedded designs.
- **Select** cases on theoretical grounds and defend the selection.
- **Maintain** a chain of evidence from research question to conclusion.
- **Argue** analytic generalisation without claiming statistical generalisation.

🔗 Before You Start

Chapter 2 — case study research sits most comfortably in critical realism or interpretivism, and the difference changes what you claim. Chapter 3 for how-and-why questions.

📖 Key Terms in This Chapter

case study • unit of analysis • holistic and embedded design • single and multiple case • proposition • theoretical replication • literal replication • chain of evidence • triangulation • pattern matching • analytic generalisation • case study database

12.1 The Most Abused Term in Computing Research

“We evaluated our approach with a case study” usually means “we tried it once”. That is a demonstration (Chapter 9), not a case study, and the confusion costs authors credibility with anyone who knows the difference.

📖 Case study

An empirical investigation of a contemporary phenomenon *in its real-life context*, especially where the boundary between phenomenon and context is not clear, using multiple sources of evidence.

The italicised clause is the point. If you can separate the phenomenon from its context, you should — that is an experiment, and it will give you a cleaner answer. The case study exists for situations where the context is part of what you are studying, so removing it would destroy the phenomenon.

It is not	Because	Which would be
An example	One illustration of your method working	A demonstration in a DSR evaluation
A pilot	Trying something once to see what happens	A feasibility study, honestly labelled
A small survey	Asking a few people questions	An underpowered survey (Chapter 13)
An experiment with N=1	No manipulation, no control, no randomisation	Nothing — N=1 experiments require special designs

Anchor Reading: The reference for our field

Per Runeson and Martin Höst, *Guidelines for conducting and reporting case study research in software engineering*, Empirical Software Engineering 14(2), 2009 [51].  OA

Why it matters. It translates Yin's general methodology into software engineering, with a reporting structure and a checklist. It is open access and it is what reviewers in our field expect you to have followed.

What to extract. The five process steps, the distinction between the four purposes (exploratory, descriptive, explanatory, improving), and the reporting checklist — which you should use as a table of contents for the methodology chapter.

12.2 Design: Two Choices That Determine Everything

12.2.1 The unit of analysis

The **unit of analysis** is what the case *is*. It is astonishing how often this is left implicit, and every downstream decision depends on it.

If the question is	The case is probably
How does this team adopt the practice?	The team
How does this organisation manage the migration?	The organisation
How does this project's architecture degrade?	The project
How do individual reviewers decide?	The reviewer, embedded in a team-level case

12.2.2 Four designs

	Holistic (one unit)	Embedded (sub-units within)
Single case	One organisation studied as a whole. Justified when the case is critical, unique, revelatory, or longitudinal	One organisation, with several teams analysed within it. Guards against the whole-case account floating free of evidence
Multiple case	Several organisations compared as wholes	Several organisations, each with sub-units. Richest, most expensive, and the strongest basis for cross-case conclusions

⚠ A single case needs an explicit justification

Single-case designs are legitimate, and Yin gives five grounds: the case is *critical* (it tests a well-formed theory), *extreme or unusual*, *common* (and therefore informative about ordinary situations), *revelatory* (previously inaccessible), or *longitudinal*.

“It was the only company that would let me in” is not on that list. If access drove the choice — as it often, honestly, does — then find which of the five grounds *also* applies and argue that, or reframe the study as exploratory. Do not leave the reader to infer that the case was chosen by convenience, since they will.

12.3 Propositions and the Chain of Evidence

Propositions are what you expect to find and why. They are not hypotheses to be tested statistically; they direct attention, and they make the study falsifiable in the sense that matters here — you can be surprised.

💡 Exploratory case studies have questions, not propositions

If the study is genuinely exploratory, say so and state what you will look at instead of what you expect. What you must not do is write propositions after data collection and present them as though they guided it. That is HARKing (Chapter 26) in qualitative clothing.

The **chain of evidence** is the property that a reader can follow a conclusion back through your report to the evidence, and forward from a research question to the data that addressed it, in both directions without a gap.

- | | |
|---|--|
| 1 | The report cites specific evidence — “Interview 7, T3” — not “participants said”. |
| 2 | That evidence exists in a case study database : transcripts, notes, documents, logs, kept separately from your narrative. |
| 3 | The database records the circumstances of collection: when, where, who, under what protocol. |
| 4 | The protocol links the collection back to the research questions. |

⚠ The case study database is what makes this auditable

Keeping the raw evidence separate from your interpretation of it is the qualitative equivalent of publishing your data (Chapter 25). It is also what **PhD Regs 2024 §14(i)** requires when it states that “the raw data will have to be submitted to the supervisor along with a copy of the thesis”, and it is what lets a supervisor or examiner check that a conclusion is actually supported.

12.4 Selecting Cases, and Multiple-Case Logic

Cases are chosen *theoretically*, never randomly. A random sample of four organisations is neither representative — four is far too few — nor informative, because nothing connects them to your question.

Logic	Choose cases so that
Literal replication	They are similar in the respects your theory says matter, and you predict similar results. Confirmation across them strengthens the theory
Theoretical replication	They differ in a respect your theory says matters, and you predict <i>different</i> results for a stated reason. This is far stronger: it is a genuine test

💡 Theoretical replication is the case study’s version of an experiment

If your theory says the practice will help co-located teams and hinder distributed ones, then studying one of each and finding exactly that is powerful evidence — and finding the opposite refutes the theory. Predicting the difference *in advance* is what converts a descriptive study into a test, and it costs nothing but the discipline of writing the prediction down before collecting.

12.5 Analysis

Pattern matching	Compare an empirically observed pattern with the one your propositions predicted. Correspondence supports the proposition; divergence is a finding
Explanation building	Iteratively build an explanation of the case, revising as evidence accumulates. Powerful and dangerous — record the revisions, or it becomes unfalsifiable
Time-series / process	Trace the sequence of events, testing a predicted process against the observed one
Cross-case synthesis	Treat each case as a whole and compare. Word tables per case, then compare across
Rival explanations	For each conclusion, state the best competing explanation and the evidence that discriminates. This is the single most persuasive thing in a case study report

12.5.1 Triangulation

Data triangulation	Multiple sources: interviews, repository history, documents, observation
Observer triangulation	More than one researcher collecting or coding
Methodological triangulation	Qualitative and quantitative strands (Chapter 15)
Theory triangulation	Interpreting through more than one theoretical lens

❗ Triangulation is not corroboration-hunting

The purpose is not to accumulate sources that agree. It is to expose disagreement, because disagreement is informative: when developers say review is thorough and the repository shows a median review time of ninety seconds, you have found something worth the whole study.

A case study reporting that all sources agreed on everything has usually not looked hard enough, or has looked only where agreement was likely.

12.6 What You May Claim

Statistical generalisation	From a sample to a population, by probability. Case studies cannot do this , and should not pretend to
Analytic generalisation	From findings to a theory, which is then applicable to other cases sharing the relevant conditions. This is what a case study offers

💡 How to write the generalisation paragraph

Not: “these findings apply to software companies in Pakistan”. That is a statistical claim four cases cannot support.

Instead: “the mechanism identified here — that review checklists are abandoned when review is time-boxed below the checklist’s completion time — should be expected wherever those two conditions co-occur. We observed it in four organisations differing in size and domain, which increases confidence that it is not an artefact of any one setting. It would not be expected where review is untimed, and testing that is the natural next study.”

That paragraph states a mechanism, its scope conditions, the evidence for transferability, and a falsifiable prediction. It is stronger than the statistical claim would have been, and it is honest.

🔗 Four Traditions — the case study Across Ways of Knowing

📊 Empirical

Uneasy: no randomisation, no statistical inference. Its contribution is to generate the theory an experiment later tests, and to establish that a phenomenon occurs in the field at all.

🔧 Design Science

Provides the naturalistic summative evaluation of the FEDS grid (Chapter 9) — what happened when the artefact met real use.

🗣️ Qualitative

Its home. Thick description, multiple sources, context as constitutive, and the researcher acknowledged as part of the instrument.

√* Formal / Analytical

Contributes little directly, but case studies are where the assumptions of formal models are checked against practice — often unfavourably.

🔧 Try It Yourself: Design a two-case study with theoretical replication

1. State the research question as a how or why question.
2. Name the unit of analysis in one noun phrase.
3. Write two propositions.
4. Choose two cases for *theoretical* replication: name the dimension on which they differ and predict, in writing, how the results will differ.
5. List at least three data sources per case and say which proposition each addresses.
6. For your main expected finding, write the strongest rival explanation and the evidence that would discriminate between it and yours.

Step 6 is the one examiners notice. Feed this into portfolio piece P3.

❓ Checkpoint — Test Yourself

1. Why is a case study preferable to an experiment when phenomenon and context cannot be separated? What is given up?
2. A study of four companies concludes that “agile adoption in Pakistan faces cultural barriers”. What is wrong with the claim, and how would you rewrite it?
3. Explain why theoretical replication provides stronger evidence than literal replication.

4. Your interview data and repository data disagree. What are the three things this could mean, and how would you tell them apart?

✓ Chapter Summary

- A case study investigates a phenomenon in a context that cannot be separated from it. Trying something once is a demonstration, not a case study.
- Name the unit of analysis explicitly; choose holistic or embedded, single or multiple.
- A single case needs one of the five justifications. Access alone is not one.
- Propositions direct attention and let you be surprised. Exploratory studies state questions instead — and never invent propositions afterwards.
- The chain of evidence and the case study database make the work auditable, and §14(i) requires the raw data anyway.
- Choose cases theoretically. Predicting a difference in advance — theoretical replication — turns description into a test.
- Triangulate to find disagreement, not to accumulate agreement.
- Claim analytic generalisation: a mechanism with scope conditions and a falsifiable prediction. Never statistical generalisation.

✍ Review Questions

1. Case study research is criticised for lacking rigour and permitting biased interpretation. Identify the specific methodological devices in this chapter that answer that criticism, and say which of them is hardest to verify from a published report.
2. Explanation building revises the explanation as evidence accumulates, which resembles the practice Chapter 26 condemns as HARKing in quantitative work. Are they the same thing? Distinguish carefully.
3. You have access to one organisation and a strong theory. Design a single-case study that constitutes a genuine test of that theory, and state what result would refute it.
4. **PhD Regs 2024 §14.3(A)** asks whether “the sample produce[s] the type of knowledge necessary to understand the structures and processes within which the individuals or situations are located”. Contrast this criterion with statistical representativeness, and say why applying the wrong one to a case study is a category error.

Survey Research

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Separate** population, sampling frame and achieved sample, and say what each limits.
- **Choose** a sampling method and state honestly what inference it supports.
- **Diagnose** nonresponse bias, and explain why response rate alone does not measure it.
- **Write** items free of the six common wording faults.
- **Decide** whether to adopt a validated instrument or build one.
- **Report** a survey to a recognised standard.

🔗 Before You Start

Chapter 3 for constructs and operationalisation — a questionnaire is an operationalisation, and most survey failures are measurement failures. Chapter 17 follows this chapter and completes it.

📖 Key Terms in This Chapter

population • sampling frame • coverage error • probability sampling • stratified sampling • cluster sampling • convenience sampling • snowball sampling • response rate • nonresponse bias • Likert item and Likert scale • double-barrelled question • social desirability bias • cognitive interview • pilot

13.1 What a Survey Can and Cannot Do

A survey measures what a population reports about itself at one point in time. That is genuinely valuable — attitudes, practices, perceptions and self-reported behaviour are often exactly what you need to know, and no other method gets them at scale.

It is also a narrow instrument, and three limits are worth stating at the outset.

A survey cannot	Because
Establish causation	Nothing is manipulated and nothing is randomised. Cross-sectional survey data supports association (Chapter 11)
Measure behaviour	It measures <i>reported</i> behaviour. People misremember, round, and present themselves favourably. Where a trace exists, mine it (Chapter 11) and treat the survey as complementary
Rescue a vague construct	If “code quality” is undefined, asking 300 people to rate it produces 300 different constructs averaged into one number (Chapter 17)

13.2 Population, Frame, Sample

☰ Three different things, routinely conflated

The **population** is who you want to say something about — “professional software developers in Pakistan”.

The **sampling frame** is the list you can actually draw from — members of three professional groups on a messaging platform. The gap between population and frame is **coverage error**, and it is usually the largest error in a computing survey, larger than sampling error by a wide margin.

The **sample** is who you invited; the **achieved sample** is who responded. The gap between them is nonresponse.

⚠ The confidence interval does not know about coverage error

A survey of 400 developers recruited from one online community can report a margin of error of about five percentage points. That number describes sampling error *within the frame* only. It says nothing about the developers who are not in that community, who may be the majority and may differ systematically.

Reporting the interval without discussing the frame implies a precision the study does not have. State the frame, state who it excludes, and say in which direction that would bias each main result.

13.3 Sampling

Method	How	Inference it supports
Simple random	Every frame member equally likely	To the frame, with quantified uncertainty. Rarely achievable in our field
Stratified	Divide the frame into strata, sample within each	As above, with better precision, and lets you report by stratum — valuable when a subgroup is small but important
Cluster	Sample groups, then units within them	As above, but with a design effect: correlated units inside clusters mean the effective sample size is smaller than the count
Systematic	Every k th member	As simple random, unless the frame has a periodicity matching k
Convenience	Whoever responds	<i>To nobody beyond the respondents.</i> Descriptive only
Purposive	Deliberately chosen for a characteristic	Analytic, not statistical (Chapter 12)
Snowball	Respondents recruit others	Reaches hidden populations; biased towards the well-connected
Quota	Fill preset category counts non-randomly	Looks representative on the quota variables, is not on anything else

💡 Non-probability sampling is not disqualifying; pretending otherwise is

Almost every software engineering survey uses convenience or snowball sampling, because no sampling frame for “software developers” exists. That is a real constraint, not a personal failing.

What separates a good such survey from a bad one is the write-up. A good one says: this is a convenience sample recruited from these three channels; here is the respondent profile in detail; here is who is likely under-represented; the results describe these respondents and generalise only as far as the reader judges them typical. A bad one reports a margin of error and calls the sample representative.

PhD Regs 2024 §14.3(B) asks about external validity — “can the results be generalized to other settings and to other populations?” The honest answer for a convenience sample is “to a limited degree, and here is why”, which is an acceptable answer. A dishonest “yes” is not.

13.4 Nonresponse

⚠️ Response rate is a weak proxy for the thing you actually care about

The standard reflex is to worry about response rate. What matters is **nonresponse bias** — whether responders differ from non-responders *on the variables you are measuring*.

These come apart in both directions. A 20% response rate with responders and non-responders alike on the measured variables produces an unbiased estimate. A 70% response rate where the 30% who declined are precisely the developers who found the topic uncomfortable produces a badly biased one. Response rate is worth reporting, but it does not settle the question.

Check	How
Compare with frame characteristics	If you know the frame’s distribution on any variable, compare it with your respondents’
Wave analysis	Compare early responders with late ones. Late responders resemble non-responders more closely; a trend across waves suggests bias
Follow up a sample of non-responders	Even 20 short interviews with people who declined is far more informative than any rate
Report the funnel	Invited, opened, started, completed, usable. Each drop-off is a place bias can enter

13.5 Writing Items That Measure Something

Fault	Example	Why it breaks
Double-barrelled	“Our tests are fast and reliable”	A respondent who thinks they are fast but flaky cannot answer
Leading	“How much has our tool improved your workflow?”	Presupposes improvement; no option expresses “it did not”
Vague quantifier	“Do you review code regularly?”	Regularly means daily to one person and monthly to another. Ask for a frequency
Assumes knowledge	“Do you use mutation testing?”	Respondents who do not know the term will guess rather than admit it

Fault	Example	Why it breaks
Socially loaded	“Do you write unit tests for your code?”	There is a right answer and everyone knows it. Ask what proportion of last week’s commits had tests
Recall-heavy	“How many bugs did you fix last year?”	Nobody knows. Shorten the window or use a trace

13.5.1 Likert items, and one distinction that matters

A **Likert item** is a single statement with an ordered agreement response. A **Likert scale** is several items summed to measure one construct. The distinction matters because it determines what analysis is legitimate: a single item is ordinal, and a summed multi-item scale is closer to interval. Chapter 17 and Chapter 19 take this up; the design consequence is that **if you intend to treat a construct as continuous, you must write several items for it**, not one.

Points	5 or 7. More points do not add information once respondents stop discriminating
Midpoint	Include one unless you have a specific reason to force a choice; forcing produces arbitrary answers, not decisive ones
Labels	Label every point, not just the ends
Direction	Reverse-word a few items to detect straight-lining — but avoid negations, which are misread
Don’t know	Offer it separately from the midpoint. They mean different things and conflating them corrupts the mean

💡 Adopt a validated instrument if one exists

If you are measuring technology acceptance, usability, workload or job satisfaction, validated instruments exist — UTAUT2 [62], SUS, NASA-TLX and others. Use them, unmodified, and cite them.

The gain is substantial: the construct validity work is done, you can compare your results with the literature, and reviewers do not have to take your word for the measurement. Modifying items — even lightly, even to fix awkward wording — discards the validation, so if you must adapt, say so explicitly and re-establish reliability and validity on your own data (Chapter 17, Chapter 21).

📅 Template: C8 — Survey instrument checklist

Work through this before the pilot, not after.

- ☐ Every item traces to a construct, and every construct to a research question
- ☐ Multi-item scales for any construct to be treated as continuous
- ☐ No double-barrelled, leading or vague-quantifier items
- ☐ Sensitive items behaviourally anchored rather than asked directly
- ☐ Response options exhaustive and mutually exclusive
- ☐ “Don’t know” distinct from the midpoint
- ☐ Demographics at the end, and only those actually analysed
- ☐ Estimated completion time, measured on the pilot not guessed
- ☐ Consent screen: purpose, storage, withdrawal, contact (Chapter 27)
- ☐ Anonymous or identifiable — decided, stated, and consistent with the data handling plan
- ☐ Cognitive interviews with 3–5 people from the target population
- ☐ Full pilot, with the analysis script run on pilot data
- ☐ Instrument to be published as an appendix (Chapter 25)

13.5.2 Cognitive interviewing

Sit with three to five people from the target population and have them complete the questionnaire while thinking aloud. Ask what each question meant to them and how they arrived at the answer. This is the single highest-value hour in survey design. It reliably reveals items that are read in a sense you never intended — and unlike a pilot, which tells you the distribution of answers, it tells you *why* people answered as they did.

13.6 Reporting

Population and frame	Who you wanted, what list you had, who is missing
Recruitment	Every channel, with dates
The funnel	Invited, started, completed, usable, with exclusions and reasons
Respondent profile	Enough that a reader can judge typicality
Instrument	In full, as an appendix or a deposit
Reliability	For each multi-item scale (Chapter 17)
Analysis	Including how ordinal items were treated and why
Nonresponse	Which checks you ran and what they showed

For internet surveys the CHERRIES checklist [16] is a usable reporting standard, and Kitchenham and Pfleeger’s series remains the software engineering reference [35].

! The survey that measured nothing

The characteristic failure: 300 responses, a 4.1 out of 5 mean agreement that “documentation is important”, and no finding.

It happens because the items were written without a research question behind them, so no answer could have been surprising. Before writing any item, apply the null test from Chapter 10: what distribution of answers would have made you conclude the opposite? If no distribution would have, delete the item.

Four Traditions — the survey Across Ways of Knowing

Empirical

Estimates a population quantity with quantified uncertainty — when probability sampling is possible. When it is not, the survey is descriptive and the write-up must say so.

Design Science

Surveys practitioners to characterise the problem before designing, and to gather perceived-utility evidence afterwards — which is formative naturalistic evaluation, and weak as summative evidence.

Qualitative

Open-ended responses are qualitative data and deserve real analysis (Chapter 23), not a word cloud. Often the most informative part of the instrument.

Formal / Analytical

Little role, except that survey sampling theory is itself a formal subject and design effects are computed, not guessed.

Try It Yourself: Build and cognitively test an instrument

1. List your constructs. For each, decide: validated instrument, or your own items?
2. Write items. Multi-item scales for anything continuous.
3. Work template C8 end to end.
4. Run cognitive interviews with three people from the target population. Record every item that was misread.
5. Revise, then pilot with 10–15 people and run your full analysis script on the pilot data.
6. Write the population, frame and coverage paragraph — including who is excluded and in which direction it biases each main result.

This is portfolio piece P4.

Checkpoint — Test Yourself

1. Distinguish population, sampling frame and achieved sample for a survey of Pakistani software professionals. Where is the largest error likely to be?
2. Why can a 70% response rate be more biased than a 20% one?
3. Rewrite: “Do you agree that our new CI pipeline has made builds faster and more reliable?”
4. You need to treat “perceived usefulness” as continuous in a regression. What does that require of your instrument design?

Chapter Summary

- Surveys measure reported attitudes and practices at one time. They do not establish causation and do not measure behaviour.
- Population, frame and achieved sample are three things. Coverage error usually dominates sampling error in our field, and the confidence interval does not know about it.
- Non-probability sampling is normal and acceptable; claiming representativeness for it is not.
- Nonresponse bias, not response rate, is what matters. Check it with wave analysis, frame comparison or follow-up.

- Six wording faults account for most bad items. Behaviourally anchor anything socially loaded.
- Several items per construct if you want to treat it as continuous. Adopt validated instruments unmodified where they exist.
- Cognitive interviews are the highest-value hour in survey design.
- Report the funnel, the profile and the instrument in full.

Review Questions

1. No sampling frame exists for “software developers”. Given that, is probability sampling ever achievable in software engineering survey research? If so, describe a setting where it is; if not, say what the field should do about its collective reliance on convenience samples.
2. Self-report and trace data on the same behaviour frequently disagree. Design a study that would establish the size and direction of that gap for one specific practice, and say what it would be worth knowing.
3. Adopting a validated instrument imports its construct definition, which may have been developed in a different culture and language. Discuss the trade-off for research in Pakistan, and describe what re-validation would require.
4. **PhD Regs 2024 §14.3(B)** asks for external validity in both ecological and population senses. For a convenience-sampled survey, what is the most defensible thing a dissertation can claim, and how should the threats section be written?

Qualitative Methods and Coding

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 3 — Research Designs for Computing

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Design** an interview guide that produces analysable data.
- **Choose** between grounded theory and thematic analysis on principled grounds.
- **Code** qualitative data through open, axial and selective stages, with memos.
- **Judge** when inter-coder reliability is appropriate and when it is a category error.
- **Defend** a claim of saturation with evidence rather than assertion.

🔗 Before You Start

Chapter 2 — the interpretivist and critical realist positions, and why importing positivist criteria into this work is a mistake. Chapter 12 shares much of the data collection machinery.

📖 Key Terms in This Chapter

semi-structured interview • interview guide • probe • focus group • purposive sampling • theoretical sampling • saturation • grounded theory • constant comparison • open, axial and selective coding • memo • thematic analysis • codebook • inter-coder reliability • Krippendorff's alpha • member checking • reflexivity

14.1 What Qualitative Work Is For

Quantitative methods tell you how much and how often. Qualitative methods tell you what something *is*, how it works, and why people do what they do. When the categories themselves are unknown — when you cannot yet write the questionnaire because you do not know what to ask — there is no quantitative substitute.

In computing this arises constantly. Why do teams abandon a practice they believe in? What does “technical debt” actually mean to the people who use the phrase? How do reviewers decide what deserves a comment? These are not questions with numeric answers.

⚠️ The most common error is importing the wrong criteria

A quantitatively trained reader asks a qualitative study: is the sample representative? What is the margin of error? Have you controlled for confounders?

These are category errors. A study seeking to understand meaning in context is not estimating a population parameter, and 14 purposively chosen participants are not a failed attempt at 400 random ones. The right questions are different — and **PhD Regs 2024 §14.3(A)** asks them: does the sample produce the *type* of knowledge needed to understand the structures involved, is the description detailed enough to interpret, are different sources compared and contrasted, and does the analysis move from quotation to interpretation of significance?

Learn to ask those, and to answer them when asked the others.

14.2 Interviews

Type	Structure	Use when
Structured	Fixed questions, fixed order	You know the categories and want comparability. Close to an orally administered survey
Semi-structured	A guide of topics, freely ordered, with probing	The default in computing research. Comparable across participants, and still able to follow the unexpected
Unstructured	A single opening prompt and listening	Genuinely exploratory work, or life-history style research

📄 Template: C9 — Interview guide and codebook

Interview guide

Opening (rapport, consent, recording permission)

Warm-up: role and context

Topic 1 — addresses RQ____
probes

Topic 2 — addresses RQ____
probes

Critical incident prompt

Closing: anything I did not ask?

Codebook — one row per code, maintained as coding proceeds

Code	Definition	Include when	Exclude when
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

💡 Four habits that separate usable interviews from unusable ones

1. **Ask about specific incidents, not general policy.** “Walk me through the last code review you did” produces data. “What is your code review process?” produces the official answer, which is often not what happens.

2. **Tolerate silence.** The three seconds after an answer is where the qualification comes. Interviewers who fill it lose their best material.

3. **Probe with the participant’s own words.** “You said it was messy — what did messy look like that day?” Never substitute your term for theirs; that is how the researcher’s categories replace the participant’s.

4. **Record and transcribe.** Notes reconstruct what you already believed. Budget four to six hours of transcription per interview hour, or a transcription tool plus correction.

14.3 Sampling and Saturation

Qualitative sampling is **purposive**: participants are chosen because they can inform the question. In grounded theory it becomes **theoretical sampling** — each round of participants is chosen based on what the previous round revealed, which means you cannot fix the sample in advance and should not pretend to.

 **Saturation**

Saturation is the point at which additional data stops producing new codes, categories or theoretical insight.

It is the standard justification for a qualitative sample size, and it is routinely asserted rather than demonstrated [53]. It is also frequently misapplied: saturation belongs to grounded theory’s iterative logic, where sampling and analysis alternate. A study that conducts all 15 interviews before analysing any of them cannot know where saturation occurred.

Instead of asserting it	Show it
“Saturation was reached at 14”	Report new codes per interview: a table or curve showing 9 new codes in interview 1, 2 in interview 10, 0 in interviews 13 and 14
“No new themes emerged”	State the stopping rule <i>in advance</i> : two consecutive interviews yielding no new codes
Silence on the point	If sampling was constrained by access rather than saturation, say so. It is a limitation, not a disqualification

14.4 Two Analytic Approaches

14.4.1 Grounded theory

Grounded theory builds a theory from data, through iterative coding and constant comparison, with sampling directed by the emerging theory. It is demanding, and it is invoked far more often than it is done.

 **Anchor Reading: Grounded theory as it is actually practised in our field**

Klaas-Jan Stol, Paul Ralph and Brian Fitzgerald, *Grounded theory in software engineering research: a critical review and guidelines*, ICSE 2016 [59].

Why it matters. They reviewed papers claiming grounded theory and found most used it as a synonym for “we coded some interviews” — without theoretical sampling, without constant comparison, and without producing a theory. If you are going to claim grounded theory, read this first and make sure you can meet the claim.

What to extract. Which variant you are following — Glaserian, Straussian or Charmaz’s constructivist [7] — and their reporting guidelines. The variants differ on questions that matter, particularly whether you may review the literature first.

Stage	What you do	Output
Open coding	Line-by-line, naming what is happening. Codes stay close to the data — gerunds help: “deferring the refactor”, not “technical debt”	Many codes, deliberately unrefined
Axial coding	Relate codes to each other: conditions, actions, consequences	Categories with structure
Selective coding	Identify the core category and relate everything to it	A theory with a centre
Memoing	Throughout: write your thinking as it happens, dated	The audit trail, and most of the eventual discussion chapter

⚠️ Memos are not optional

Memoing is what distinguishes grounded theory from coding. A memo records why you created a code, how two categories seem to relate, what surprised you, and what you now want to sample for.

Without memos, the theory appears at the end with no traceable derivation, and neither you nor an examiner can reconstruct how you got there. With them, the audit trail is complete and the discussion chapter is half-written. Date every memo; they are also your evidence of when an idea occurred, which matters if anyone asks whether it preceded or followed the data.

14.4.2 Thematic analysis

Braun and Clarke’s six phases [5] are more accessible and, for most graduate projects with a defined research question, more appropriate than grounded theory. Thematic analysis identifies patterns; it does not claim to build a theory, and it does not require theoretical sampling.

1	Familiarisation	Read everything, repeatedly, before coding anything
2	Initial codes	Systematic coding across the whole dataset
3	Searching for themes	Collate codes into candidate themes
4	Reviewing themes	Check themes against coded extracts and the whole dataset; some collapse, some split
5	Defining themes	Name each and write what it is and is not. If you cannot define it in two sentences, it is not a theme
6	Writing up	Analytic narrative with data extracts as evidence, not as decoration

❗ Themes do not “emerge”

The word is a habit and it hides a claim. Themes are constructed by an analyst making choices — which codes to group, what to call them, what to leave out. Saying they emerged implies they were sitting in the data waiting to be found, which removes the analyst from a process they in fact drove.

Write instead: “we constructed four themes”. Then say who did the coding, what their background was, and how disagreements were resolved. That is reflexivity, and it is what §14.3(A) means by treating subjective perceptions as knowledge in their own right — including the researcher’s.

14.5 Reliability, and When It Does Not Apply

Whether two coders should agree is a genuinely contested question, and the answer depends on your paradigm (Chapter 2).

Position	Argument
Reliability is required	If codes are meant to be descriptive categories applied consistently, two trained coders should apply them alike. Report Cohen's κ (two coders) or Krippendorff's α (any number, any scale) [36]
Reliability is a category error	Under a constructivist position, interpretation is the researcher's contribution. Demanding that two people converge asks for a consistency the method does not claim, and rewards shallow, safe codes

💡 A defensible position for a thesis

Split the difference along the analysis, because the two halves have different characters.

Descriptive coding — “mentions a code review tool”, “describes a deadline conflict” — should be reliable, and reporting α is appropriate and cheap.

Interpretive theme construction — what the pattern means — is argued rather than measured. Here the appropriate evidence is a documented audit trail, negative case analysis (actively seeking data that contradicts your theme), peer debriefing, and member checking with participants.

Say in the methods chapter which parts of the analysis are which, and apply the right standard to each. That is a stronger position than either extreme, and it survives examination by a committee containing both kinds of reader.

🌐 Four Traditions — qualitative inquiry Across Ways of Knowing

📊 Empirical

Regards it as hypothesis-generating: the interviews tell you what to measure, and the measurement comes later. Useful, and an incomplete account of what qualitative work contributes.

🔧 Design Science

Supplies the problem characterisation before design and the naturalistic formative evaluation after — what did people do with the artefact that you did not anticipate?

🗣️ Qualitative

Its home. Meaning is constructed, context is constitutive, the researcher is part of the instrument, and rigour comes from transparency and audit rather than from replication.

√ Formal / Analytical

Rarely engaged, but qualitative study of how practitioners reason about formal methods is one of the more revealing literatures about why formal techniques are not adopted.

🛠️ Try It Yourself: Code one interview three ways

Obtain a transcript — conduct one interview yourself, or use a published transcript from an open dataset.

1. Open-code it line by line using gerunds. Aim for 30 or more codes.
2. Have a colleague code the same transcript independently with your codebook. Compute agreement on the descriptive codes only.

3. Discuss every disagreement. Revise the codebook definitions — the include and exclude columns of template C9 — rather than negotiating individual segments.
4. Group your codes into three or four candidate themes, and write a two-sentence definition of each saying what it is *and is not*.
5. Write one memo, dated, on the most surprising thing in the transcript.

Use Taguette (open source) or QualCoder if you want tool support; a spreadsheet is entirely adequate for one transcript.

Checkpoint — Test Yourself

1. Why does asking about a specific recent incident produce better data than asking about general practice?
2. A thesis reports “saturation was reached after 12 interviews”. What evidence should accompany that claim?
3. Under what circumstances is demanding inter-coder reliability a category error, and under what circumstances is omitting it a weakness?
4. Rewrite: “Three themes emerged from the data.”

Chapter Summary

- Qualitative methods answer what, how and why when the categories are not yet known. Judging them by sampling representativeness is a category error — §14.3(A) supplies the right criteria.
- Semi-structured interviews are the field’s default. Ask about specific incidents, tolerate silence, probe in the participant’s words, record and transcribe.
- Sample purposively; in grounded theory, theoretically. Demonstrate saturation with new-codes-per-interview, or admit access constrained you.
- Grounded theory requires theoretical sampling, constant comparison, memoing and an actual theory. Do not claim it otherwise.
- Thematic analysis is usually the right choice for a graduate project with a defined question.
- Themes are constructed, not emergent. Say who constructed them.
- Apply reliability to descriptive coding; apply audit trail, negative case analysis and member checking to interpretation. State which is which.

Review Questions

1. Stol and colleagues found most software engineering papers claiming grounded theory did not practise it. What incentives produce this, and what would a reviewer have to check to detect it?
2. Negative case analysis asks you to actively seek data contradicting your theme. Compare this with falsification in Chapter 2: are they the same epistemic move in different vocabularies, or genuinely different?
3. Automated coding using large language models is increasingly proposed for qualitative analysis. Given **PhD Regs 2024 §32(IV)**, which prohibits AI from “analyzing and interpreting data”, what — if anything — may legitimately be automated here? (Chapter 29 takes this further.)
4. Member checking returns findings to participants for comment. Give a case where a participant’s disagreement should change your analysis and one where it should not, and state the principle that distinguishes them.

Mixed Methods and Triangulation

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **State** a reason for mixing that is specific to your question.
- **Select** among convergent, explanatory sequential, exploratory sequential and embedded designs.
- **Notate** a design so a reader knows the sequence and the priority.
- **Integrate** strands with a joint display rather than juxtaposing them.
- **Recognise** when mixing adds nothing and should be dropped.

🔗 Before You Start

Chapter 2 for pragmatism, Chapter 10 and Chapter 13 for the quantitative strands, Chapter 14 and Chapter 12 for the qualitative ones.

📖 Key Terms in This Chapter

mixed methods • convergent design • explanatory sequential • exploratory sequential • embedded design • priority • integration • joint display • meta-inference • triangulation • divergence

15.1 Mixing Needs a Reason

“We conducted a survey and also some interviews” is not mixed methods research. It is two studies in one document. Mixed methods means the strands are *integrated* — one informs, explains, or is explained by the other, and the combined inference is something neither could have produced alone.

Legitimate reason	What it looks like
Explanation	The quantitative result is puzzling; qualitative work explains the mechanism
Development	Qualitative work identifies the constructs; a quantitative instrument measures them at scale
Triangulation	Two methods addressing the same question, where agreement increases confidence and disagreement is informative
Complementarity	Different facets of one phenomenon: how often it happens, and what it is like
Initiation	Deliberately seeking contradiction, to provoke reframing

⚠️ If you cannot name the reason, do not mix

Mixed methods costs roughly twice as much as a single-method study and requires competence in both traditions. A graduate scholar with limited time who mixes without a reason produces two mediocre strands instead of one good one.

The test: write the sentence “the qualitative strand is necessary because the quantitative strand cannot _____”. If you cannot complete it specifically, drop a strand and do the remaining one properly.

15.2 Four Designs and Their Notation

Notation makes the design unambiguous. Capitals indicate the dominant strand, a plus sign indicates concurrent strands, an arrow indicates sequence.

Design	Notation	Sequence	Use when
Convergent parallel	QUAN + QUAL	Both at once, merged at interpretation	You want corroboration or a fuller picture, and neither strand needs the other's results first
Explanatory sequential	QUAN → qual	Quantitative first, qualitative to explain it	A result is surprising, or you need the mechanism behind an effect. The most common design in computing
Exploratory sequential	QUAL → quan	Qualitative first, then measure at scale	The constructs are unknown; you must find out what to measure before you can measure it
Embedded	QUAN(qual)	A small strand inside a larger design	Interviews inside an experiment to understand why participants behaved as they did

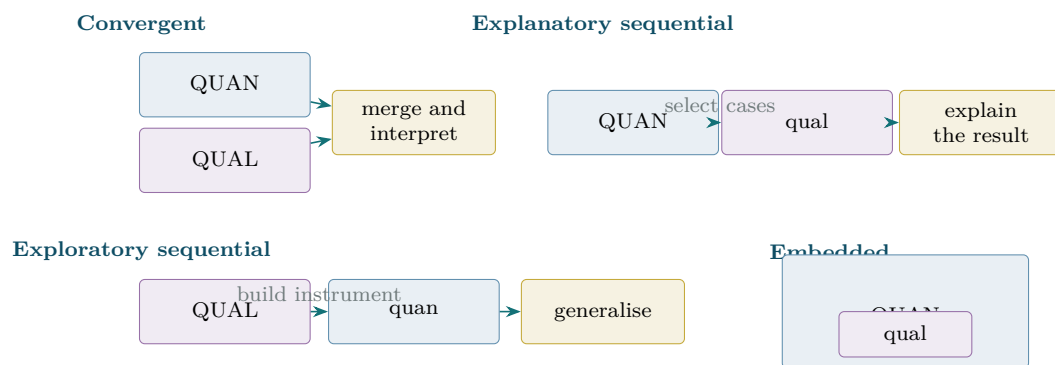


Figure 15.1: Four mixed methods designs. The gold box is the integration point — the part that distinguishes mixed methods from two studies bound together, and the part most often missing.

Worked Example: An explanatory sequential design

Strand 1 (QUAN). A survey of 280 developers finds that teams using a mandatory code review checklist report *lower* satisfaction with review quality than teams without one — the opposite of the prediction.

The integration decision. The surprising result determines who is interviewed: not a fresh sample, but a purposive selection from the survey respondents. Specifically, the extreme cases — checklist users with the lowest satisfaction scores — plus a contrasting group of checklist users with high scores. This selection *is* the integration; it could not have been made without the quantitative result.

Strand 2 (qual). Twelve interviews. The explanation that emerges: where review is time-boxed below the time the checklist takes to complete, reviewers satisfy the checklist mechanically and stop looking for anything else. Satisfaction is low because reviewers know they are performing compliance rather than reviewing.

Meta-inference. Neither strand supports this alone. The survey establishes the association across 280 people; the interviews supply the mechanism and the scope condition. The combined claim — checklists reduce perceived review quality *when review time is constrained below checklist completion time* — is testable, and Chapter 10 could now test it.

All figures illustrative.

15.3 Integration Is the Hard Part

Where	How
Design level	One strand's sampling or instrument is determined by the other's results
Method level	Building the questionnaire from interview codes; selecting interviewees from survey responses
Analysis level	Quantitising qualitative data (counting code occurrences) or qualitising quantitative data (creating profiles from clusters)
Reporting level	The joint display, and the meta-inference paragraph

15.3.1 The joint display

A **joint display** is a table or figure placing the two strands side by side so the reader can see the integration rather than being told about it.

Finding	Quantitative	Qualitative	Meta-inference
Checklist and satisfaction	Mean satisfaction 3.1 vs 3.8, $d = -0.41$	Reviewers describe “ticking boxes” under time pressure	Confirms and explains: the effect is mediated by time-boxing
Experience	No moderation by years of experience	Senior reviewers describe ignoring the checklist entirely	Divergence: the null moderation may mask two opposite behaviours cancelling out. Warrants re-analysis

💡 Divergence is the most valuable row in the table

When the strands disagree, resist the urge to explain it away. Four things it can mean, and they are distinguishable:

1. **The construct differs.** The survey and the interviews were about subtly different things.
2. **Different populations.** Respondents and interviewees were not comparable.

3. **One strand is wrong.** An underpowered quantitative result, or a qualitative interpretation over-fitted to a vivid case.
4. **Genuine complexity.** Two mechanisms operate in opposite directions, and the aggregate hides both — as in the experience row above.

The fourth is the interesting case, and it is the reason mixing was worth doing. Report the divergence and your reasoning about which of the four it is.

! Four ways mixed methods goes wrong

- **No integration.** Chapter 4 quantitative, Chapter 5 qualitative, a conclusion mentioning both. This is the default failure.
- **Unstated priority.** The reader cannot tell which strand carries the claim, so cannot judge whether its evidence suffices.
- **Sequence violated.** An exploratory sequential design in which the questionnaire was actually written before the interviews, which then served to justify it.
- **Incompatible paradigms unacknowledged.** Treating interview counts as a measure of prevalence, which imports statistical logic into purposively sampled data. Counting codes is legitimate as a descriptive device within the sample; it is not an estimate of anything.

🔗 Four Traditions — mixing methods Across Ways of Knowing

📊 Empirical

Values the qualitative strand for explaining a mechanism and for generating hypotheses. Insists that the quantitative strand's inference is not weakened by the qualitative one's sampling.

🔧 Design Science

Naturally mixed: qualitative problem characterisation, artefact construction, quantitative evaluation, qualitative study of use. FEDS (Chapter 9) is effectively a mixed methods evaluation plan.

🗣️ Qualitative

Cautions that integration can subordinate qualitative work to a supporting role — “quotes to illustrate the numbers”. Priority notation exists partly to make that subordination visible when it happens.

√* Formal / Analytical

Little direct role; but formal analysis plus empirical measurement of whether the modelled effect matters at realistic scale is a mixing of a different kind.

🔧 Try It Yourself: Design a mixed study, or decide not to

1. Complete the sentence: “the second strand is necessary because the first cannot _____”. If you cannot, stop here and note that as your finding.
2. Choose a design and write its notation, including which strand dominates.
3. Specify the integration point precisely: what exactly does one strand determine about the other?
4. Draft an empty joint display with your expected finding rows.
5. Write what you would do if the strands diverge, using the four categories above — *before* you have any data.

? Checkpoint — Test Yourself

1. Why is “we did a survey and interviews” not a mixed methods study?
2. You find a counter-intuitive quantitative result. Which design follows, and how does the first strand determine the second’s sampling?
3. What does QUAL → quan tell a reader that “mixed methods” does not?
4. Your strands disagree. Give the four possible explanations and one piece of evidence that would discriminate among them.

✓ Chapter Summary

- Mixed methods means integrated strands, not two studies in one document. Name the reason for mixing or drop a strand.
- Four designs: convergent, explanatory sequential, exploratory sequential, embedded. Notate them with capitals for priority and arrows for sequence.
- Integration happens at design, method, analysis or reporting level. Design-level integration — one strand determining the other’s sampling — is the strongest.
- The joint display shows the integration instead of asserting it.
- Divergence between strands is the most valuable finding, and there are four distinguishable reasons for it.
- Counting codes describes your sample; it does not estimate prevalence.

✍ Review Questions

1. Pragmatism is offered as the paradigm that licenses mixing. Is it a principled position or a way of avoiding the incompatibility problem? Argue both, with reference to Chapter 2.
2. Mixed methods doubles the cost. For a two-semester MS thesis, when is that ever the right allocation? Give a concrete example and say what would be sacrificed.
3. Priority notation makes subordination of the qualitative strand visible. Does making it visible make it acceptable? Discuss with reference to a QUAN(qual) design.
4. Design a study of your own topic in which the strands are genuinely likely to diverge, and say in advance what each of the four explanations would imply for your conclusion.

Benchmarking, Simulation and Performance Evaluation

Learning Outcomes

By the end of this chapter you will be able to:

- **Design** a benchmark whose workload is defensibly representative.
- **Measure** performance with a methodology that survives non-determinism.
- **Choose** the correct summary statistic, and explain why the arithmetic mean of ratios is wrong.
- **Report** performance results with uncertainty rather than as point values.
- **Validate** a simulation model, and distinguish verification from validation.

Before You Start

Chapter 10 for design and randomisation — a benchmark run is an experiment whose subjects are configurations. Chapter 3 for the comparison element, which benchmarking makes concrete.

Key Terms in This Chapter

benchmark • workload characterisation • representativeness • warm-up and steady state • non-determinism • repetition and replication • geometric mean • speedup • confidence interval • verification and validation • transient removal • benchmark overfitting

16.1 A Benchmark Is an Argument About Representativeness

Every performance claim has the same structure: on *this* workload, on *this* platform, under *this* measurement methodology, X was faster than Y by *this much*. Each qualifier is a place the claim can fail to transfer, and the benchmark is the argument that the qualifiers do not matter as much as one might fear.

Question	What a good answer looks like
Representative of what?	A named target population of real workloads, characterised on dimensions that matter — request mix, data size and skew, concurrency, arrival pattern
How do you know?	Evidence: production traces, published characterisations, a survey of practitioners. Not intuition
What does it exclude?	The workload regimes deliberately not covered, and what would happen there
Can it be re-run?	Version-pinned software, published configuration, released harness and data (Chapter 25)

Building the benchmark that your system wins

The characteristic failure, and it is usually unconscious. You develop the system, testing on workloads that exercise the parts you optimised. Those workloads become the evaluation.

The comparison is real, the measurement is careful, and the result is uninformative — because the workload was selected, implicitly, by which cases the system handles well.

Two defences. Fix the workload set *before* building, from an external characterisation, and write it into the protocol. And include at least one workload where you expect your approach to lose, then report that it did. A paper that shows where its own contribution fails is dramatically more credible than one that wins everywhere, and reviewers know it.

16.2 Measurement Under Non-Determinism

Modern systems are not deterministic. Just-in-time compilation, garbage collection, address space layout randomisation, CPU frequency scaling, cache state, background processes and network variability all move the number. A single run tells you almost nothing.

 **Anchor Reading: Why one run is not a measurement**

Andy Georges, Dries Buytaert and Lieven Eeckhout, *Statistically rigorous Java performance evaluation*, OOPSLA 2007 [21].

Why it matters. They showed that the then-standard practice — report the best of a few runs — produced conclusions that flatly contradicted statistically sound analysis of the same data. The paper changed measurement practice in the systems community, and its argument generalises well beyond Java.

What to extract. The distinction between start-up and steady-state performance, the need for multiple JVM invocations rather than multiple iterations within one, and the practice of reporting confidence intervals rather than a single number.

Concern	What to do	Why
Warm-up	Discard iterations until the metric stabilises; report how you decided	Early iterations measure compilation and cache filling, not the system
Steady state vs start-up	Decide which you are measuring, and say so	They are different questions. A serverless workload cares about start-up; a long-running service does not
Repetitions	Many iterations <i>within</i> a process, and many process invocations	Within-process variation and between-process variation have different sources; only the latter captures layout and JIT-decision randomness
Interleaving	Randomise or alternate the order of configurations	Thermal drift and background load change over a session and would otherwise confound whichever configuration ran last
Environment control	Pin frequency, disable turbo, isolate cores, record kernel and library versions	Otherwise the machine is a nuisance variable you have not controlled
Outliers	Decide the rule in advance and report how many were removed	Removing slow runs after the fact biases towards your preferred result

16.3 Summarising: the Mean That Is Usually Wrong

⚠️ Never take the arithmetic mean of speedups

This is the most common statistical error in systems papers, and it is easy to demonstrate. Two benchmarks. On the first, your system takes 1 second where the baseline takes 2 — a speedup of 2.0. On the second, yours takes 2 seconds where the baseline takes 1 — a speedup of 0.5. The arithmetic mean of the speedups is $(2.0 + 0.5)/2 = 1.25$, suggesting a 25% improvement. But total time is 3 seconds for both systems. There is no improvement at all. Worse, had you defined speedup the other way round — baseline over yours — the arithmetic mean would give the same 1.25 in favour of the baseline. A summary statistic whose answer depends on which system you put on top is not measuring anything. The geometric mean, $(\prod_{i=1}^n s_i)^{1/n}$, gives $\sqrt{2.0 \times 0.5} = 1.0$ either way round. For ratios, use it.

Quantity	Summary to use
Ratios, speedups, normalised scores	Geometric mean
Times, when total time is what matters	Arithmetic mean, or the sum
Rates (operations per second)	Harmonic mean of the rates, equivalently the reciprocal of mean time
Latency distributions	Median and high percentiles — p95, p99, p99.9. The mean of a latency distribution is close to meaningless, because the tail is what users experience

💡 Report an interval, and the whole picture changes

A table of point values invites the reader to treat a 3% difference as real. A table with confidence intervals lets them see that the intervals overlap entirely, and that the honest conclusion is “no detectable difference on this workload”. This is the same discipline as Chapter 19, applied to systems measurement, and it has the same consequence: many published performance improvements would not survive it. Publish the raw per-run measurements alongside the paper so that others can check — it costs nothing and it is what Chapter 25 asks of you.

16.4 Simulation

When the real system cannot be built, cannot be perturbed, or cannot be run at the scale of interest, simulation substitutes a model. The model’s credibility is then the whole question.

Verification	Did I build the model <i>right</i> ? Does the code implement the intended model? Debugging, unit tests, conservation checks, degenerate cases with known answers
Validation	Did I build the <i>right</i> model? Does it correspond to the real system well enough for the question being asked? Comparison with measurements from the real system, expert review, sensitivity analysis

⚠ Validation is not optional, and it is what gets skipped

A verified but unvalidated simulation is a correct implementation of an arbitrary model. Its output is a property of your assumptions, not of the world, and any conclusion drawn from it is conditional on assumptions nobody checked.

At minimum: compare the model’s output against measurements from a real system in at least one regime where both can be run, and report the discrepancy. Then run a sensitivity analysis — vary each assumed parameter and report which ones the conclusion depends on. If the conclusion flips when an unmeasured parameter moves within its plausible range, you have found the study’s real limitation, and it belongs in the abstract.

Random seeds	Fix and publish them; run enough independent replications to estimate variability across seeds
Transient removal	Discard the initial period before the simulated system reaches steady state; state the method used to decide
Replications	Independent runs with different seeds. Report the number and the resulting interval
Run length	Long enough that the estimate stabilises; show that it does

16.5 Benchmark Suites, and Their Failure Mode

Standard suites — SPEC, TPC, YCSB, MLPerf and others — give comparability and save enormous effort. They also create an incentive problem.

💡 Benchmark overfitting

When a suite becomes the measure of merit, systems get optimised for the suite rather than for the workloads it was meant to represent. Over time the suite stops being a proxy for reality and becomes a target in its own right — and improvements on it stop transferring.

This is the same phenomenon as test-set overfitting in machine learning (Chapter 22), and it has the same remedies: hold out workloads the community has not tuned against, refresh the suite periodically, and report results on a workload of your own alongside the standard one. If your system wins on the suite and loses on a realistic workload you constructed, say so — that is a genuine finding about the suite.

🕒 Four Traditions — performance evaluation Across Ways of Knowing

📊 Empirical

A benchmark run is an experiment: configurations are treatments, runs are units, and the analysis needs intervals and repetitions like any other.

🔧 Design Science

Benchmarking is summative artificial evaluation of an artefact. The FEDS caution applies: strong on technical efficacy, weak on whether anyone benefits in practice.

💬 Qualitative

Asks what practitioners actually optimise for, which is frequently not what benchmarks measure — tail latency under partial failure, or predictability rather than throughput.

√* Formal / Analytical

Provides the asymptotic analysis that measurement complements. Neither substitutes for the other: an $O(n \log n)$ algorithm can lose to an $O(n^2)$ one at every n anyone runs.

🔧 Try It Yourself: Measure something properly

Take any two implementations of the same task — two sorting libraries, two JSON parsers, two database drivers.

1. Write the measurement protocol *first*: warm-up rule, iterations, process invocations, interleaving, environment controls, outlier rule.
2. Run it. Record every individual measurement, not summaries.
3. Compute the mean, median, and a 95% confidence interval per configuration. Plot the distributions, not just the summaries.
4. Now do the wrong thing deliberately: report the single best run of each. How different is the conclusion?
5. Add a workload where you expect the winner to lose. Report the geometric mean across both.

Step 4 is the point of the exercise, and it is worth doing once so that you never again believe a paper that reports one number.

❓ Checkpoint — Test Yourself

1. Why is the arithmetic mean of speedups wrong, and what test exposes it immediately?
2. Distinguish verification from validation, and say which one an unvalidated simulation still passes.
3. Why are multiple process invocations necessary rather than multiple iterations within one process?
4. Your system wins on SPEC and loses on a production trace. What do you report, and what have you actually discovered?

📋 Chapter Summary

- A benchmark is an argument about representativeness. Fix the workload set before building, from external evidence, and include a case you expect to lose.
- One run is not a measurement. Handle warm-up, distinguish start-up from steady state, repeat within and across processes, interleave configurations, and control the environment.
- Geometric mean for ratios; arithmetic for times; harmonic for rates; percentiles for latency. The arithmetic mean of speedups is not a statistic.
- Report intervals, not point values, and publish the raw measurements.
- Verification asks whether the model was built right; validation asks whether it is the right model. Unvalidated simulation reports your assumptions back to you.
- Standard suites become targets. Report a workload of your own alongside the suite.

🔪 Review Questions

1. Georges and colleagues showed that a widely used methodology produced conclusions contradicting sound analysis of the same data. What does this imply about the published performance literature predating it, and what should a reviewer today do with such results?
2. Benchmark suites both enable comparison and distort optimisation. Design a governance scheme for a suite in your area that mitigates overfitting, and identify who would have to bear its cost.
3. Simulation validation requires comparison with a real system, but simulation is often used precisely because the real system cannot be built. How should a study proceed when validation is impossible in principle, and what may it then claim?

4. Tail latency matters more than mean latency to users, yet means dominate published results. Give one methodological and one incentive-based explanation, and say which you find more persuasive.

PART IV

Measurement, Data and Analysis

A design produces numbers or text; neither speaks for itself. This part covers the two halves of turning observation into a defensible conclusion: making a construct observable without distorting it, and reasoning from the result without claiming more than it supports. The statistics here are chosen for what graduate computing research actually uses, and every technique is introduced with the mistake it is most often used to commit.

What Part 4 Covers

- **Chapter 17 — Measurement and Instrument Validity.** Scales, construct validity, reliability, and why Cronbach's alpha is not the reassurance it appears to be.
- **Chapter 18 — Sampling and Study Power.** Probability and purposive sampling, a-priori power, and the arithmetic that decides feasibility.
- **Chapter 19 — Statistical Inference Done Properly.** What a p-value is not, effect sizes with intervals, assumptions, and multiple comparisons.
- **Chapter 20 — Regression and Modelling.** Linear, logistic and mixed-effects models; inference versus prediction; interpreting coefficients honestly.
- **Chapter 21 — Structural Equation Modelling.** Latent constructs, CB-SEM and PLS-SEM, discriminant validity and common method bias.
- **Chapter 22 — Evaluating Machine Learning Research.** Leakage, validation that matches deployment, comparing classifiers, and documenting models and data.
- **Chapter 23 — Qualitative Data Analysis in Practice.** Coding with open-source tools, the audit trail, and trustworthiness criteria.
- **Chapter 24 — Threats to Validity.** The four-way taxonomy applied to computing, and how to write the section examiners look for.

Measurement Theory and Instrument Validity

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 4 — Measurement, Data and Analysis

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Classify** a measure by scale type and say what that permits.
- **Distinguish** the forms of validity and name a threat to each.
- **Compute** and **interpret** a reliability coefficient, and explain why Cronbach's alpha is weaker evidence than it appears.
- **Derive** measures from goals using GQM rather than measuring what is convenient.
- **Defend** a proxy measure, or abandon it.

🔗 Before You Start

Chapter 3 for operationalisation. This chapter is where that becomes a technical matter with tests attached. Chapter 13 if you are building an instrument.

📖 Key Terms in This Chapter

construct • measure • nominal, ordinal, interval, ratio • face validity • content validity • construct validity • convergent and discriminant validity • criterion validity • reliability • test-retest • internal consistency • Cronbach's alpha • McDonald's omega • GQM • proxy measure

17.1 The Weakest Link

A study is only as good as its measures. This is stated often and acted on rarely, because measurement is unglamorous and its failures are invisible in the output: a regression on a bad measure produces coefficients, standard errors and p-values exactly as it would on a good one. Nothing complains.

⚠️ Analysis cannot repair measurement

No statistical technique recovers a construct that was never captured. If your measure of “code quality” is comment density, then a sophisticated mixed-effects model on comment density tells you about comment density. The model is not wrong; the inference from it to code quality is.

This is why **PhD Regs 2024 §14.3(B)** places validity — “does it measure what it says?” — alongside reliability as a criterion your dissertation is assessed on, and why it is listed *before* the analysis questions.

17.2 Scales, and What Each Permits

Stevens' four scale types [58] determine which operations are meaningful.

Scale	Structure	Example	Meaningful operations
Nominal	Categories, no order	Programming language; bug type	Counts, mode, chi-square
Ordinal	Ordered, unequal or unknown intervals	Severity: low / medium / high; a single Likert item	Median, percentiles, rank-based tests
Interval	Equal intervals, arbitrary zero	Calendar date; a summed multi-item scale, approximately	Mean, standard deviation, most parametric tests
Ratio	Equal intervals, true zero	Response time; lines of code; defect count	All of the above, plus ratios: “twice as fast” is meaningful

💡 The Likert controversy, settled practically

Is it legitimate to take the mean of responses coded 1 to 5? Strictly, no: the distance between “disagree” and “neutral” is not known to equal that between “neutral” and “agree”.

In practice the field has reached a workable position:

- A **single Likert item** is ordinal. Report the median and the distribution; analyse with rank-based methods. Do not report a mean of 3.7.
- A **summed or averaged multi-item scale** with several items measuring one construct is conventionally treated as interval. Simulation work has repeatedly found parametric tests robust to this, provided the scale has enough items and the distribution is not badly skewed.
- When in doubt, **do both** and report that the conclusion does not depend on the choice. If it does depend on the choice, that is important and you must say so.

The design consequence, stated in Chapter 13, bears repeating: if you intend to treat a construct as continuous, write several items for it.

17.3 Validity: Five Questions About One Measure

Type	Asks	How you establish it
Face	Does it look right?	Judgement. The weakest form, and not evidence — but its absence is a warning
Content	Does it cover the whole construct?	Expert panel against a definition of the construct’s domain. A usability measure covering only efficiency has poor content validity
Construct	Does it measure the intended construct and not a neighbouring one?	<i>Convergent</i> : correlates with other measures of the same thing. <i>Discriminant</i> : does <i>not</i> correlate too highly with measures of different things (Chapter 21)
Criterion	Does it predict or agree with an accepted standard?	<i>Concurrent</i> : agrees with a gold standard now. <i>Predictive</i> : predicts a future outcome
Ecological	Does it behave the same outside the study setting?	Comparison with field data; often only arguable, not demonstrable

 Anchor Reading: Questionable measurement practices

Jessica Kay Flake and Eiko I. Fried, *Measurement Schmeasurement: Questionable Measurement Practices and How to Avoid Them*, *Advances in Methods and Practices in Psychological Science* 3(4), 2020 [18].  OA

Why it matters. It does for measurement what the replication literature did for analysis: names the specific practices — using a measure with no validity evidence, silently modifying items, selecting among several measured scales after seeing results — that quietly invalidate studies.

What to extract. Their six questions every author should be able to answer about every measure in their study. Answer all six for each of yours, in writing, before data collection.

17.4 Reliability, and the Alpha Problem

Reliability is consistency: would the measure give the same answer again? It is necessary but not sufficient for validity — a bathroom scale reading three kilograms heavy is perfectly reliable and invalid.

Type	Question	Estimated by
Test–retest	Stable over time?	Correlation between two administrations. Needs a gap long enough to prevent recall, short enough that the construct has not changed
Internal consistency	Do the items hang together?	Cronbach’s α [10], McDonald’s ω
Inter-rater	Do two observers agree?	Cohen’s κ , Krippendorff’s α (Chapter 14)
Parallel forms	Do two versions agree?	Correlation between forms

 What Cronbach’s alpha does not tell you

$\alpha \geq 0.7$ is quoted as though it certified a scale. It does not, for three reasons that are well established and widely ignored.

It is not a measure of unidimensionality. A scale measuring two distinct constructs can have a high α . Alpha tells you the items correlate, not that they correlate because of *one* underlying thing. Establishing dimensionality requires factor analysis (Chapter 21).

It increases with the number of items. Twenty mediocre items will reach 0.9. Adding items is not improving the scale.

It assumes tau-equivalence — that every item relates equally strongly to the construct — which is almost never true. When it is false, α underestimates or overestimates reliability unpredictably.

McDonald’s ω makes weaker assumptions and is available in the R packages `psych` and `MBESS` [50]. Report ω ; report α too if you wish, since reviewers expect it, but do not let it be your only evidence.

Listing 17.1: Reliability done properly: dimensionality first, then omega, then alpha for the reviewers who expect it.

```
library(psych)

# 1. Is it one dimension? Parallel analysis before anything else.
fa.parallel(items, fa = "fa") # how many factors are indicated?
```

```
fa(items, nfactors = 1, rotate = "none") # do all items load on it?

# 2. Reliability. omega_total is the one to report.
omega(items, nfactors = 1)

# 3. Alpha, with the item-dropped column: an item whose removal
# raises alpha substantially is measuring something else.
alpha(items)
```

17.5 Deriving Measures from Goals

The alternative to measuring what is convenient is to derive measures from what you want to know. The Goal–Question–Metric approach [2] does this in three levels.

Goal	Conceptual. <i>Analyse the code review process for the purpose of improving it with respect to defect detection from the viewpoint of the team lead in the context of a small product team</i>
Question	Operational. How many defects are found in review versus after release? Does review coverage vary by module criticality?
Metric	Quantitative. Defects found in review per KLOC; proportion of changed files reviewed; time from submission to first comment

The value of the goal template is that it forces the viewpoint and context to be stated. The same metric means different things to a team lead and to a developer, and a metric collected without a viewpoint tends to be used by whoever finds it convenient — usually as a performance measure, which is how measurement programmes get subverted.

Worked Example: Operationalising “developer productivity”

A candidate for the least valid construct in common use. Watch it fail four times.

Attempt 1: lines of code per week. Rewards verbosity, punishes deletion and reuse. A developer who removes 500 lines of duplication has negative productivity. *Construct validity: absent.*

Attempt 2: commits per week. Rewards splitting work into small commits, which is good practice — and therefore measures commit style, not output. *Confounded with a practice you may also be studying.*

Attempt 3: story points completed. Points are estimated by the team, often by the same people being measured, and inflate under measurement. A measure that changes when you start using it is not measuring a stable property. *Reactivity.*

Attempt 4: features delivered per quarter. Better — closer to value — but features differ enormously in size, and this ignores maintenance, review, mentoring and incident response, which is most of what senior developers do. *Content validity: poor, systematically biased against senior staff.*

The honest conclusion. “Productivity” as a single number for an individual developer may not be a valid construct at all. The defensible moves are: measure something narrower and well-defined (cycle time for a change of a stated class), measure at the team level where the output is more meaningful, or reframe the research question so it does not require the construct.

Discovering this in week 10 is a good outcome. Discovering it at the viva is not.

! Proxy measures: three questions before you accept one

Most computing measures are proxies. That is acceptable if you can answer:

1. **What is the evidence the proxy tracks the construct?** A citation, or a validation in your own data. Not plausibility.
2. **What else does it track?** Cyclomatic complexity correlates with lines of code, so “complexity predicts defects” may be “size predicts defects”.
3. **Does it change when people know it is measured?** If yes, Goodhart’s law applies: it will stop measuring the construct precisely when it starts mattering.

🔗 Four Traditions — measurement Across Ways of Knowing

📊 Empirical

Validity and reliability are technical properties with tests. Report both for every measure, and treat unvalidated measures as a threat.

🔧 Design Science

Evaluation criteria are measures too, and inherit every problem here. An artefact evaluated on an invalid measure has not been evaluated.

🗣️ Qualitative

Rejects the framing: the analogue is trustworthiness — credibility, transferability, dependability, confirmability — established by audit trail and member checking rather than by coefficients (Chapter 23).

✓✗ Formal / Analytical

Measurement is definitional: the quantity is what the definition says. The validity question reappears as whether the formal definition captures the informal notion.

🔧 Try It Yourself: Audit your own measures

For every measure in your alignment table (Chapter 3):

1. Name the construct and the measure separately.
2. State the scale type, and check that your planned analysis permits it.
3. Answer Flake and Fried’s question: what is the evidence this measures the construct? If the answer is “it seems reasonable”, mark it red.
4. For multi-item scales, state how you will report reliability — dimensionality check, then ω .
5. For each proxy, answer the three questions above.
6. Write a GQM goal statement for your main measure, including viewpoint and context.

Any measure still marked red needs either validation work or replacement. This is portfolio piece P4.

? Checkpoint — Test Yourself

1. A scale of 20 items has $\alpha = 0.91$. What does this establish, and what does it not?
2. Why can a measure be perfectly reliable and completely invalid? Give a computing example.
3. You report the mean of a single 5-point Likert item. What is the objection, and what would you report instead?

4. Give a construct in your area where measurement reactivity would be a serious threat, and say how you would detect it.

✓ Chapter Summary

- Analysis cannot repair measurement, and bad measures produce respectable-looking output. §14.3(B) assesses validity explicitly.
- Scale type determines permissible operations. Single Likert items are ordinal; multi-item scales are treated as interval, which is why constructs need several items.
- Five validity questions; construct validity, split into convergent and discriminant, is the one that matters most and is checked least.
- Reliability is necessary and not sufficient. Cronbach's alpha does not establish unidimensionality, rises with item count, and assumes tau-equivalence. Check dimensionality, then report McDonald's omega.
- GQM derives metrics from goals with a stated viewpoint and context, instead of measuring what is available.
- Some constructs in common use — individual developer productivity among them — may not be validly measurable. Narrow the construct, raise the level, or change the question.
- For every proxy: what is the evidence, what else does it track, and does it change under observation?

✍ Review Questions

1. Flake and Fried argue that unvalidated measures are as damaging as p-hacking, and far less discussed. Assess this claim for computing research, and identify which of our subfields is most exposed.
2. Goodhart's law says a measure that becomes a target ceases to be a good measure. What follows for research that studies organisations already using the metric you intend to measure?
3. Construct a case for the position that “developer productivity” is a coherent construct that we simply measure badly, and then the case that it is incoherent. Which do you find stronger, and what evidence would decide?
4. **PhD Regs 2024 §14.3(B)** lists reliability and validity as separate criteria. Design a dissertation-level demonstration that a new measure in your area satisfies both, and estimate what it would cost in time.

Sampling and Study Power

Learning Outcomes

By the end of this chapter you will be able to:

- **Relate** the four quantities — α , power, effect size, N — and solve for any one.
- **Choose** a smallest effect size of interest and defend it.
- **Run** an a-priori power analysis and report it.
- **Explain** why post-hoc power is uninformative and why underpowered studies inflate published effects.
- **Estimate** sample size for qualitative work without pretending it is a power calculation.

Before You Start

Chapter 10 for designs, Chapter 13 for sampling frames, and Chapter 17 — power depends on measurement precision, so an unreliable measure costs you sample size.

Key Terms in This Chapter

type I and type II error • α • power • effect size • Cohen’s d • Cliff’s δ • smallest effect size of interest • a-priori power analysis • post-hoc power • winner’s curse • design effect • saturation

18.1 Four Quantities, Three Free

α	Probability of a false positive: concluding there is an effect when there is none. Conventionally 0.05, which is a convention, not a law
Power $1 - \beta$	Probability of detecting an effect that is really there. Conventionally 0.80, meaning you accept a one-in-five chance of missing a real effect
Effect size	How big the effect is, in standardised or natural units
N	Sample size, at the unit of randomisation

Fix any three and the fourth is determined. In practice α and power are set by convention, so the real question is always the trade between effect size and N : *how small an effect do I need to be able to detect, and can I afford the sample that requires?*

Do this before the study, not after

An a-priori power analysis is a design decision. A post-hoc one — computing power from the effect you observed — is uninformative, and specifically so: it is a deterministic function of the p-value. A non-significant result always yields low “observed power”, so reporting it adds nothing beyond restating that the result was not significant.

If you want to say something useful after a null result, report the confidence interval on the effect, or run an equivalence test (Chapter 19). Those answer the question post-hoc power is usually reached for — “was my study capable of detecting anything worth detecting?” — and they answer it correctly.

18.2 Choosing an Effect Size, Honestly

This is where power analysis is usually corrupted. The scholar needs N to come out affordable, so they choose a large expected effect, and the calculation obligingly returns a small sample.

Basis	Assessment
Meta-analytic estimate from the literature	Best available. Note that published effects are inflated by publication bias, so shade downwards
A pilot study	Dangerous. Effect estimates from small pilots are extremely noisy, and if the pilot happened to over-estimate, you power for an effect that does not exist. Pilots are for feasibility and instrumentation, not effect size
Smallest effect size of interest	Best practice. Not what you expect, but the smallest effect that would matter to anyone. Below it you do not care whether the effect exists
Cohen's conventions (0.2 / 0.5 / 0.8)	A last resort. Cohen offered them for fields with no information and warned against routine use

The smallest effect size of interest turns statistics into an argument about substance

Instead of “what effect do I expect?”, ask “how large would the effect have to be for anyone to act on it?”

If a code review technique saved 30 seconds per review, no team would change their process; a saving of 10 minutes would matter. That judgement makes the target effect a claim about practice, defensible in the thesis and arguable by an examiner — which is exactly what you want. It also protects you: a study powered for the smallest interesting effect gives a meaningful null result, because failing to find an effect that size is worth knowing.

18.2.1 Common effect sizes

Measure	For	Notes
Cohen's d	Difference between two means	$d = (\bar{x}_1 - \bar{x}_2) / s_{\text{pooled}}$. Assumes roughly normal, similar variances
Hedges' g	As d , small samples	Corrects d 's upward bias when N is small. Prefer it below about 20 per group
Cliff's δ	Ordinal or non-normal data	Probability one group exceeds the other, minus the reverse. Distribution-free; the right default for software engineering data, which is rarely normal
Odds ratio / risk ratio	Binary outcomes	Report the risk ratio too: odds ratios are routinely misread as risk ratios and overstate effects when outcomes are common
r , R^2	Association, variance explained	R^2 is sample-dependent; report adjusted R^2 and an interval

Worked Example: Powering the code review experiment

The design. Two review techniques compared on defects identified. Between-subjects, $\alpha = 0.05$, two-tailed, power 0.80.

Step 1: smallest effect of interest. Reviews currently find about 6 defects on average, with a standard deviation of roughly 2.5. The team lead judges that fewer than 1 extra

defect per review would not justify changing the process. So the SESOI is 1 defect, giving $d = 1/2.5 = 0.40$.

Step 2: required N.

```
library(pwr)
pwr.t.test(d = 0.40, sig.level = 0.05, power = 0.80,
           type = "two.sample", alternative = "two.sided")
# n = 99.08 per group -> 100 per group, 200 total
```

Step 3: confront reality. You can recruit 30 developers. At $n = 15$ per group, power to detect $d = 0.40$ is about 0.14 — a 14% chance of finding a real effect of the size that matters. Running this study is close to pointless.

Step 4: the four honest responses.

- **Change the design.** Within-subjects: each developer reviews under both techniques. This removes between-subject variance and dramatically increases power. With a correlation of about 0.6 between a developer's two scores, roughly 40 to 50 developers-worth of paired observations suffice — and 30 developers each doing four reviews may reach it, if the nesting is modelled (Chapter 20).
- **Change the outcome.** A more precise measure — fewer missed defects on a seeded-defect task, where the ground truth is known — shrinks the standard deviation and raises d for the same real effect.
- **Change the question** to an exploratory or qualitative one (Chapter 14), which 30 people amply supports.
- **Run it as a registered pilot** whose declared purpose is to estimate parameters for a later multi-site study, and report it as such.

What is not honest is running it at $n = 15$, obtaining $p = 0.22$, and writing “no significant difference was found between the techniques”. That sentence implies evidence of no difference, and the study had almost no capacity to find one.

Figures illustrative.

18.3 Why Underpowered Studies Damage the Literature

An underpowered study is not merely uninformative. It is actively harmful, and the mechanism is worth understanding [6].

Consequence	Mechanism
Inflated published effects (the winner's curse)	With low power, only unusually large sample effects reach significance. So the effects that get published are systematically over-estimates — sometimes by a factor of two or more
Low positive predictive value	Among all significant results in a low-powered literature, a large proportion are false positives, even with $\alpha = 0.05$
Failed replications that teach nothing	A replication powered for the inflated published effect is itself underpowered for the true one
Wasted participants	Subjects gave time to a study that could not answer its question — which is an ethical issue, not only a methodological one (Chapter 27)

💡 The consolation for a small field

Software engineering research often cannot recruit hundreds of professionals. That is a real constraint, and the answer is not to give up on quantitative work. It is to:

- prefer within-subject designs, which buy power for free;
- use precise measures, since precision substitutes for sample size;
- report effect sizes with confidence intervals rather than dichotomous significance verdicts, so that a small study contributes an estimate even when it cannot settle a question;
- and design so the result can be *pooled*. Three well-reported small studies with compatible measures can be meta-analysed (Chapter 5); three studies reporting only p-values cannot.

That last point is the most important one in this chapter. A small study that reports its effect size, its interval and its raw data has contributed permanently to the evidence base. One that reports only “no significant difference” has contributed nothing.

18.4 Sample Size in Qualitative Work

Power analysis does not apply, and pretending otherwise is a category error. But “as many as I could get” is not an answer either.

Basis	How to state it
Saturation	With a stopping rule declared in advance and evidence of new codes per interview (Chapter 14)
Information power	Fewer participants are needed when the aim is narrow, the sample is specific, the theory is established and the dialogue is strong. State which of these apply
Design requirement	A cross-case design needs enough cases for the comparison: theoretical replication requires at least one of each type (Chapter 12)
Pragmatic	Access limited you. Say so plainly, and discuss what it means for the claim. This is an acceptable answer honestly given

! Five power mistakes

- **Post-hoc power** reported after a null result.
- **Powering from a pilot’s observed effect**, which is noise.
- **Powering at the wrong unit** — 200 reviews from 20 developers is not $N = 200$ (Chapter 10).
- **Ignoring the design effect** in clustered sampling: correlated units mean the effective sample is smaller than the count.
- **Ignoring measurement unreliability**. An unreliable measure attenuates the observed effect, so the N you need is larger than the calculation assuming perfect measurement suggests.

🔗 Four Traditions — sample size Across Ways of Knowing

📊 Empirical

Determined by a-priori power analysis against the smallest effect of interest, at the unit of randomisation, and reported in full.

🔗 Design Science

Evaluation sample size follows the FEDS strategy: a naturalistic summative evaluation may involve one organisation, and the claim is scoped accordingly (Chapter 9).

 Qualitative

Determined by saturation or information power, demonstrated rather than asserted. More participants is not automatically better; depth competes with breadth for the same hours.

 Formal / Analytical

Not applicable. A proof holds for all instances, which is precisely its advantage.

 Try It Yourself: Power your own study, then face the answer

1. State your smallest effect size of interest in *natural units*, and justify it with a sentence about practice.
2. Convert to a standardised effect using a variance estimate from the literature or your pilot.
3. Compute required N with **pwr** for your design.
4. Compare with what you can actually recruit.
5. If short, work through the four responses in the worked example and choose one. Write down which, and why.

Step 5 is the deliverable. Portfolio piece P3 asks for exactly this decision, and Appendix H gives credit for abandoning an infeasible design on evidence.

 Checkpoint — Test Yourself

1. Why is post-hoc power uninformative, and what should be reported instead after a null result?
2. Explain the winner's curse and why it makes an underpowered literature worse than no literature.
3. Why is powering from a pilot's observed effect size dangerous?
4. Your measure has reliability 0.6 rather than 0.9. Which direction does that move your required sample size, and why?

 Chapter Summary

- α , power, effect size and N : fix three, the fourth follows. The real trade is effect size against N .
- Do power analysis before, not after. Post-hoc power is a function of the p-value and adds nothing.
- Power for the smallest effect size of interest, not the expected one. That makes the target a defensible claim about practice.
- Cliff's δ is a better default than Cohen's d for software engineering data, which is rarely normal.
- Underpowered studies inflate published effects and fill the literature with false positives.
- When you cannot recruit: within-subject designs, precise measures, effect sizes with intervals, and reporting that permits later pooling.
- Qualitative sample size rests on saturation or information power, demonstrated — or on honestly stated access constraints.

 Review Questions

1. Power of 0.80 accepts a one-in-five failure to detect a real effect, while $\alpha = 0.05$ accepts one-in-twenty false positives. Why are these asymmetric, and is the asymmetry defensible for software engineering research?
2. Suppose every software engineering experiment were required to be powered at 0.80 for a defensible SESOI. What would happen to the field's published output, and would the field be better off?
3. Design a multi-site study for your topic that would achieve adequate power by combining several institutions' small samples, and identify the methodological problems that combination introduces.
4. Underpowered studies waste participants' time, which Chapter 27 treats as an ethical matter. Should ethics review boards assess statistical power? Argue both sides.

Statistical Inference Done Properly

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 4 — Measurement, Data and Analysis

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **State** what a p-value is, and identify the common misinterpretations of it.
- **Report** a result as an effect size with a confidence interval rather than a significance verdict.
- **Check** the assumptions a test actually makes, and choose a robust alternative when they fail.
- **Correct** for multiple comparisons, choosing between family-wise and false-discovery control.
- **Test** for equivalence when your claim is that there is no meaningful difference.

🔗 Before You Start

Chapter 18 for power and effect sizes — inference without power is uninterpretable. Chapter 17 for scale types, which constrain the tests available.

📖 Key Terms in This Chapter

null hypothesis significance testing • p-value • confidence interval • effect size • assumption checking • normality of residuals • non-parametric test • bootstrap • family-wise error rate • false discovery rate • Bonferroni • Holm • Benjamini–Hochberg • equivalence testing • TOST • garden of forking paths

19.1 What a p-Value Is

📖 The p-value

The probability of obtaining data *at least as extreme* as what you observed, **if the null hypothesis were true** and all other assumptions of the model held.

That final clause is routinely dropped and is doing real work. A small p-value indicates that the data are unusual under the *entire* model — null hypothesis plus independence, plus the distributional assumptions, plus the assumption that this test was the one you planned to run. Any of those can be what is wrong.

📖 Anchor Reading: The two documents to read on this

Ronald L. Wasserstein and Nicole A. Lazar, *The ASA Statement on p-Values: Context, Process, and Purpose*, The American Statistician 70(2), 2016 [64].  OA

Sander Greenland and colleagues, *Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations*, European Journal of Epidemiology 31, 2016 [22].  OA

Why they matter. The first is the American Statistical Association's official position — the professional body of statisticians stating that the practice its own members taught for

decades is widely misused. The second enumerates twenty-five specific misinterpretations, with corrections.

What to extract. From the ASA, the six principles. From Greenland, find the misinterpretations you personally hold; most researchers find two or three, and finding them is the point.

The misinterpretation	The correction
“ p is the probability the null hypothesis is true”	It is a probability computed <i>assuming</i> the null is true. It says nothing directly about the probability of a hypothesis
“ $p > 0.05$ means there is no effect”	It means this study did not detect one. With low power, that is expected even for real effects
“ $p < 0.05$ means the effect is important”	Significance is not magnitude. With $N = 10,000$, a trivial effect is significant
“ $p = 0.04$ and $p = 0.06$ are meaningfully different”	They are almost identical evidence. The threshold is a convention, not a boundary in nature
“A significant result will replicate”	The probability of replication depends on the true effect and the replication’s power, not on p
“ $1 - p$ is the probability the alternative is true”	No. This is the same error as the first, restated

⚠ The ASA’s six principles, compressed

1. P-values can indicate how incompatible the data are with a specified model.
2. P-values do not measure the probability that the hypothesis is true, or that the data were produced by chance alone.
3. Scientific conclusions should not be based only on whether a p-value passes a threshold.
4. Proper inference requires full reporting and transparency — every analysis run, not the one that worked.
5. A p-value does not measure the size of an effect or the importance of a result.
6. By itself, a p-value does not provide a good measure of evidence.

19.2 Report the Estimate, Not the Verdict

Instead of	Write
“The difference was significant ($p < 0.05$).”	“Checklist review found 1.3 more defects on average (95% CI [0.4, 2.2]), $d = 0.52$, $p = 0.004$. ”
“No significant difference was found.”	“The estimated difference was 0.2 defects (95% CI [−1.1, 1.5]). The interval includes effects large enough to matter, so this study does not settle the question.”
“Group A outperformed group B.”	“Group A found more defects, by a median difference of 2 (Cliff’s $\delta = 0.41$, 95% CI [0.12, 0.63]).”

💡 Why the interval carries more information

A confidence interval tells you the effect size, the precision, and the significance verdict all at once — if the interval excludes zero, $p < 0.05$. Nothing is lost by reporting it, and two things are gained.

First, a null result becomes interpretable. An interval of $[-0.1, 0.1]$ says the effect is small; an interval of $[-3.0, 3.2]$ says you learned almost nothing. Both give $p > 0.05$, and they are entirely different findings.

Second, your study becomes poolable. A meta-analysis (Chapter 5) needs estimates and their uncertainty, not verdicts. Reporting an interval is how a small study contributes permanently to the evidence base.

19.3 Assumptions, and What to Do When They Fail

Assumption	What it really says	If it fails
Independence	Each observation contributes fresh information	The most serious failure and the least often checked. Model the structure: mixed-effects (Chapter 20), or analyse at the unit of randomisation
Normality	The <i>residuals</i> are approximately normal — not the raw data	With N moderate, tests are robust. Otherwise use a rank-based test, bootstrap the interval, or transform
Equal variances	Groups have comparable spread	Use Welch's t-test, which does not assume it. Welch should arguably be the default rather than the fallback
Linearity	The relationship has the form the model assumes	Plot residuals against fitted values. Add terms, or use a model that does not assume it
No influential outliers	No single point drives the conclusion	Report the analysis with and without. If they disagree, that is the finding

❗ Testing for normality is usually the wrong move

The reflex is to run Shapiro–Wilk and choose a test based on the outcome. This is doubly unhelpful.

With a small sample the normality test has no power, so it passes almost anything — exactly when non-normality would matter most. With a large sample it detects trivial deviations and rejects, exactly when the central limit theorem has made the parametric test robust anyway. The test's sensitivity runs opposite to your need.

Look at a Q–Q plot instead, and decide on the size of the departure. Better still, use methods that do not require the assumption: rank-based tests, or bootstrap confidence intervals, which make no distributional assumption at all and are two lines of R.

Question	Parametric	Robust alternative
Two independent groups	Welch's t-test	Mann–Whitney U; bootstrap
Two paired measurements	Paired t-test	Wilcoxon signed-rank
Three or more groups	ANOVA	Kruskal–Wallis
Association, two continuous	Pearson r	Spearman ρ ; Kendall τ
Two categorical	Chi-square	Fisher's exact for small counts

19.4 Multiple Comparisons

Run 20 independent tests at $\alpha = 0.05$ on data with no real effects, and you expect one significant result. This is not a subtle problem: it is arithmetic, and it is why a paper reporting one significant finding out of many tested comparisons has usually found nothing.

Method	Controls	Use when
Bonferroni	Family-wise error rate: α/m	Few comparisons, and a false positive is costly. Simple, and conservative to the point of destroying power when m is large
Holm	Family-wise error rate, sequentially	Strictly more powerful than Bonferroni with the same guarantee. There is no reason to prefer Bonferroni over Holm [28]
Benjamini–Hochberg	False discovery rate: expected proportion of false positives among rejections	Many comparisons, exploratory work. You accept some false positives in exchange for finding real effects [4]
None, declared	Nothing	One pre-registered primary outcome. Everything else is explicitly exploratory and labelled as such

💡 Declaring a primary outcome is the cleanest solution

The strongest option is not a correction at all. Nominate one primary outcome and one primary comparison in advance (Chapter 26). Test it at 0.05. Report everything else as exploratory, without significance claims.

This is honest, it preserves power for the question you care about, and it removes the awkward question of what constitutes a “family” of comparisons — a question with no principled answer, which is why the choice of correction is so often argued about after the fact.

⚠️ The garden of forking paths

Multiplicity is not only about the tests you ran. It is also about the tests you *would have* run had the data looked different.

If you would have analysed the subgroup had the main effect failed; if you would have log-transformed had the residuals looked worse; if you would have excluded that outlier had it gone the other way — then your effective number of comparisons is far larger than the number you performed, and no correction applied to the tests you actually ran can fix it. This holds even if you ran exactly one test and reported it honestly.

The only real defence is to specify the analysis before seeing the data, which is what Chapter 26 is about, and what portfolio piece P5 asks of you.

19.5 Claiming There Is No Difference

Failing to reject the null is not evidence for the null. To claim two things are equivalent, you must test for equivalence.

💡 Two one-sided tests

TOST specifies equivalence bounds — the range within which any difference is too small to matter, which is your SESOI from Chapter 18 — and tests whether the effect is significantly *inside* both bounds [38].

The four possible outcomes are genuinely informative:

- Significant difference, not equivalent: a real, meaningful effect.
- Not significant, equivalent: real evidence of no meaningful effect — a conclusion NHST alone can never deliver.
- Not significant, not equivalent: inconclusive. The study was underpowered, and this is the honest label for it.
- Significant, and equivalent: a real effect that is too small to matter. Common with large samples, and worth reporting plainly.

Software engineering has a genuine need for this. “Our approach is as accurate as the state of the art but ten times faster” is an equivalence claim on accuracy, and it is normally supported by a non-significant p-value, which does not support it at all.

Listing 19.1: A complete two-group analysis, reported as an estimate.
Note that the significance test is the least of what is computed.

```
library(effsize); library(boot); library(TOSTER)

# 1. Look at the data before testing anything.
boxplot(defects ~ condition, data = d)
qqnorm(residuals(lm(defects ~ condition, data = d)))

# 2. Estimate with an interval that does not assume normality.
t.test(defects ~ condition, data = d)           # Welch by default in R
cliff.delta(defects ~ condition, data = d)      # robust effect size + CI

# 3. If the claim is "no meaningful difference", test that claim.
#   Bounds are the smallest effect of interest, not a convention.
TOSTtwo(m1 = 6.1, m2 = 5.9, sd1 = 2.4, sd2 = 2.6, n1 = 60, n2 = 60,
        low_eqbound_d = -0.40, high_eqbound_d = 0.40)

# 4. Correct only across the comparisons declared in the protocol.
p.adjust(c(p1, p2, p3), method = "holm")
```

19.6 A Word on Bayesian Methods

Bayesian analysis answers the question researchers actually want answered — how probable is this hypothesis given my data — rather than the inverted one NHST answers. It handles optional stopping gracefully, quantifies evidence for the null rather than merely failing to reject it, and produces credible intervals that mean what people wrongly think confidence intervals mean.

The honest caveats: it requires specifying a prior, which must be justified and which reviewers in our field will question; conventions for reporting are less settled; and your examiners may not know it. If your question genuinely calls for evidence *for* a null, a Bayes factor is the natural tool, and equivalence testing is the frequentist alternative that will be more readily accepted. Choose deliberately, and be able to defend the choice either way.

Four Traditions — statistical inference Across Ways of Knowing

Empirical

Central. The discipline is to report estimates with uncertainty, check assumptions, pre-specify comparisons, and treat significance as one piece of information rather than a verdict.

Design Science

Applies to evaluation results. A performance improvement demonstrated without an interval is an anecdote (Chapter 16).

Qualitative

Not applicable, and importing it is the category error of Chapter 14. Counting codes describes a sample; it estimates nothing.

Formal / Analytical

Probability appears as an object of study rather than a tool of inference. Randomised algorithms have provable bounds, not confidence intervals.

Try It Yourself: Analyse one dataset three ways

Take a two-group dataset — yours, or any open dataset.

1. Report only “significant / not significant”.
2. Report the effect size with a bootstrap confidence interval.
3. Run an equivalence test against a SESOI you justify in one sentence.
4. Write the three conclusions side by side. Which would a practitioner act on?
5. Now list every analysis decision you made — exclusions, transformations, which test — and ask how many you would have made differently had the data looked different. That count is your real multiplicity.

Checkpoint — Test Yourself

1. State precisely what $p = 0.03$ means, including the clause everyone omits.
2. Two studies both report $p > 0.05$. One has CI $[-0.05, 0.05]$, the other $[-2.0, 2.1]$. What does each tell you?
3. Why is testing for normality with Shapiro–Wilk usually unhelpful?
4. You want to claim your method is as accurate as the state of the art. Why is a non-significant t-test insufficient, and what would suffice?

Chapter Summary

- A p-value is the probability of data this extreme *under the whole model*, not the probability that a hypothesis is true.
- Report effect sizes with confidence intervals. A null result is interpretable only through its interval, and only interval-reporting studies can be pooled.
- Check assumptions by looking at residuals, not by testing normality. Prefer Welch, rank-based methods and the bootstrap.
- Correct for multiplicity, preferring Holm to Bonferroni and Benjamini–Hochberg for exploratory work — or declare a single primary outcome and label the rest exploratory.
- The garden of forking paths means your effective multiplicity includes analyses you did not run. Only pre-specification defends against it.

- To claim no difference, test for equivalence against a justified bound. Non-significance is not evidence of absence.

Review Questions

1. The ASA states that scientific conclusions should not be based on whether a p-value passes a threshold, yet thresholds remain universal. Identify the institutional forces sustaining this and assess whether any proposed reform could overcome them.
2. Equivalence testing requires specifying a bound, which is a judgement about practical importance. Critics say this makes it subjective. Is that a weakness, given that the alternative is a threshold nobody chose either?
3. Software engineering data is frequently skewed and heteroscedastic, and parametric tests dominate anyway. What accounts for the gap between the field's data and its methods, and what would change it?
4. Bayesian methods answer the question researchers want answered but require a prior. For a specific question in your area, specify a prior you could defend to an examiner, and say what you would do if they rejected it.

Regression and Modelling for Inference

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 4 — Measurement, Data and Analysis

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Interpret** a regression coefficient correctly, including the conditioning clause everyone omits.
- **Choose** among linear, logistic, Poisson and negative binomial models by the shape of the outcome.
- **Specify** a mixed-effects model when observations are nested.
- **Distinguish** modelling for explanation from modelling for prediction, and stop using the tools of one for the other.
- **Report** a model completely enough to be reproduced.

🔗 Before You Start

Chapter 19 for assumptions and intervals; Chapter 3 for variable roles — a regression coefficient means something quite different depending on whether the covariate is a confounder or a mediator.

📖 Key Terms in This Chapter

linear model • coefficient • residual • logistic regression • odds ratio • Poisson and negative binomial • overdispersion • mixed-effects model • random intercept and slope • collinearity • VIF • stepwise selection • explanation versus prediction

20.1 What a Coefficient Means

📖 The conditioning clause

In $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$, the coefficient β_1 is the expected change in y for a one-unit increase in x_1 **holding x_2 fixed**.

That clause is not decoration. It means β_1 answers a different question depending on what else is in the model, and adding or removing a covariate changes its value and sometimes its sign. There is no such thing as “the effect of x_1 ” independent of the model it sits in.

⚠️ Which covariates to include is a causal question, not a statistical one

The habit of including every available variable “to control for them” is actively harmful, and which variables belong depends on their causal role (Chapter 3).

- **Confounders** — causes of both treatment and outcome — *must* be included, or the estimate is biased.
- **Mediators** — on the causal path from treatment to outcome — *must not* be included if you want the total effect. Controlling for a mediator removes exactly the effect you were trying to measure.

- **Colliders** — common effects of treatment and outcome — must not be included: conditioning on one *creates* an association where none existed.

Draw the assumed causal structure before choosing the model. “I controlled for everything I had” is not a defence; it is a description of a model whose coefficients cannot be interpreted.

20.2 Choosing a Model by the Shape of the Outcome

Outcome	Model	Coefficient means
Continuous, roughly symmetric	Linear regression	Change in y per unit of x , in y 's units
Binary	Logistic regression	Change in <i>log odds</i> . Exponentiate for an odds ratio — which is <i>not</i> a risk ratio, and is routinely misread as one
Count, variance $\approx$ mean	Poisson regression	Multiplicative change in the expected count, after exponentiating
Count, variance $>$ mean	Negative binomial	As Poisson. This is the usual case in software data — defects, commits and issues are almost always overdispersed
Count with excess zeros	Zero-inflated or hurdle model	Two processes: whether any occur, and how many given some do
Time to event	Cox proportional hazards	Hazard ratio; handles censoring, which matters for time-to-fix data
Ordinal	Ordinal (proportional odds) regression	Respects ordering without assuming equal spacing (Chapter 17)

! Fitting Poisson to overdispersed counts

Defect counts, commit counts and issue counts nearly always have variance substantially greater than their mean, because the units are heterogeneous — some modules are simply more defect-prone than others.

Fitting a Poisson model to such data does not produce wrong coefficients so much as *far too narrow* standard errors. The result is confidence intervals that are too tight and p-values that are too small, and the entire inference is wrong in the direction of over-confidence.

Check: compute the residual deviance divided by its degrees of freedom. If it is much above 1, use a negative binomial model. In R this is one function call — `MASS::glm.nb` in place of `glm(family = poisson)`.

20.3 Nested Data and Mixed-Effects Models

Chapter 10 warned that analysing 200 reviews from 20 developers as $N = 200$ inflates significance. The mixed-effects model is the principled fix.

Fixed effect	The population-level effect you want to estimate — the treatment
Random intercept	Each developer has their own baseline. Removes the assumption that all developers are alike
Random slope	The treatment effect itself differs by developer. Fit this when it is plausible, which it usually is

Listing 20.1: A mixed-effects model for the code review experiment, with reviewer and the reviewed artefact as crossed random effects.

```
library(lme4); library(lmerTest)

# Reviewers and artefacts are crossed, not nested: every reviewer sees
# several artefacts and every artefact is seen by several reviewers.
m <- lmer(defects_found ~ technique + experience +
          (1 + technique | reviewer) + (1 | artefact),
          data = reviews)

summary(m)
confint(m, method = "profile")      # intervals, not just p-values

# Does the random slope earn its place?
m0 <- lmer(defects_found ~ technique + experience +
          (1 | reviewer) + (1 | artefact), data = reviews)
anova(m0, m)
```

💡 When you must use a mixed model

Whenever observations share something that makes them more alike than observations from different groups:

- Several measurements from the same participant.
- Several files from the same repository.
- Several students from the same class or supervisor.
- Repeated measures over time on the same unit.

Ignoring this structure is one of the most consequential errors available, and it is invisible in the output: the model runs, the coefficients look fine, and the standard errors are wrong. Decide the random-effects structure from the *design*, before fitting — it is a description of how the data was generated, not a modelling choice to be selected by fit statistics [3].

20.4 Explanation Is Not Prediction

📖 Anchor Reading: The distinction that resolves most modelling arguments

Galit Shmueli, *To Explain or to Predict?*, Statistical Science 25(3), 2010 [56].  OA

Why it matters. Explanatory and predictive modelling have different goals, and therefore different criteria, different variable selection, and different definitions of a good model. Confusing them is behind a great deal of muddled work — especially in computing, where scholars trained on machine learning apply its habits to inferential questions.

What to extract. That a model can be excellent for prediction and useless for explanation, and the reverse. Decide which you are doing, in writing, before you fit anything.

	Explanation	Prediction
Goal	Estimate an effect and its uncertainty	Minimise error on new data
Variables	Chosen by causal role	Chosen by predictive contribution; causality irrelevant
Judged by	Unbiasedness; correct standard errors	Held-out performance (Chapter 22)
Collinearity	A serious problem: it destroys the interpretability of individual coefficients	Largely harmless: predictions stay accurate
Overfitting	Less central, though still real	The central concern
Report	Coefficients with intervals	Performance on a test set, with the validation scheme

⚠ Stepwise selection has no place in explanatory modelling

Automated forward, backward or stepwise selection on p-values produces biased coefficients, standard errors that are far too small, p-values that are meaningless, and R^2 that is inflated. It also gives unstable results — resample the data and a different model is selected.

The reason is simple: the selection process is itself an analysis of the data, and the reported inference does not account for it. This is the garden of forking paths (Chapter 19) automated.

For explanation, specify the model from the causal structure and report it. For prediction, use regularisation — lasso, ridge, elastic net — with performance judged on held-out data, and do not interpret the resulting coefficients as effects.

20.5 Collinearity

When predictors are strongly correlated, the model cannot separate their contributions. Coefficients become unstable and their standard errors inflate, even though overall fit is unaffected.

Diagnose	Variance inflation factor. Above 5 warrants attention; above 10 usually indicates a real problem
Accept it	If the collinear variables are controls and you do not interpret their coefficients, collinearity among them is not a problem
Combine	If two predictors measure the same construct, combine them into one scale (Chapter 17) rather than fighting the model
Drop	Only with a substantive reason. Dropping the one with the larger p-value is a data-driven decision that invalidates the inference

Note the classic case in software engineering: cyclomatic complexity and lines of code correlate strongly. A model containing both cannot tell you which predicts defects, and a paper claiming complexity predicts defects while controlling for size is making a much weaker claim than it sounds.

20.6 Reporting a Model

The full specification	Every term, including random effects, in a form that could be retyped
Why these covariates	The causal reasoning, not “available in the dataset”
Coefficients	With confidence intervals, in interpretable units. Exponentiate log-odds and count-model coefficients
Model checks	Residual plots, overdispersion, VIF, influential points
Sample size	At every level: how many observations, how many developers, how many repositories
Missing data	How much, and how handled. Listwise deletion is a choice with consequences, not a neutral default
Software	Package and version; the model syntax verbatim (Chapter 25)

! Three interpretation errors that reach print

Reading an odds ratio as a risk ratio. An odds ratio of 2.0 does not mean the outcome is twice as likely. When the outcome is common the two diverge sharply, and the odds ratio always looks more dramatic.

Interpreting a coefficient as an effect in observational data. Regression adjusts for what you measured; it does not create the counterfactual (Chapter 11).

Comparing standardised coefficients across models or samples. Standardisation divides by that sample’s standard deviation, so a “standardised” coefficient is not comparable across populations with different variances.

🔗 Four Traditions — regression modelling Across Ways of Knowing

📊 Empirical

The workhorse. Model specification follows from the causal structure, and the report is coefficients with intervals plus diagnostics.

🔧 Design Science

Used to model the relationship between artefact configuration and outcome, and to identify which design parameters matter.

💬 Qualitative

Not used. The nearest analogue — explaining variation by tracing mechanisms across cases — is Chapter 12’s explanation building.

√* Formal / Analytical

Regression models have properties that are proved: identifiability, consistency, asymptotic normality. Knowing which of these fail under your data is what distinguishes fitting from modelling.

🔧 Try It Yourself: Fit, then break, a model

Take a dataset with a nested structure — multiple observations per unit.

1. Draw the assumed causal structure, and mark each variable as confounder, mediator or collider.
2. Fit an ordinary regression ignoring the nesting. Record the standard error on your main coefficient.
3. Fit the mixed-effects model with the correct random structure. Record the standard error again.

4. Compare. The ratio is how much the naive model overstated your precision.
5. Now add a mediator to the model and watch the main coefficient move. Explain in one sentence why it moved.

Checkpoint — Test Yourself

1. Why does the meaning of β_1 depend on what else is in the model?
2. Your outcome is a defect count with variance four times its mean. Which model, and what goes wrong if you use Poisson?
3. When must you fit a random slope rather than only a random intercept?
4. Why is stepwise selection unacceptable for explanation but tolerable for prediction?

Chapter Summary

- A coefficient means change in the outcome *holding the other terms fixed*. Which covariates to include is a causal question: include confounders, exclude mediators and colliders.
- Choose the model from the outcome's shape. Software counts are overdispersed; use negative binomial, not Poisson.
- Nested observations require a mixed-effects model, and the random structure follows from the design, not from fit statistics.
- Explanation and prediction have different goals, criteria and variable selection. Decide which you are doing before fitting.
- Stepwise selection invalidates inference. Specify from causal structure, or regularise and do not interpret coefficients.
- Report the full specification, intervals in interpretable units, diagnostics, sample sizes at every level, missing data handling, and software versions.

Review Questions

1. Controlling for a mediator removes the effect you wanted to measure, yet “controlling for more variables” is widely treated as more rigorous. Explain the misconception and give a software engineering example where it would reverse a conclusion.
2. Shmueli argues explanation and prediction require different models. Computing research increasingly uses machine learning methods for inferential questions. What specifically goes wrong, and how would you detect it as a reviewer?
3. Complexity metrics correlate strongly with size. Design a study that could establish whether complexity predicts defects independently of size, and say what data it would require.
4. Mixed-effects models are standard in psychology and rare in software engineering, despite our data being at least as nested. Account for the gap, and say what would close it.

Structural Equation Modelling and Latent Constructs

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 4 — Measurement, Data and Analysis

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Explain** what a latent variable is and why measurement error matters for estimating relationships.
- **Separate** the measurement model from the structural model.
- **Choose** between covariance-based and partial least squares SEM on defensible grounds.
- **Assess** convergent and discriminant validity, including HTMT.
- **Detect** and address common method bias.
- **Report** an SEM study to the standard reviewers expect.

🔗 Before You Start

Chapter 17 for validity and reliability — SEM is measurement theory made operational. Chapter 13 if your data comes from a questionnaire, which it usually will. Chapter 20 for the modelling ideas SEM extends.

📖 Key Terms in This Chapter

latent variable • indicator • measurement model • structural model • reflective and formative indicators • confirmatory factor analysis • CB-SEM • PLS-SEM • composite reliability • average variance extracted • Fornell–Larcker • HTMT • common method bias • fit index

21.1 Why This Chapter Exists for an IT Course

A large share of MS IT theses study technology acceptance, user satisfaction, system quality or adoption intention. These are latent constructs measured by questionnaire, related to one another in a model — which is precisely what structural equation modelling is for. Done well it is powerful; done as it commonly is, it produces impressive diagrams supporting conclusions the data do not carry.

📖 Latent variable

A **latent variable** is a construct that cannot be observed directly and is inferred from multiple observed **indicators**.

“Perceived usefulness” is latent; the four questionnaire items measuring it are indicators. SEM estimates relationships *between the latent constructs*, explicitly modelling the fact that each indicator measures its construct imperfectly.

💡 What SEM buys over regression on scale means

Averaging four items and running a regression treats the average as if it were the construct, measured without error. It is not, and the consequence is systematic: measurement error *attenuates* estimated relationships, biasing them towards zero.

SEM models the error explicitly and estimates the relationship between the error-free constructs. For a scale with reliability 0.7, the attenuation is substantial — so this is not a refinement, it is a correction.

The cost is that SEM requires more data, more assumptions, and considerably more reporting. Do not use it because it looks sophisticated; use it because your constructs are latent and measured with error.

21.2 Two Models in One

Measurement model	How each latent construct relates to its indicators. Assessed <i>first</i> , and the whole analysis stops here if it fails
Structural model	How the latent constructs relate to each other. The hypotheses live here

⚠ Assess the measurement model first, and be willing to stop

If the indicators do not measure the constructs adequately, the structural paths between them are meaningless — you would be estimating relationships between things you have not measured.

The order is non-negotiable, and so is the willingness to act on it. A thesis that reports poor discriminant validity and then proceeds to interpret path coefficients anyway has told the examiner exactly what to ask about.

21.2.1 Reflective and formative indicators

	Reflective	Formative
Direction	Construct → indicators	Indicators → construct
Meaning	Indicators are effects of the construct; they should correlate highly and are interchangeable	Indicators are components that constitute the construct; they need not correlate
Example	Satisfaction: four items all reflecting how satisfied someone is	Socioeconomic status: income, education, occupation. Raising income raises status; it does not raise education
Assessment	Reliability, AVE, discriminant validity	Collinearity among indicators, and the significance of each indicator’s weight

Misspecifying the direction is a common and serious error: applying reflective criteria to a formative construct will make a perfectly good construct look invalid, and dropping “poorly loading” formative indicators removes part of the construct’s definition.

21.3 CB-SEM or PLS-SEM?

	CB-SEM (lavaan)	PLS-SEM (semnr)
Estimates	Parameters reproducing the observed covariance matrix	Composite scores maximising explained variance in the outcomes
Purpose	Theory <i>testing</i> and confirmation	Prediction and theory <i>development</i>

	CB-SEM	PLS-SEM
Sample size	Larger; a common rule of thumb is 10 observations per estimated parameter	Smaller samples tolerated — though the “10 times the largest number of paths” rule is widely criticised as far too permissive
Distribution	Assumes multivariate normality for maximum likelihood	Distribution-free; inference by bootstrapping
Global fit	Yes: χ^2 , CFI, TLI, RMSEA, SRMR	No true global fit test. SRMR is reported as an approximate check
Formative constructs	Awkward; requires additional identification constraints	Handles them naturally

! PLS-SEM is not a way to escape a small sample

It is chosen for that reason constantly, and the justification does not hold. PLS tolerates smaller samples than CB-SEM, but small-sample PLS estimates are still imprecise, and the bootstrap intervals will show it if you report them.

Choose PLS because your model contains formative constructs, because your aim is prediction, or because the theory is still being developed. Choose CB-SEM because you are testing an established theory and want a global fit test. Then say which reason applied, in one sentence, in the methods chapter. “PLS-SEM was used due to the small sample size” is the sentence reviewers have learned to distrust.

21.4 Assessing the Measurement Model

Criterion	Conventional threshold	What it checks
Indicator loading	≥ 0.70	Each indicator relates strongly to its construct
Composite reliability	0.70 to 0.95	Internal consistency. <i>Above</i> 0.95 suggests redundant items, not an excellent scale
Average variance extracted (AVE)	≥ 0.50	The construct explains more than half the variance in its indicators
Fornell–Larcker	$\sqrt{\text{AVE}}$ exceeds correlations with other constructs	Discriminant validity. Now known to have poor sensitivity
HTMT	< 0.85 , or < 0.90 for conceptually similar constructs	Discriminant validity, and substantially better at detecting its absence [24]

! Report HTMT, and understand what failing it means

Fornell–Larcker was the standard for two decades and has been shown to miss discriminant validity problems in exactly the situations where they matter. HTMT — the heterotrait-monotrait ratio of correlations — detects them reliably, and reviewers in information systems now expect it.

Failing HTMT means two of your constructs are not empirically distinguishable. Your respondents did not separate them, whatever the theory says. The honest responses are to merge the constructs and rewrite the model, or to redesign the items so they diverge — not to report the failure and interpret the path between them anyway.

21.5 Common Method Bias

When one respondent supplies both the predictors and the outcome, in one questionnaire, at one sitting, some of the observed correlation may be produced by the method rather than by the constructs [47]. Since this describes nearly every acceptance study ever run, it deserves attention rather than a formulaic paragraph.

Procedural remedies (better)	Separate the measurement of predictor and outcome in time or source; guarantee anonymity; counterbalance item order; avoid ambiguous items; use different response formats for different constructs
Statistical detection (weaker)	Harman's single-factor test — widely used, widely criticised as insensitive; a marker variable approach; a common latent factor in the model

⚠ Harman's single-factor test proves almost nothing

It is reported in a large fraction of IS theses, usually as a single sentence concluding that common method bias is not a concern. The test has very low power to detect bias, so passing it is weak evidence, and the sentence is performing reassurance rather than analysis.

Design the bias out procedurally — temporal separation between predictor and outcome measurement is the single most effective step — and then report what you did in the design, not only what you tested afterwards. That is a stronger answer, and it is available to you only before data collection.

Listing 21.1: A PLS-SEM assessment in the order that matters: measurement model first, structural model only if it passes.

```
library(semnir)

measurements <- constructs(
  reflective("PU", multi_items("PU", 1:4)), # perceived usefulness
  reflective("PEOU", multi_items("PEOU", 1:4)), # perceived ease of use
  reflective("INT", multi_items("INT", 1:3)) # behavioural intention
)

structure <- relationships(
  paths(from = c("PU", "PEOU"), to = "INT"),
  paths(from = "PEOU", to = "PU")
)

m <- estimate_pls(data = survey, measurements, structure)
s <- summary(m)

# 1. Measurement model. Stop here if it fails.
s$loadings; s$reliability # loadings, rhoC, AVE
s$validity$htmt # discriminant validity -- the one to report

# 2. Structural model, with bootstrap intervals rather than p-values alone.
b <- bootstrap_model(m, nboot = 5000)
summary(b)$bootstrapped_paths
s$paths # R-squared per endogenous construct
```

21.6 Reporting

Why SEM, and which variant	The substantive reason, not the sample size
Every item	In an appendix, with its source or its adaptation stated (Chapter 13)
Measurement model	Loadings, composite reliability, AVE, HTMT matrix — as tables, in full
Any items dropped	Which, and why. Dropping items to reach a threshold is a data-driven decision and must be disclosed
Structural model	Path coefficients with bootstrap confidence intervals, R^2 , and effect sizes
Fit (CB-SEM)	χ^2 with df, CFI, TLI, RMSEA, SRMR — all of them, not the flattering subset
Common method bias	What was done procedurally, then what was tested
Data and code	Deposited (Chapter 25)

! Four things examiners ask about SEM theses

- **“Why these paths?”** Every path must come from theory stated in advance. A model with paths between everything is not a hypothesis.
- **“What did you drop?”** Items removed to improve AVE, and modification indices followed to improve fit, are data-driven decisions. Fit obtained that way is not evidence for the theory.
- **“Is this causal?”** Cross-sectional SEM estimates associations. Arrows in the diagram are the theory’s claims, not the design’s warrant (Chapter 11).
- **“Would a different model fit as well?”** Usually yes. Equivalent models with reversed paths often fit identically, and acknowledging this is far better than being shown it.

🔗 Four Traditions — latent variable modelling Across Ways of Knowing

📊 Empirical

Its natural extension: measurement error is modelled rather than ignored, and theory is tested against a covariance structure.

🔧 Design Science

Used to evaluate an artefact’s perceived qualities and their relation to intention to use — formative naturalistic evaluation, and weak evidence of actual use.

💬 Qualitative

Would object that constructs like “perceived usefulness” are imposed on respondents rather than derived from them, and that an exploratory sequential design (Chapter 15) should establish the constructs first.

√* Formal / Analytical

Identification — whether parameters can be uniquely determined — is a formal question with a formal answer, and an unidentified model is not a statistical problem but an algebraic one.

🔧 Try It Yourself: Assess a measurement model

Using an open dataset, or the pilot data from your own instrument (Chapter 13):

1. Specify three constructs with at least three reflective indicators each, and the paths between them, from theory.
2. Estimate the measurement model. Report loadings, composite reliability and AVE.
3. Compute both Fornell–Larcker and HTMT. Do they agree? If not, which do you trust and why?
4. Only if the measurement model passes, estimate the structural model with 5,000 bootstrap resamples and report intervals.
5. Write the paragraph you would put in the thesis about common method bias — describing design steps first.

🔍 Checkpoint — Test Yourself

1. Why does measurement error bias regression coefficients towards zero, and how does SEM address it?
2. Give a formative construct from IT research and explain why reflective assessment criteria would misjudge it.
3. Composite reliability of 0.97 is reported as excellent. What is the concern?
4. Two constructs fail HTMT. What are your two legitimate options?

📋 Chapter Summary

- Latent constructs measured with error attenuate relationships; SEM corrects for this by modelling the error.
- Measurement model first, structural model only if it passes.
- Reflective and formative indicators require different assessment; misspecifying the direction invalidates the assessment.
- Choose CB-SEM or PLS-SEM for a substantive reason. Small sample size is not one.
- Report HTMT, not only Fornell–Larcker. Failing it means merging constructs or redesigning items.
- Design common method bias out procedurally; Harman’s single-factor test is weak evidence.
- Disclose every dropped item and every modification index followed. Fit obtained by data-driven revision is not evidence for the theory.

🔧 Review Questions

1. Technology acceptance models are among the most replicated in IS and among the most criticised as theoretically thin. Assess that criticism, and say what a genuinely novel contribution using UTAUT2 would have to add.
2. Equivalent models can fit identically while asserting opposite causal directions. What does this imply about the evidential value of a well-fitting cross-sectional SEM, and what design would break the tie?
3. PLS-SEM’s sample size rules of thumb have been shown to be far too permissive. Why do they persist, and what should replace them?
4. **PhD Regs 2024 §14.2(v)** requires that “quantitative research proposals must include a valid statistical design for data analysis”. What would a valid design section for an SEM study contain that most contain now?

Evaluating Machine Learning Research

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 4 — Measurement, Data and Analysis

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Audit** a pipeline for the eight forms of data leakage.
- **Choose** a validation scheme that matches how the model would be deployed.
- **Select** metrics appropriate to the base rate and the costs of error.
- **Compare** classifiers across datasets without running twenty t-tests.
- **Design** an ablation that shows which component earns the gain.
- **Document** a model and a dataset to a recognised standard.

🔗 Before You Start

Chapter 19 for multiple comparisons and intervals; Chapter 17 for construct validity, which applies to labels as much as to questionnaires; Chapter 16 for measurement under non-determinism.

📖 Key Terms in This Chapter

data leakage • train-validation-test split • temporal split • group split • baseline • class imbalance • precision and recall • ROC-AUC and PR-AUC • calibration • decision threshold • ablation • Friedman test • test set reuse • model card • datasheet

22.1 The Claim a Model Makes

A machine learning paper claims that a model will perform at some level on data it has not seen, drawn from some population, under some deployment conditions. Everything in this chapter is about whether the evaluation actually supports that claim — and the dominant finding of the last few years is that very often it does not.

📖 Anchor Reading: Leakage across an entire literature

Sayash Kapoor and Arvind Narayanan, *Leakage and the reproducibility crisis in machine-learning-based science*, Patterns 4(9), 2023 [32].  OA

Why it matters. They surveyed 17 fields using ML for scientific claims, found leakage affecting 294 papers across them, and showed that correcting it frequently eliminated the reported advantage of machine learning over much simpler methods. This is not a warning about a rare bug; it is evidence of a systematic failure mode.

What to extract. Their taxonomy of eight leakage types, and their model info sheet. Fill the sheet in for your own work — it takes an hour and is the cheapest insurance available to a computational thesis.

22.2 Eight Ways Information Leaks

Data leakage

Leakage occurs when information that would not be available at prediction time is used during training or model selection. The result is performance that is real on your test set and unattainable in deployment.

Type	What happens	Fix
No test set at all	Performance reported on training data	Hold out a test set and touch it once
Preprocessing on all data	Scaling, imputation or encoding fitted before splitting	Fit every transform inside the training fold. Use a pipeline object so this cannot be forgotten
Feature selection on all data	Features chosen using the full dataset's labels	Select inside the cross-validation loop. This one routinely produces dramatic apparent performance from pure noise
Duplicates	Near-identical records in train and test	Deduplicate before splitting, including near-duplicates
Temporal leakage	Training on data from after the test period	Split by time, always, whenever the deployment is a forecast
Group leakage	The same patient, developer or repository in both splits	Split by group, not by row
Sampling bias in the test set	The test set is not drawn from the deployment population	Construct it to match deployment, and say how
Tuning on the test set	Hyperparameters or thresholds chosen by test performance	Three-way split, or nested cross-validation

Worked Example: A leakage audit that changes the conclusion

The study. Predicting defect-prone files from code metrics across 30 repositories. Reported: 0.91 AUC with a gradient boosting model, “substantially outperforming logistic regression”.

Audit finding 1 — group leakage. Rows were split randomly. Files from the same repository appear in both training and test. Repositories differ enormously in defect base rate, so the model can identify the repository from its metric profile and predict that repository's base rate. It is not predicting defects; it is memorising repositories.

Fix: split by repository. Train on 24, test on 6, and rotate.

Audit finding 2 — temporal leakage. Files were labelled defective if a defect was ever found in them, including after the metrics snapshot. The model uses present metrics to predict labels partly determined by the future, which is unavailable at deployment.

Fix: fix a snapshot date; label from defects found in a defined window after it.

Audit finding 3 — preprocessing leakage. Features were standardised across the full dataset before splitting, so test-set statistics informed the training scaling. Small effect here, but free to fix.

Audit finding 4 — no trivial baseline. With a 12% defect rate, “always predict not defective” achieves 88% accuracy. The paper reports AUC, which is better, but never compares against predicting from file size alone — which in defect prediction is a notoriously strong baseline.

After correction. Grouped, temporal split, pipeline-encapsulated preprocessing, and a size-only baseline included. Illustrative outcome: AUC falls to about 0.68, the size-only baseline reaches 0.65, and logistic regression on the same features reaches 0.67. The honest conclusion is that gradient boosting offers little over a simple model, and that most of the signal is size.

That is a publishable finding — it is a negative result about a widely assumed advantage — and it is a far better thesis than the original.

22.3 Validation Must Match Deployment

Deployment	Correct scheme	What random k -fold gets wrong
Predict for new instances from the same population	Random k -fold	Nothing. This is the case it is designed for
Predict the future	Time-series split; train on past, test on future, rolling forward	Trains on the future, which inflates results and cannot be reproduced in deployment
Predict for a new group — new project, hospital, user	Grouped k -fold, or leave-one-group-out	Lets the model memorise group identity
Predict under distribution shift	Test explicitly on a shifted set	Reports performance for a world that will not occur

⚠ The question that settles the scheme

Ask: *at prediction time in the real system, what would the model know?*

Then construct the validation so it knows exactly that and no more. Every leakage type above is a case of the model knowing something the deployed system would not. This single question catches most of them without needing to memorise the taxonomy.

22.4 Metrics, and the Base Rate

Metric	Answers	Fails when
Accuracy	What fraction is correct?	Classes are imbalanced. At a 1% base rate, 99% accuracy is achieved by predicting nothing
Precision	Of those flagged, how many are real?	Used alone — it says nothing about what was missed
Recall	Of the real ones, how many were found?	Used alone — flagging everything gives recall 1.0
F_1	Harmonic mean of the two	Precision and recall have different costs, which they usually do. Weight them by cost instead
ROC-AUC	Ranking quality across thresholds	Heavy imbalance — ROC looks optimistic because true negatives are plentiful [52]
PR-AUC	Ranking quality, precision against recall	Preferred under imbalance. Note the baseline is the positive rate, not 0.5
Calibration	Are the probabilities meaningful?	Ignored almost always, and essential if a probability feeds a decision

💡 The threshold is a decision, not a default

Classifiers output scores; 0.5 is a convention with no claim to optimality. The right threshold follows from the costs of the two errors.

In defect prediction, a false negative means a defect ships and a false positive means a developer spends twenty minutes checking clean code. Those are not equal, and the ratio between them determines the threshold. State the cost ratio you assume, choose the threshold from it on *validation* data, and report performance at that threshold on test data.

A paper reporting only threshold-free metrics has evaded the question its users must answer.

22.5 Baselines and Ablations

Trivial baseline	Majority class; random; predict the mean. Establishes the floor, and occasionally beats the proposed method
Simple baseline	Logistic regression, a decision tree, or one strong feature. In software engineering, file size is a famously strong baseline for defect prediction
Current best	The actual state of the art, tuned with the same budget you gave your own method. Tuning yours for a week and the baseline for an hour is not a comparison
Ablation	Your method with one component removed at a time. This is what identifies which part earns the gain

❗ The unablated improvement

A paper proposes a method combining a new architecture, a new loss, a new augmentation scheme and a longer training schedule, and reports a 2-point gain. Without ablations, no reader — including the authors — knows which of the four produced it, or whether it was the longer schedule alone.

This matters beyond credit assignment: an unablated improvement cannot be transferred, because nobody knows what to transfer. Run the ablations, and report the ones where removing a component changed nothing. That is information, and omitting it is what makes a paper unreproducible in the useful sense.

22.6 Comparing Across Datasets

Comparing k classifiers over n datasets with pairwise t-tests is wrong on two counts: the multiplicity is uncorrected (Chapter 19), and the normality assumption is untenable for performance scores across heterogeneous datasets.

💡 The standard procedure

- Demšar’s recommended approach [11] is now conventional and is two steps:
1. **Friedman test** on the ranks of the classifiers across datasets. Non-parametric, and it asks the right global question: are these classifiers performing equivalently overall?
 2. If significant, a **post-hoc test** — Nemenyi for all-pairs comparison, or Bonferroni–Dunn when comparing everything against one control method, which is usually what you actually want.

Report the result as a **critical difference diagram**: classifiers on a rank axis, with bars joining those not significantly different. It communicates the whole comparison in one figure and makes over-claiming difficult.

Listing 22.1: Comparing four classifiers across twelve datasets, the way it should be done.

```
# rows = datasets, columns = classifiers, cells = performance
friedman.test(as.matrix(results))

# If significant, post-hoc pairwise comparison on ranks
library(PMCMRplus)
frdAllPairsNemenyiTest(as.matrix(results))

# Report mean ranks alongside, not raw means: ranks are what was tested
colMeans(t(apply(-results, 1, rank)))
```

22.7 Test Set Reuse

Every time you look at test performance and change something, you leak a little information. Over a project — or over a community’s decade of using one benchmark — this accumulates into overfitting the test set, and reported gains stop transferring.

Three-way split	Train, validation, test. All decisions on validation; test touched once, at the end
Nested cross-validation	When data is scarce: an inner loop for tuning, an outer loop for estimating performance
Count the looks	Record how many times you evaluated on test. If it is more than once, say so — the number is meaningful
A fresh test set	The only real remedy for a community-level problem. Where a new one exists, report on it

22.8 Documenting Models and Data

Datasheet [20]	For a dataset: motivation, composition, collection process, preprocessing, uses, distribution, maintenance. Crucially, its known biases and the uses for which it is <i>not</i> appropriate
Model card [41]	For a model: intended use, out-of-scope use, training data, evaluation data, metrics <i>disaggregated by relevant subgroup</i> , ethical considerations, caveats

Disaggregated evaluation is the part most often skipped and most often consequential: an aggregate accuracy of 0.90 can conceal 0.95 on one subgroup and 0.60 on another, and only reporting by subgroup reveals it.

🔗 Four Traditions — evaluating a model Across Ways of Knowing

 **Empirical**
A model evaluation is an experiment. It needs repetitions across seeds, intervals rather than point estimates, a baseline, and correction for the comparisons made.

 **Design Science**
The model is the artefact. Held-out performance is artificial summative evaluation; whether anyone benefits from deploying it is a separate and usually unasked question (Chapter 9).

 Qualitative

Asks what the labels mean and who produced them. A defect label is a human judgement, and label noise is a measurement validity problem (Chapter 17), not merely a data quality nuisance.

 Formal / Analytical

Provides generalisation bounds and guarantees under stated assumptions — typically far looser than observed performance, but true for reasons that do not depend on a test set.

 Try It Yourself: Audit a pipeline

Take any ML pipeline — yours, a published replication package, or a public notebook.

1. Work the eight leakage types. For each, state where in the code it would occur and whether it does.
2. Ask the deployment question and check the validation scheme matches.
3. Add a trivial baseline and one strong simple baseline. Record the gap.
4. Re-run with the leakage fixed. Record how much performance falls.
5. Write the model card, including disaggregated metrics for at least one meaningful subgroup.

Step 4 is the one to report in the seminar. A drop is not a failure; it is the study working.

 Checkpoint — Test Yourself

1. Why does fitting a scaler before splitting constitute leakage, and how large is the problem in practice?
2. You are predicting whether a developer will abandon a project. What split do you use, and why is random k -fold wrong?
3. At a 2% positive rate, why is ROC-AUC misleading, and what should you report?
4. Why are pairwise t-tests wrong for comparing five classifiers across fifteen datasets, and what replaces them?

 Chapter Summary

- Leakage is systematic, not rare, and correcting it frequently eliminates reported advantages of complex methods over simple ones.
- Ask what the deployed system would know at prediction time, and build validation to match. That question catches most leakage.
- Accuracy fails under imbalance; ROC-AUC flatters it; PR-AUC and calibration are usually what you want. The threshold follows from costs.
- Include a trivial baseline, a strong simple baseline, and a fairly tuned state of the art. Ablate, and report the components that changed nothing.
- Compare across datasets with Friedman plus a post-hoc test, reported as a critical difference diagram.
- Touch the test set once, and count the times you did not.
- Publish a datasheet and a model card, with metrics disaggregated by subgroup.

 Review Questions

1. Kapoor and Narayanan found leakage across 17 fields. What features of how ML is taught and published produce this, and which would be easiest to change?

2. Benchmark test sets are reused by whole communities for years. Propose a workable governance mechanism for a benchmark in your area, and identify who bears its cost.
3. Defect labels come from issue trackers and are noisy and biased (Chapter 11). What does label noise do to a reported AUC, and how would you estimate its magnitude?
4. **PhD Regs 2024 §14.3(B)** asks whether results are replicable. For a deep learning dissertation, enumerate exactly what would have to be deposited for a third party to replicate the reported numbers, and estimate what fraction of published work meets that bar.

Qualitative Data Analysis in Practice

Learning Outcomes

By the end of this chapter you will be able to:

- **Organise** a qualitative analysis so it can be audited.
- **Use** open-source coding tools without letting the tool shape the analysis.
- **Establish** trustworthiness using criteria appropriate to qualitative work.
- **Map** your practices onto the criteria your dissertation is formally assessed against.
- **Write** up qualitative findings so that evidence supports interpretation visibly.

Before You Start

Chapter 14 for coding, saturation and the reliability debate. This chapter is the practical and evaluative half: tools, audit trail, and the standard your work will be judged by.

Key Terms in This Chapter

audit trail • trustworthiness • credibility • transferability • dependability • confirmability • thick description • negative case analysis • peer debriefing • member checking • reflexivity • CAQDAS

23.1 Rigour Without Replication

Quantitative rigour rests on replication and quantified uncertainty. Neither is available here: another researcher analysing your transcripts would not produce identical themes, and under an interpretivist position they are not expected to.

That does not leave qualitative work without standards. It relocates them. Rigour becomes **transparency**: a reader must be able to see how you got from data to claim, and to judge whether the route was reasonable.

Criterion	Quantitative analogue	Established by
Credibility	Internal validity	Prolonged engagement, triangulation, member checking, negative case analysis, peer debriefing
Transferability	External validity	Thick description of context, so a reader can judge applicability to theirs. <i>You</i> do not claim transfer; you enable the reader to assess it
Dependability	Reliability	Audit trail: decisions, dated, with reasons, so the process can be followed
Confirmability	Objectivity	Findings traceable to data rather than to the researcher's preferences; reflexivity about position

💡 These map onto the regulations almost line by line

PhD Regs 2024 §14.3(A) is, in substance, a trustworthiness checklist, and reading it that way tells you what your examiner is looking for:

The clause asks	Which is
“Does the research illuminate the subjective meaning, actions and contexts of those being researched?”	Credibility
“Is there evidence of adaption and responsiveness of the research design to the circumstances of real-life social settings?”	Dependability, and the theoretical sampling of Chapter 14
“Is the description detailed enough to allow the researcher or reader to interpret the meaning and context?”	Transferability, via thick description
“How are the different sources of knowledge about the same issue compared and contrasted?”	Triangulation
“How does the research move from a description of the data through quotation or examples to an analysis and interpretation of the meaning and significance of it?”	Confirmability, and the analytic ladder below

23.2 The Audit Trail

Raw data	Recordings, transcripts, documents, field notes — preserved unaltered and separately from analysis
Data reduction	Coded transcripts, each code applied and traceable
Data reconstruction	Category structures, theme definitions, and the versions they went through
Process notes	Dated memos: decisions, reasons, changes of mind
Materials	Interview guide, consent form, recruitment text, protocol revisions
Reflexivity notes	Your position, prior beliefs, and where you noticed them influencing interpretation

⚠️ Version control your qualitative analysis

Keep the trail in a Git repository, exactly as you would code. Commit after each coding session with a message saying what changed and why.

This gives you a dated, immutable record of when every code was created and every theme revised — which is precisely what dependability requires, and precisely what a supervisor or examiner needs to see. Keep the repository private and be careful about what goes into it: raw transcripts are personal data (Chapter 27), so commit de-identified transcripts and hold the key separately.

23.3 Tools

Tool	Licence	Notes
Taguette	Open source	Highlight-and-tag, browser-based, exports to CSV and HTML. Sufficient for most thesis-scale work, and free
QualCoder	Open source	Desktop; handles text, image, audio and video; supports coder comparison and simple matrices
NVivo, ATLAS.ti, MAXQDA	Proprietary	Powerful, expensive. Check what your institution licenses before paying
Spreadsheet	—	Entirely adequate for one or two transcripts. Do not let tool selection delay analysis

! The tool is not the method

Software stores codes, retrieves segments and counts occurrences. It does not interpret, and three specific distortions follow from forgetting that:

- **Code proliferation.** Creating a code is one click, so analyses end up with 300 codes and no structure. Codes should be consolidated deliberately, and the consolidation is analysis.
- **Counting as evidence.** “This code appeared 47 times” invites a quantitative reading of purposively sampled data. Frequency describes your corpus; it estimates nothing (Chapter 15).
- **Fragmentation.** Retrieving all segments for a code strips them from the narrative they belonged to. Re-read whole transcripts periodically, or you will build themes out of decontextualised fragments.

23.4 From Code to Claim

The move that §14.3(A) asks about — “from a description of the data through quotation to an analysis and interpretation of the meaning and significance” — is the hardest thing in qualitative writing. It has four rungs, and weak theses stop at the second.

1	Data	“I just approve it if the tests pass, honestly.”
2	Code	<i>Deferring to automated checks</i>
3	Theme	<i>Automation as licence to disengage</i> — reviewers treat passing CI as sufficient evidence of correctness, substituting the tool’s judgement for their own
4	Interpretation	Where automated checks are visible in the review interface, they reframe review from evaluation to confirmation. This suggests the placement of automated signals is a design decision with consequences for review depth — testable, and actionable

💡 Using quotations as evidence rather than decoration

- **Quote to support a claim you have just made**, not to make it. The analytic sentence comes first; the quotation demonstrates it.
- **Attribute** — P7, senior developer, Firm B — so a reader can see whether one participant is carrying the whole theme.
- **Report distribution honestly.** “Three of fourteen participants” where that is the case. A theme resting on one vivid quotation should say so.

- **Include disconfirming quotations.** A theme presented with only supporting evidence looks selected, because it may have been. Showing where it did not hold, and saying why, is what negative case analysis produces.

23.5 Techniques That Strengthen Credibility

Negative case analysis	Actively hunt for data contradicting your theme. Either the theme is revised, or its scope condition is stated. This is falsification (Chapter 2) in qualitative form, and it is the most persuasive single practice available
Member checking	Return findings to participants. Their disagreement is data — but it does not automatically override your analysis, since participants may not share your analytic frame. Report what they said and what you did about it
Peer debriefing	A colleague outside the study challenges your interpretations. Record the challenges and your responses in the audit trail
Prolonged engagement	Enough time in the setting to distinguish the typical from the incidental
Reflexivity	State your position: prior experience, relationship to participants, expectations. Not confession — a description of the instrument, which in this method is you

📍 Four Traditions — qualitative analysis Across Ways of Knowing

-  **Empirical**
Sees it as generating hypotheses and defining constructs for later measurement. A partial reading, but a real contribution.
-  **Design Science**
Uses it to characterise the problem before designing and to understand appropriation afterwards — especially uses the designers did not anticipate.
-  **Qualitative**
Its home. Trustworthiness replaces validity and reliability, and transparency replaces replication.
-  **Formal / Analytical**
Sceptical by disposition, but qualitative studies of how practitioners reason about formal notation have explained non-adoption better than any formal argument has.

🔧 Try It Yourself: Build an auditable analysis

- Using the transcript you coded in Chapter 14:
1. Set up a Git repository with the six audit-trail components as directories. De-identify before committing anything.
 2. Import into Taguette or QualCoder and apply your codebook.
 3. Consolidate to at most 25 codes. Write one memo explaining each merge.
 4. Build one theme up all four rungs of the ladder, ending with an interpretation that predicts something.
 5. Conduct negative case analysis on that theme: find contradicting data, or state the scope condition it required.

6. Write the trustworthiness paragraph, mapping each practice onto the matching clause of §14.3(A).

🔍 Checkpoint — Test Yourself

1. Why does qualitative rigour rest on transparency rather than replication?
2. “This code appeared 47 times” — what is wrong with offering this as evidence, and what would be legitimate?
3. Give an example from your area of moving from theme to interpretation, where the interpretation predicts something.
4. A participant disagrees with your central finding during member checking. Give a case where you should revise and one where you should not.

📋 Chapter Summary

- Rigour here is transparency: credibility, transferability, dependability, confirmability — which §14.3(A) asks for almost clause by clause.
- Keep an audit trail in version control, de-identified, committed after each session.
- Open-source tools are sufficient. The tool stores and retrieves; it does not interpret, and it encourages proliferation, counting and fragmentation.
- Climb all four rungs: data, code, theme, interpretation. Stopping at themes is the commonest weakness.
- Quote to support a claim, attribute, report distribution honestly, and include disconfirming evidence.
- Negative case analysis is the single most persuasive practice available.

🔧 Review Questions

1. Trustworthiness criteria were proposed as parallels to validity and reliability. Does the parallel help qualitative research gain acceptance, or does it concede the quantitative framing? Argue a position.
2. Version-controlled audit trails would make qualitative analysis far more transparent, yet almost nobody does it. Identify the practical and epistemic obstacles, and say which is the real one.
3. Member checking assumes participants are well placed to validate analysis. When is that assumption false, and what should replace it there?
4. Large language models are now used to code qualitative data at scale. Given **PhD Regs 2024 §32(IV)**, which forbids AI from analysing and interpreting data, what — precisely — could be automated without breaching it, and would the result still be qualitative analysis? (Chapter 29 continues this.)

Threats to Validity and How to Report Them

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **Classify** a threat into one of the four validity types.
- **Identify** the threats characteristic of each research design.
- **Distinguish** a threat you mitigated, one you measured, and one you merely acknowledged.
- **Write** a threats section that increases rather than decreases a reader’s confidence.
- **Recognise** the trade-off between validity types, and defend your position on it.

🔗 Before You Start

Everything in Parts III and IV. This chapter is where the designs and the analysis are audited, and it is the chapter to reread before writing up.

🔑 Key Terms in This Chapter

internal validity • external validity • construct validity • conclusion validity • confounding • selection • maturation • instrumentation • mono-operation bias • hypothesis guessing • ecological validity • mitigation

24.1 Four Questions About One Study

Cook and Campbell’s four-way taxonomy [9] remains the standard, and each type asks a different question about the chain from your treatment to your conclusion.

Type	Asks	If it fails
Conclusion validity	Is there a relationship in the data at all?	You claim an effect that is noise, or miss one that is there
Internal validity	Is the relationship causal?	Something other than your treatment produced the effect
Construct validity	Do the operationalisations capture the intended constructs?	You measured or manipulated something other than what you claim
External validity	Does it hold beyond this setting?	The result is true here and nowhere anyone cares about

💡 Read them in order, because they depend on each other

The order is not arbitrary. If conclusion validity fails there is no relationship to be causal about. If internal validity fails, generalising a non-causal relationship is pointless. If construct validity fails, you do not know what you generalised.

A threats section that opens with external validity — “our sample was only from one company” — and never addresses whether the effect exists or is causal has skipped the two questions a reviewer asks first.

24.2 The Threats, by Type

Threat	What it is	Typical mitigation
Conclusion validity		
Low power	Too small to detect the effect of interest	A-priori power analysis (Chapter 18); within-subject design
Violated assumptions	The test's assumptions do not hold	Check residuals; use robust or rank-based methods (Chapter 19)
Fishing	Many comparisons, one reported	Pre-specify; correct; label exploratory analyses as such
Unreliable measures	Noise attenuates the relationship	Establish reliability; use multi-item scales (Chapter 17)
Random irrelevancies	Uncontrolled setting variation adds noise	Standardise conditions; block on known sources
Internal validity		
Selection	Groups differed before treatment	Randomise; otherwise an identification strategy (Chapter 11)
History	Something else happened during the study	Control group experiencing the same period
Maturation	Subjects changed anyway — learning, fatigue	Control group; counterbalance order
Instrumentation	The measuring process changed mid-study	Fix the instrument; if human coders, check drift over time
Mortality	Drop-out differs between conditions	Report attrition per condition; intention-to-treat analysis
Diffusion	The control group learns the treatment	Separate groups; ask afterwards whether it happened
Construct validity		
Inadequate definition	The construct was never pinned down	Define before operationalising (Chapter 3)
Mono-operation bias	One task, one system, one instantiation of the treatment	Multiple tasks or systems; report per-object results
Mono-method bias	One kind of measure for everything	Triangulate methods (Chapter 15)
Hypothesis guessing	Subjects infer the expected result and comply	Blind subjects to condition and to hypothesis
Experimenter expectancy	The researcher's expectation influences conduct or coding	Blind coding; independent second coder
Evaluation apprehension	Subjects perform because they are watched	Anonymity; assurance that individuals are not assessed
External validity		
Population	Subjects unlike the target population	Characterise subjects fully; state the target; argue the specific task (Chapter 10)
Setting	Laboratory conditions unlike practice	Realistic tasks and time pressure; a field study alongside
Time	The result is period-specific	State the technological context; expect decay
Object	The systems studied are unrepresentative	Select objects on stated criteria, and report which

24.3 The Trade-off Is Real and Must Be Owned

Increasing one kind of validity often decreases another, and no design escapes this.

Move	Gains	Costs
Tighter laboratory control	Internal, conclusion	External — setting realism
Field study in one company	External — setting	Internal — no randomisation
More measures per construct	Construct	Conclusion — more comparisons; and participant fatigue
Larger, more diverse sample	External — population; conclusion — power	Internal — more heterogeneity to control; and cost

💡 State your position deliberately

There is no design that maximises all four, so a competent study chooses. What distinguishes a good threats section is that the choice is stated as a choice.

“We prioritised internal validity: a controlled experiment with randomised assignment on a standardised task. This limits external validity, since the task is smaller than industrial code and the subjects are graduate students. We regard this as the right trade at this stage, because no prior work has established that the effect exists at all; establishing it under controlled conditions is a precondition for studying it in the field, which we propose as the natural follow-up.”

That paragraph tells an examiner you understood the trade-off rather than fell into it.

24.4 Writing the Section

⚠️ The three grades of threat, and why the distinction matters

Not all threats are handled equally, and conflating them is the fastest way to lose a reader’s trust.

1. **Mitigated.** You designed it out. “Assignment was randomised, removing selection.” Strongest.
2. **Measured.** You could not remove it but you quantified it. “Attrition was 8% and 9% across conditions, and drop-outs did not differ on baseline experience.” Nearly as strong, and often more honest.
3. **Acknowledged.** You can do neither. “Subjects were graduate students; professionals may show a smaller effect, since the checklist likely substitutes for experience they already have.” Acceptable *when it states a direction*.

A section made entirely of grade 3 reads as a list of excuses. A section that shows several grade 1s and 2s, and reserves grade 3 for genuine limits, reads as competence — and it is the difference **PhD Regs 2024 §14.2(iv)** is reaching for when it asks that shortcomings “have been addressed”.

Do	Do not
Order by validity type, conclusion first	Open with “our sample was small” and stop
Give each threat a direction	“This may have affected the results” — which way, and by how much?
Name the most serious threat explicitly and treat it first	Bury it in the middle, where a reviewer will find it and note that you did not
Distinguish mitigated, measured and acknowledged	Present everything as an equal limitation
Say what would resolve it	Leave the reader to design the follow-up study

 Worked Example: Two versions of the same paragraph

Weak. “There are several threats to validity in this study. The sample size was small, which may have affected the results. The participants were students, which may limit generalisability. The study was conducted in one university, which is another limitation. Future work should address these issues.”

Four sentences, no directions, no mitigations, no follow-up. It reads as though the section were required rather than meant, and it invites every one of the questions it avoids.

Strong. “*Conclusion validity.* With 34 participants in a within-subjects design the study had 0.81 power to detect $d = 0.35$, our pre-registered smallest effect of interest; smaller effects would likely have been missed. Residuals were right-skewed, so we report Cliff’s δ with bootstrap intervals alongside the parametric test; both agree.

Internal validity. Treatment order was counterbalanced by Latin square, removing order and material effects. Two participants withdrew, one from each condition, so attrition is unlikely to be differential.

Construct validity. Defects found is a partial measure of review quality: it ignores false positives raised, which we did not record. If the checklist increases false positives, our measure overstates its benefit. This is the study’s most serious limitation and we recommend recording both in replications.

External validity. Participants were MS students with a median of 14 months of industrial experience — closer to junior professionals than to senior reviewers. We expect the checklist effect to be *smaller* among senior reviewers, who plausibly already apply an internalised equivalent, so the reported effect should be read as an upper bound for experienced populations. The code artefacts were 200–400 line modules; behaviour on large legacy systems is untested.”

Same study. The second is believable, and it names its own worst problem before a reviewer can.

 Naming your own worst threat is a strength

Scholars fear that identifying a serious limitation invites rejection. The opposite is closer to true. A reviewer will find it regardless; the only question is whether you found it first.

An author who names it, sizes it, and says what would resolve it demonstrates command of their own work. An author who omits it is either unaware — which is worse — or hoping

the reviewer will be. **PhD Regs 2024 §14.3(D)** puts this plainly: “the assessment of the results performed by the author must not be superficial and lacking substance”.

24.5 Threats by Design

Design	Strongest	Watch most carefully
Controlled experiment	Internal	External — setting and population; mono-operation bias
Quasi-experiment	External — realistic setting	Internal — the identifying assumption (Chapter 11)
Repository mining	Conclusion — large samples	Construct — what the trace actually measures; external — open source only
Survey	External — if sampling permits	Construct — self-report; internal — no causal warrant
Case study	Construct — rich, contextual	External — analytic not statistical (Chapter 12)
Design science	Relevance	Conclusion — demonstration mistaken for evaluation (Chapter 9)
Benchmarking	Conclusion — if repeated properly	External — workload representativeness (Chapter 16)
ML evaluation	Conclusion — large test sets	Internal — leakage; construct — label validity (Chapter 22)

📋 Template: C11 — Threats to validity table

Fill one row per threat. Anything you cannot grade is a threat you have not thought about yet.

Type	Threat	Grade	Direction and response
Conclusion	_____	_____	_____
Internal	_____	_____	_____
Construct	_____	_____	_____
External	_____	_____	_____

Grade: M = mitigated by design, Q = quantified, A = acknowledged.
Direction: which way would this bias the result, and by roughly how much?
Most serious threat overall: _____

🔄 Four Traditions — threats to validity Across Ways of Knowing

 **Empirical**
The four-way taxonomy, reported by type, with each threat graded and given a direction.

 **Design Science**
Reframed as evaluation adequacy: does the evaluation support the claimed contribution, and what would falsify the design knowledge claim?

 Qualitative

Replaced by trustworthiness — credibility, transferability, dependability, confirmability (Chapter 23). Do not force the four-way taxonomy onto qualitative work.

 Formal / Analytical

Validity is soundness of proof plus realism of assumptions. The second is where formal results usually fail to transfer, and it deserves a paragraph.

 Try It Yourself: Grade your own threats

For your own study design:

1. Complete template C11 with at least two threats per validity type.
2. Grade each M, Q or A.
3. For every A, write the direction of bias in one sentence. If you cannot state a direction, you do not yet understand the threat.
4. Identify the single most serious threat and write the paragraph that treats it first.
5. Count your Ms. If there are none, your design has not been engineered — go back to Chapter 10 or Chapter 11 and add a control.

 Checkpoint — Test Yourself

1. Why should conclusion validity be addressed before external validity?
2. Distinguish mitigated, measured and acknowledged threats with an example of each from your own work.
3. Why is “this may have affected the results” worthless, and what replaces it?
4. Give a design change that would improve external validity while damaging internal validity, and say when that trade is right.

 Chapter Summary

- Four validity types, addressed in order: conclusion, internal, construct, external. They depend on each other.
- Each design has characteristic threats; know your design's and treat them first.
- Validity types trade against each other. Choose deliberately and say that you chose.
- Grade every threat as mitigated, measured or acknowledged. A section that is all acknowledgement reads as excuses.
- Every acknowledged threat needs a direction, and ideally a magnitude.
- Name your own worst threat before a reviewer does. §14.3(D) requires that the self-assessment not be superficial.
- Do not apply this taxonomy to qualitative work; use trustworthiness.

 Review Questions

1. The four-way taxonomy was developed for field experiments in social science. Which computing designs does it fit poorly, and what would a taxonomy built for our field look like?
2. Threats sections are near-universal and near-universally formulaic. What incentive produces the formulaic version, and what reviewing practice would change it?

3. Mono-operation bias — one task, one system — affects most software engineering experiments, because using several is expensive. Is the field's tolerance of it justified? Propose a practical standard.
4. **PhD Regs 2024 §14.3(B)** asks about ecological and population external validity separately. Give a study strong in one and weak in the other, and say which matters more for a dissertation claiming practical relevance under §14.3(D).

PART V

Rigour, Reproducibility and Ethics

Everything to this point concerns getting an answer. This part concerns whether anyone should believe it — including you. Reproducibility, pre-registration, ethics and integrity are usually taught as compliance, which is why they are usually ignored. They are better understood as the practices that make a finding survive contact with a sceptic. The rules in this part are also the ones you can be failed on without ever having been taught them, so they are quoted here exactly.

What Part 5 Covers

- **Chapter 25 — Reproducibility and Artefact Evaluation.** Badging, containers, seeds, FAIR data, and packaging work so it survives you.
- **Chapter 26 — Pre-registration and Questionable Practices.** HARKing, p-hacking, forking paths, publication bias, and registered reports.
- **Chapter 27 — Research Ethics and Human Subjects.** Consent, data protection, re-identification risk, and the ethics of scraping and mining.
- **Chapter 28 — Academic Integrity and the Similarity Test.** The numeric thresholds that bind you, how to read an originality report, and authorship.
- **Chapter 29 — Generative AI in the Research Process.** The IUB position, read exactly; what is permitted, what is prohibited, and how to disclose.

Reproducibility and Artefact Evaluation

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 5 — Rigour, Reproducibility and Ethics

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **TODO** — learning outcomes not yet written.

Pre-registration, Registered Reports and Questionable Practices

- Part 1
- Part 2
- Part 3
- Part 4
- Part 5
- Part 6

Part 5 — Rigour, Reproducibility and Ethics

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **TODO** — learning outcomes not yet written.

Research Ethics and Human Subjects in IT Research

- Part 1
- Part 2
- Part 3
- Part 4
- Part 5
- Part 6

Part 5 — *Rigour, Reproducibility and Ethics*

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **TODO** — learning outcomes not yet written.

Academic Integrity, Plagiarism and the Similarity Test

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 5 — Rigour, Reproducibility and Ethics

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **TODO** — learning outcomes not yet written.

Generative AI in the Research Process

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 5 — Rigour, Reproducibility and Ethics

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **TODO** — learning outcomes not yet written.

PART VI

Producing and Defending Research

Research that is not written is not research. This part is procedural by design: the synopsis in the form your Departmental Research Committee expects, the thesis and the paper, the submission and the reviewers, and the defence. It is placed last because it depends on everything before it, and it is taught in one workshop week because by this point you should be executing decisions already made rather than making them.

What Part 6 Covers

- **Chapter 30 — The Synopsis.** The IUB structure, populated, and the approval path from supervisor to AS&RB.
- **Chapter 31 — Writing the Thesis and the Paper.** IMRaD, the abstract, tables and figures, and the reporting quality criteria you are judged on.
- **Chapter 32 — Publishing.** Venue selection, submission, responding to reviewers, and surviving rejection.
- **Chapter 33 — Presenting and Defending.** The comprehensive examination, the synopsis defence, the viva, and the four verdicts.

The Synopsis and the Research Proposal

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 6 — Producing and Defending Research

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **TODO** — learning outcomes not yet written.

Writing the Thesis and the Paper

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 6 — Producing and Defending Research

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **TODO** — learning outcomes not yet written.

Publishing: Venue, Submission and Peer Review

Part 1

Part 2

Part 3

Part 4

Part 5

Part 6

Part 6 — Producing and Defending Research

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **TODO** — learning outcomes not yet written.

Presenting and Defending Your Research

- Part 1
- Part 2
- Part 3
- Part 4
- Part 5
- Part 6
- Part 6 — Producing and Defending Research*

🎯 Learning Outcomes

By the end of this chapter you will be able to:

- **TODO** — learning outcomes not yet written.

Appendices

IUB Compliance Quick Reference: MS IT and PhD IT

Methodology does not exist in a vacuum. A design that is impeccable in the abstract can still fail you if it cannot be completed inside your degree's time limit, if its similarity report crosses a threshold, or if the paper it produces lands in a journal category that does not count. This appendix gathers the rules that actually bind you into one place, so that the rest of the book can talk about method without stopping to recite regulation.

Two documents govern, and they are *different documents with different numbers*. Read the column for your own degree.

Why this course exists — PhD Degree Regulations-2024, §10(iv)

Beyond the four specialisation courses, “the PhD student will be required to take the following two courses of 3 credit hours each: (a) Advanced Statistical Analysis/Hypothetical Testing/Research Methodology. (b) Dissertation Writing including PhD Proposal writing.”

This handout is course (a). Chapter 30 and Chapter 31 overlap deliberately with course (b).

This appendix is a map, not the territory

Regulations are amended. Every figure here was transcribed from the 2024 editions approved by the Syndicate on 13 June 2024, and every clause number is given so you can check it. When a number here disagrees with the current document from the Directorate of Advanced Studies and Research, **the Directorate's document wins** — and please tell the course instructor so this appendix can be corrected.

A.1 The Two Governing Documents

	PhD IT	MS IT
Governing document	<i>Doctor of Philosophy (PhD) Degree Regulations-2024</i> [13]	<i>MPhil/MS/LLM/MSc (Hons)/MBA Degree Regulations-2024</i> [42]
NQF level	Level 8	Level 7
Written product	Dissertation	Thesis
Formatting standard	<i>IUB Theses/Synopses Guidelines-2023</i> [30], for both tracks	

A.2 Programme Structure

Item	PhD IT	MS IT
Coursework	18 CH (§41)	Research-based: 24 CH. Course-based: 36 CH (§7, §15)
Research / thesis credit	36 CH	6 CH (research-based); none (course-based)
Total degree	54 CH	30 CH or 36 CH
Load per semester	9 CH (§41)	12 CH in semesters 1–2 (§15)
Semester duration	18 weeks; 16–18 with 2 weeks of examinations (§41, §53)	18 weeks, 2 reserved for examinations (§15, §17)

Item	PhD IT	MS IT
Regular semesters	16 maximum	4
Minimum duration	3 years / 6 semesters (§16)	2 years / 4 semesters (§13)
Maximum duration	8 years / 16 semesters (§16)	4 years; 5 only on force majeure (§13)
Class strength	Minimum 5 to commence (§10(vii))	Minimum 8, maximum 40 (§19)

A.3 This Course: Assessment and Examination

The weights are *identical* across the two tracks, and they are fixed by regulation — an instructor cannot reallocate them. Appendix H maps the course’s portfolio of work onto these components rather than adding anything on top of them.

Component	Weight	Basis
Sessional	20%	Four sub-components below
Participation / behaviour / group work	5%	
Quiz (surprise, 2–5 quizzes, 5 questions each)	5%	
Assignment (at least one)	5%	
Presentation / seminar	5%	Marked on information and references included, logical reasoning, and organisation of material
Mid-term examination	30%	10–20 minutes, individual or group
Final-term examination	50%	After 7–8 weeks of teaching
		Whole syllabus; 20–30% weight to pre-midterm material

PhD Regs 2024 §51.1–51.2; MPhil/MS Regs 2024 §36.

Paper patterns

The mid-term is the same in both tracks. The final differs only in the clock.

Paper	Question type	Marks	Time (PhD)	Time (MS)
Mid-term (30 marks)	Objective, 10 or 20	10	15 min	15 min
	Short answer, 5 at 2 marks	10	25 min	25 min
	Essay, 1	10	35 min	35 min
	Total	30	1 h 15	1 h 15
Final (50 marks)	Objective, 20 at 1 mark	20	30 min	20 min
	Short answer, 7 at 2 marks	14	35 min	40 min
	Essay, 2–4	16	55 min	60 min
	Total	50	2 h	2 h

📌 Attendance — PhD §50, MPhil/MS §35

80% attendance is required *in each component* to sit the final examination. Below 80% is treated as failure in that course, with all the consequences that has for good standing. Between 70% and 80% may be made up by extra classes or assignments arranged by the teacher, but only for unavoidable circumstances.

Grading — where the two tracks genuinely diverge

This is the difference students most often get wrong. The PhD scale is harder in every respect: a higher pass mark, a higher CGPA floor, and a “C” that is already disqualifying.

	PhD IT (§55, §56)	MS IT (§38, §39)
Pass percentage	65%	60%
Failing grade	D, at 64% or below	D, at 59% or below
“C” band	65–69%, remarked <i>Poor</i>	60–69%, remarked <i>Satisfactory</i>
CGPA for the degree	3.00, with no C grade	2.70, with no D grade
Probation after sem 1	SGPA 2.70	SGPA 2.30
Dropped below	CGPA 3.00 in later semesters	CGPA 2.70 in later semesters
Per-course floor	GPA 2.75 in <i>each</i> course before the comprehensive examination (§11)	—

! Reading the grade table backwards

A PhD scholar who scores 67% in this course has passed the course and simultaneously acquired a “C” that disqualifies them from the degree until it is repaired by repeating the course. The regulation permits repeating a course to clear a C or to improve a B (§56(v)). Notice also that fractional marks round *up* to the next whole figure: 64.10 becomes 65.00, which for a PhD scholar is the difference between D and C.

A.4 Milestones After the Coursework

Milestone	PhD IT	MS IT
Supervisor	Consent at admission; appointed by the Board of Studies (§21)	Assigned by end of semester 2 (§8)
Comprehensive examination	Required. Within 60 days of completing coursework, at CGPA 3.0. Written and oral, 65% to pass each. One retake within 30 days, then dropped (§11)	None , except in the Institute of Business, Management and Administrative Sciences and the Department of Law (§12)
Synopsis submission	Not later than completion of the 6th semester (§13(iii))	To the supervisor within 8 weeks of the start of semester 3; DRC requires CGPA 2.70 first (§8, §10(i))
Title approval	Board of Studies on DRC recommendation	Fixed by end of semester 3 (§47(iii))

Milestone	PhD IT	MS IT
Synopsis defence	Before the Departmental Research Committee, then BoS/BoF, then AS&RB. Evaluated by at least one external referee from a national university, chosen from a list of 6 experts supplied by the supervisor (§13)	Presented and defended before the Departmental Research Committee (§8)
Publication required	Yes. One article in a W-category journal <i>or</i> two in X-category, for Science disciplines. Scholar must be first author, the work must be relevant to the dissertation, and it must be published <i>after</i> synopsis approval (§15)	No publication requirement for the award of the degree
Examiners	Four (five with a co-supervisor): the supervisor, two foreign externals for evaluation, and one national external for the viva (§33)	Two: the supervisor as internal, and one external chosen by the Dean from a panel of at least six (§48)
Defence	Open defence, publicly announced by DASR, after positive external reports (§33)	Viva voce in the presence of the external examiner; open defence preferred (§48)
Verdicts	Pass / Pass with minor amendments / Deferred, re-defence within 180 days / Fail (§33(vii))	Pass / Pass with minor amendments / Deferred for resubmission within six months / Fail (§48(xi))

💡 What this means for how you choose a method

The two timetables reward different designs.

An MS IT scholar has roughly two semesters of research time and no publication gate. A design that yields a defensible answer in one pass — a well-scoped case study (Chapter 12), a survey with an existing validated instrument (Chapter 13), a focused controlled experiment (Chapter 10) — is the sensible shape.

A PhD scholar must publish before submitting, which means the research has to produce a publishable increment *part-way through*, not only at the end. That favours a programme of linked studies: a systematic review that stands alone as a paper (Chapter 5), then the empirical or design-science work it motivates (Chapter 9). Plan the first paper into the synopsis.

A.5 Integrity Thresholds

These are identical in both documents, and they are numeric, so there is no room to argue about them after the fact.

🚀 Similarity — PhD §32(III), MPhil/MS §49(3)

- Overall similarity index: **19% or less**.
- From any *single* source: **less than 5%**.
- In the findings and conclusion section specifically: **not more than 9%** — “the Similarity index should be considered very seriously in the section of findings and conclusion”.

A similarity test must be run before the thesis or dissertation goes to the external experts. References, bibliography and tables of contents are removed before submission for checking, because leaving them in inflates the index (§29 IV(iv)).

💡 Interpreting the report, not just the number

The regulations are unusually careful here, and Chapter 28 takes this apart properly. A similarity index is not a plagiarism verdict: matches with proverbs, universal truths, common phrases and proper nouns are expected, and matches against your own earlier work may be ignored if the material is cited (§29 V). Conversely, an index of 8% that is a single uncited paragraph from one source is plagiarism, and an index of 18% spread over correctly quoted and cited material may not be. The instructor or supervisor delivers the verdict after interpreting the report — the software does not.

A.6 Generative AI

The IUB position is stricter than most international publisher policies and it is internally in tension, so read both clauses rather than one. Chapter 29 is devoted to working within it.

🔗 Machine-generated text — PhD §32(I)(C)(vi), MPhil/MS §49(C)(vi)

Listed among activities that “still count as academic cheating and plagiarism if done without permission from the original artists/creators”:

“(vi) Using ChatGPT and similar machine-generated text”.

🔗 Permitted use — PhD §32(IV), MPhil/MS §49(4)

“Researchers may use ChatGPT, Artificial Intelligence (AI) and AI-assisted technologies to understand basic phenomena of anything and should not replace the key researcher tasks such as producing scientific insights, analyzing and interpreting data or drawing scientific conclusions. The authors are responsible and accountable for the contents of the work and should not rely solely on AI-generated content.”

Read together, the workable reading is: AI may help you *understand*; it may not *produce* the intellectual content, and it may not generate text that then appears in your submission. You remain accountable either way. Chapter 29 gives a permitted/prohibited table for this course and a disclosure statement template, and Appendix C repeats the template in fill-in form.

A.7 Research Ethics

🔗 Academic integrity — PhD §29(I)

The regulations adopt the International Center for Academic Integrity’s 2022 definition: academic integrity is “commitment, even in the face of adversity, to six fundamental values: honesty, trust, fairness, respect, responsibility, and courage. From these values flow principles of behavior that enable academic communities to translate ideals into action.”

For publication and theses, COPE guidelines are to be followed (§29 IV(i)) [8]. Plagiarism cases are handled under the HEC Anti-Plagiarism Policy [1].

The dissertation itself is also assessed on methodological quality, and the criteria are written into the regulations rather than left to the examiner. **PhD Regs 2024 §14.3** sets out separate checklists for qualitative and quantitative work; Chapter 23 and Chapter 24 map the course’s material onto them directly, and Appendix C turns them into a pre-submission checklist.

 The four quality tests a PhD dissertation must satisfy

1. **Qualitative quality** (§14.3A): does the research illuminate the subjective meaning, actions and context of those researched? Is the design responsive to real-life settings? Is the description detailed enough for a reader to interpret meaning and context?
2. **Quantitative quality** (§14.3B): reliability, validity, internal validity, external validity — ecological and population — and replicability.
3. **Appropriateness of methods to aims** (§14.3C): the methods justified against the objectives, and the results shown to support them.
4. **Relevance to policy and practice** (§14.3D): usefulness established, implications discussed, and the assessment “must not be superficial and lacking substance”.

Note also §14.1: the research area should correspond to community needs at regional and local levels, comply with the national research agenda, and preferably coincide with the Sustainable Development Goals.

A.8 What This Appendix Does Not Cover

Admission criteria, fee schedules, migration and credit transfer, semester freezing, grievance procedure, sub-campus processing, and payment of foreign evaluators are all in the source documents and are not reproduced here. If you need them, go to the Directorate of Advanced Studies and Research rather than to a second-hand summary — including this one.

Statistical Test Selection and Formula Sheet

Templates and Checklists

Five Worked IT Studies

Exercises and Answer Sketches

Annotated Open-Access Reading List

Sixteen-Week Schedule and Assessment Map

A mapping from the 33 chapters onto a standard semester of three contact hours per week, with the assessment points that **PhD Regs 2024 §51** and **MPhil/MS Regs 2024 §36** require. Adjust it to your own calendar; the dependencies between chapters are what matter, not the week numbers.

📌 Semester shape — PhD §53, §41; MPhil/MS §15, §17

The semester is 18 weeks, of which **2 are reserved for the mid-term and final examinations**, leaving 14 to 16 weeks of teaching. This schedule assumes 16. A contingency for 14 is given at the end of this appendix.

This is a 3(3–0) course: three credit hours of theory, three contact hours of class work per week, no laboratory hour. Practical work at this level is expected to be accomplished through the research itself rather than through scheduled labs (§51.2(v)).

H.1 The Teaching Schedule

Wk	Chapters	In class	Portfolio due
1	1 Contribution 2 Philosophy	What a contribution is; the paradigms behind method choice	—
2	3 Questions 4 Publication system	FINER and PICOC on the students' own topics; HJRS categories	P1 Topic statement
3	5 SLR	Protocol workshop: search strings built and run live	—
4	6 Grey literature 7 Bibliometrics	Snowballing; Zotero and VOSviewer at the keyboard	P2 SLR protocol <i>Quiz 1</i>
5	8 Appraisal 9 Design science	Three-pass reading of one assigned paper; DSR artefact types	—
6	10 Experiments 11 Quasi-experiments	Designing a code-review experiment; confounding in repository data	—
7	12 Case study 13 Survey	Unit of analysis; instrument critique	P3 Design choice memo
8	Mid-term 14 Qualitative	Mid-term examination, Chapters 1–13 ; then coding practice	—
9	15 Mixed methods 16 Benchmarking	Joint displays; workload characterisation	—
10	17 Measurement 18 Sampling and power	Operationalising a construct; a power calculation each	P4 Draft instrument
11	19 Inference	The p-value misinterpretation clinic; effect sizes in R	<i>Quiz 2</i>
12	20 Regression 21 SEM	Model interpretation; a PLS-SEM run in semnr	—

Wk	Chapters	In class	Portfolio due
13	22 ML evaluation 23 Qualitative analysis	A leakage audit on a supplied notebook; coding reliability	P5 Analysis plan
14	24 Validity 25 Reproducibility	Writing a threats section; packaging an artefact	<i>Quiz 3</i>
15	26 Pre-registration 27 Ethics 28 Integrity 29 Generative AI	The rigour and integrity block; similarity report interpretation	P6 Pre-registration
16	30 Synopsis 31 Writing 32 Publishing 33 Defending	Production workshop and defence-style presentations	P7 Synopsis draft Presentation

H.2 The Portfolio and How It Maps onto the Regulated Weights

The seven portfolio pieces are one continuous piece of work: each is a section of the synopsis you will actually submit. They run *inside* the weights fixed by regulation, not on top of them.

Regulated component	Weight	How it is earned in this course
Participation / group work	5%	Weekly seminar; the week 5 three-pass reading and the week 11 clinic are the two graded discussions
Quiz	5%	Three surprise quizzes (weeks 4, 11, 14), 5 questions each, as §51.1(b) prescribes
Assignment	5%	The portfolio, P1–P7, marked as one submission at the end
Presentation / seminar	5%	Week 16, 15 minutes, run as a mock DRC synopsis defence with questions
Mid-term examination	30%	Week 8, Chapters 1–13, 30 marks in 1 h 15
Final-term examination	50%	Whole syllabus, 50 marks; 20–30% of the weight on pre-midterm material

The portfolio is worth more than 5%, whatever the table says

Read that table honestly and it looks perverse: the largest piece of work in the course carries the smallest weight, because the regulation caps assignments at 5% and cannot be varied by the instructor.

It is not perverse, because of where the essay questions come from. The mid-term essay and the two-to-four final essay questions are drawn from the same material the portfolio makes you work through — justify a design against alternatives, state the threats to validity of a given study, write a search string for a stated question. A student who has done P1–P7 properly has already rehearsed 80% of the marks. A student who has not will be writing those essays from memory of the lectures, which is a much worse position.

The seven portfolio pieces

Piece	What it must contain
P1 Topic statement	The problem, the gap, and why it matters here. Template C1 and C2. One page.
P2 SLR protocol	Research questions, PICOC, search strings per database, inclusion and exclusion criteria, quality checklist, extraction form. Templates C3 and C4.
P3 Design choice memo	The design you have chosen, two you rejected, and the constraint that decided it. Two pages.
P4 Draft instrument	The questionnaire, interview guide, experiment protocol or extraction sheet your design needs, with each item traced to a construct. Templates C7–C10.
P5 Analysis plan	The test or model per research question, the effect size you will report, the assumption checks, and the multiple-comparison correction. Written <i>before</i> any data exists.
P6 Pre-registration	P3–P5 assembled into the template of Chapter 26, with the deviations policy stated. Template C12.
P7 Synopsis draft	The IUB synopsis structure, populated. Template C14. This is the deliverable that outlives the course.

H.3 Two Tracks, One Course

The teaching is common; the portfolio target differs, because the two degrees demand different things of you (Appendix A).

	MS IT	PhD IT
P7 targets	A thesis synopsis deliverable within two semesters of research time	A dissertation synopsis with a first publishable increment identified
P2 scope	A focused review, roughly 25–40 primary studies	A review scoped to stand alone as a submittable paper
P6 emphasis	Feasibility: can this be finished and defended in time?	Publication plan: which venue, which category, and when relative to synopsis approval (§15)
Extra reading	Chapter 12, Chapter 13	Chapter 5, Chapter 9, Chapter 32

💡 A note for PhD scholars on the comprehensive examination

This course is examined in its own right, but its content reappears. The PhD comprehensive examination (§11) is a written and oral test of doctoral-level mastery, set by the teachers who taught the courses and marked with an external examiner, at 65% to pass. Methodology questions in that examination are drawn from exactly this syllabus. Keep the portfolio; it is the best revision material you will have.

H.4 If You Have Fourteen Teaching Weeks, Not Sixteen

PhD Regs 2024 §53 permits as few as 14. Compress as follows, and do not compress anything else:

- Merge weeks 9 and 10: Chapter 15 and Chapter 16 become assigned reading with a checkpoint, and Chapter 17 and Chapter 18 take the class time. Both merged chapters remain examinable.

- Merge weeks 12 and 13: teach Chapter 20 and Chapter 22 in class; Chapter 21 becomes a guided exercise for those whose design needs it, and Chapter 23 likewise for those doing qualitative work. Most students need one or the other, rarely both.

Do not compress Chapter 5 (week 3), the integrity block (week 15) or the production workshop (week 16). The first is where the portfolio is actually built, and the last two are where students learn the rules they can otherwise fail on without ever having been taught them.

H.5 For the Course File

PhD Regs 2024 §49 makes a course file compulsory and lists its contents. This appendix supplies several of them directly:

Required item	Where it comes from
Description of the course	The front matter of this volume
Course coding	From the departmental scheme of study
Weekly teaching schedule	The table in §1 of this appendix
Copy of material / outline distributed	This volume
Topic and evaluation record of assignments	The portfolio table, §2
Copy of the quiz / test	Instructor’s own; three per §2
Mid and final papers with key	Instructor’s own; pattern in Appendix A
Course award list	Instructor’s own

💡 Two assessment design notes for the instructor

Set at least one essay question that cannot be answered by recall. “Describe the PRISMA flow diagram” tests memory. “A student proposes to answer RQ1 with a survey of 40 developers recruited from one company. State the strongest threat to validity and the cheapest design change that would reduce it” tests the thing this course is for.

Give credit in the portfolio for a design that is honestly abandoned. P3 asks for two rejected designs and the constraint that decided it. A student who discovers in week 10 that their power calculation makes the planned experiment infeasible, and says so, has learned more than one who proceeds regardless. Mark that as a success, not a failure — it is precisely the judgement §14.3(C) asks examiners to look for in a dissertation.

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